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Algorithmes stochastiques pour l'estimation bayésienne et les processus de diffusion

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Abstract

This thesis focuses on the study of two Langevin-type algorithms designed for sampling a probability measure defined on $\mathbb{R}^d$. The first algorithm analyzed is a continuous-time version of the Stochastic Gradient Langevin Dynamics, which incorporates a stochastic sampling step into the traditional over-damped Langevin diffusion. The second algorithm is the Unadjusted Langevin Algorithm, which is the Euler discretization of the over-damped Langevin diffusion. Both methods are widely used for sampling from posterior distributions.

Our analysis is conducted within a weakly convex framework, parameterized using the Kurdyka–Łojasiewicz (KL) inequality. This approach allows us to address settings with vanishing curvature and is significantly less restrictive compared to the strongly convex case. We establish explicit non-asymptotic estimates for the simulation horizon required to achieve an ε-approximation (in terms of entropy) for each method. Special attention is given to the computational cost, considering the number of observations (n) generating the posterior distribution, the dimension of the ambient space (d) where the parameter of interest resides, and the parameter r involved in the KL inequality, which ranges from 0 (strongly log-concave case) to 1 (limiting Laplace case).

Additionally, we present and analyze a generalization of the weak log-Sobolev inequality, which was introduced in [Cattiaux et al., 2007]. This new functional inequality is applicable to a family of unbounded functions. We also investigate the quantitative behavior of the relative entropy along a diffusion semigroup, where the invariant measure satisfies this inequality.

Résumé

Cette thèse porte sur l'étude de deux algorithmes de type Langevin conçus pour l'échantillonnage d'une mesure de probabilité définie sur $\mathbb{R}^d$. Le premier algorithme analysé est une version en temps continu du Stochastic Gradient Langevin Dynamics, qui incorpore une étape d'échantillonnage stochastique dans la diffusion de Langevin suramortie traditionnelle. Le second algorithme est l'algorithme de Langevin non-ajusté, qui correspond à la discrétisation d'Euler de la diffusion de Langevin suramortie. Ces deux méthodes sont largement utilisées pour l'échantillonnage de distributions postérieures.

Notre analyse est menée dans un cadre faiblement convexe, paramétré à l'aide de l'inégalité de Kurdyka–Łojasiewicz (KL). Cette approche permet de traiter des configurations avec une courbure décroissante et est beaucoup moins restrictive que le cas fortement convexe. Nous établissons des estimations explicites non-asymptotiques du temps de simulation nécessaire pour obtenir une ε-approximation (en termes d'entropie) pour chaque méthode. Une attention particulière est accordée au coût de calcul, en tenant compte du nombre d'observations (n) générant la distribution postérieure, de la dimension de l'espace ambiant (d) où se trouve le paramètre d'intérêt, ainsi que du paramètre r impliqué dans l'inégalité KL, qui varie de 0 (cas fortement log-concave) à 1 (cas limite de Laplace).

En outre, nous présentons et analysons une généralisation de l'inégalité de Sobolev logarithmique faible, introduite dans [Cattiaux et al., 2007]. Cette nouvelle inégalité fonctionnelle est applicable à une famille de fonctions non-bornées. Nous examinons également le comportement quantitatif de l'entropie relative le long d'un semi-groupe de diffusion, dont la mesure invariante satisfait cette inégalité.

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Introduction

This work is devoted to the study of two Langevin-type algorithms conceived for sampling a probability measure defined on $\mathbb{R}^d$, which is supposed to be absolutely continuous with respect to the Lebesgue measure and whose density μ is written as

$$\mu(\theta) \propto e^{-U(\theta)}, \quad \theta \in \mathbb{R}^d, \tag{1}$$

where $U : \mathbb{R}^d \to \mathbb{R}$ is a $\mathcal{C}^2(\mathbb{R}^d)$ convex function and ∇U is $L-$Lipschitz. With a slight abuse of notation, throughout this work we refer to both the probability measure and its density function as μ.

Sampling a probability measure μ that can be written as in (1), is a fundamental problem of applied mathematics, as it is involved in Bayesian estimation, [Gelman et al., 2004], in high-dimensional statistics, [Dalalyan and Tsybakov, 2012], in machine learning, [Andrieu et al., 2003], and in partial differential equations, [Lelièvre and Stoltz, 2016], among others.

(Over-damped) Langevin dynamics (LD for short) is a popular diffusion process to solve the sampling problem. It is defined by the stochastic differential equation

$$\mathrm{d}\vartheta_t = -\nabla U(\vartheta_t)\mathrm{d}t + \sqrt{2}\mathrm{d}B_t, \quad t > 0, \tag{2}$$

where $(B_t)_{t\geq 0}$ is a $d-$dimensional standard Brownian Motion. Standard results on Markov processes guarantee the strong existence and ergodicity of the continuous-time process $(\vartheta_t)_{t\geq 0}$ which converges in many senses (for example: Wasserstein, $\mathbb{L}^2_\mu(\mathbb{R}^d)$, entropy, total variation) towards its invariant distribution μ (see [Pavliotis, 2016]).

Although LD is a popular algorithm in various applications, there are several practical settings in which computing ∇U is computationally infeasible, or even intractable. For example, in Bayesian estimation, μ represents a posterior distribution, that is, we have n i.i.d. observations $X_1, \ldots, X_n$ distributed according to a density function p_θ where $\theta \in \mathbb{R}^d$ is the parameter of interest and then μ is the density function of $(\theta|X_1, \ldots, X_n)$. In this framework, the function U, is now rewritten as $\frac{1}{n}\sum_{i=1}^n U_i$, where U_i is a function associated with the observation X_i, $1 \leq i \leq n$. If the number of observations n and the dimension d in which the parameter lies are large values, then computing ∇U becomes problematic.

In this work, we manage to deal with a continuous-time version of the Stochastic Gradient Langevin Dynamics (SGLD) algorithm, whose discrete-time version was proposed by [Welling and Teh, 2011] to study situations like the one described for Bayesian estimation. Instead of using the function U in the LD equation, the continuous-time SGLD includes a sampling strategy among the set of observations

$X_1, \ldots, X_n$. Later, we provide a formal definition of this method and investigate its convergence properties.

Another problem that arises when we use the LD to sample the measure μ is that, in general, simulating a continuous-time diffusion is not practical, since it is almost never possible to find the explicit solution of (2). In the past ten years, a myriad of papers have studied some discretisation strategies of (2) and, among them, the most popular approach is certainly the first-order Euler-Maruyama scheme, known as the Unadjusted Langevin Algorithm (ULA, for short). With a fixed step size $h > 0$, ULA is defined as

$$\theta_{(k+1)h} = \theta_{kh} - h\nabla U(\theta_{kh}) + \sqrt{2h}\xi_{k+1}, \quad k \geq 0,$$

where $(\xi_k)_{k\geq 1}$ are standard Gaussian random variables in $\mathbb{R}^d$, mutually independent and independent of θ_0.

Most works analyzing the convergence of ULA and SGLD require restrictive assumptions such as strong convexity of the function U. In our work, we have chosen to study a weakly convex framework and parameterize this lack of strong convexity with the help of the Kurdyka-Łojasiewicz inequality [Kurdyka, 1998, Lojasiewicz, 1963], which is a standard tool in optimization to describe the transition between convexity and strong convexity and allows to describe explicit bounds.

For a complete understanding, in this introductory chapter we first present the mathematical background, then the state-of-the-art and at the end an overview of the contributions of our work and the structure of the thesis.

General background

One of the main objectives of inferential statistics is to estimate a characteristic of interest from a sample. Assume, for example, that we have n i.i.d. observations $X_1, X_2, \ldots, X_n$, sampled according to a probability distribution p_θ, where $\theta \in \mathbb{R}^d$ is the parameter of interest. In other words, we would be interested in finding the *real value* θ^* of θ that generated the observations.

There are essentially two approaches to estimate this parameter. The first is the classical or frequentist approach, which proposes, for example, methods such as maximum likelihood estimation (MLE, for short), where an estimator of θ is obtained using the optimization problem

$$\widehat{\theta}_{\text{MLE}} := \arg\max \prod_{i=1}^{n} p_\theta(X_i).$$

This is the most popular technique for deriving estimators and, under mild assumptions, the maximum likelihood estimator is proved to be consistent and asymptotically normal, see [Casella and Berger, 2024]. However, since $\widehat{\theta}_{\text{MLE}}$ is found by a maximization problem, it is susceptible to the problems associated with that process, among them that of numerical instability.

We focus our attention on the Bayesian approach, which is fundamentally different from the classical approach. In Bayesian inference, θ is considered a random variable whose probability distribution is commonly called the prior distribution and

is denoted by π_0. This prior distribution could be chosen on the basis of previous knowledge and even before observing the data.

After specifying the prior distribution π_0, then the posterior distribution is computed by Bayes' rule as

$$\mu(\theta) \propto \pi_0(\theta) \prod_{i=1}^{n} p_\theta(X_i),$$

and this is the probability distribution of θ conditional to $(X_1, X_2, \ldots, X_n)$. Note that, for simplicity, we omitted the dependence of μ on n and $(X_1, X_2, \ldots, X_n)$. A classical estimator of θ is the Bayesian estimator, also known as the posterior mean,

$$\widehat{\theta}_{\mathrm{B}} := \int_{\mathbb{R}^d} \theta \mathrm{d}\mu(\theta). \tag{3}$$

This estimator is usually consistent, but, generally, it is not explicit. Therefore, it is usual to define an algorithm to sample the probability measure μ and then estimate θ using that sample. We focus our attention on Langevin-type algorithms.

Langevin sampling

For the sake of completeness, we redefine some concepts that were already briefly introduced earlier. Assume that the measure μ for sampling can be written as

$$\mu(\theta) \propto e^{-U(\theta)}, \quad \theta \in \mathbb{R}^d,$$

where $U \in \mathcal{C}^2(\mathbb{R}^d)$ is a convex function and ∇U is L-Lipschitz. Under these conditions, μ is the unique invariant distribution of the over-damped Langevin Dynamics (LD), which is defined by

$$\mathrm{d}\vartheta_t = -\nabla U(\vartheta_t)\mathrm{d}t + \sqrt{2}\mathrm{d}B_t, \quad t \geq 0, \tag{4}$$

where $(B_t)_{t\geq 0}$ is a $d-$dimensional standard Brownian Motion. Thus, the probability measure μ can be approximated using the long-time ergodic convergence of $(\vartheta_t)_{t\geq 0}$ towards μ. A popular estimator of the posterior mean, which requires a single path of (4), is the Césaro average

$$\widehat{\theta}_t := \frac{1}{t}\int_0^t \vartheta_s \mathrm{d}s, \qquad t > 0.$$

This estimator depends on t and is based on the exact solution of (4). In practice, we use a numerical approximation of the LD (ULA, for example) and then its Césaro average. That is, let $(\theta_{kh})_{k\geq 0}$ be a discrete-time approximation of $(\vartheta_t)_{t\geq 0}$ with step size $h > 0$, then the Césaro average of $(\theta_{kh})_{k\geq 0}$ is defined as

$$\widetilde{\theta}_{kh} := \frac{1}{k+1}\sum_{j=0}^{k} \theta_{jh}, \qquad k \geq 0.$$

As we discussed earlier, there are several situations where the use of LD is

problematic and, although other Langevin-type algorithms will be discussed later, we first present how to evaluate the convergence of an algorithm conceived to sample the measure μ.

Convergence rate and distances

Once again, we are interested in sampling the measure μ using an algorithm which produces observations $(\theta_{kh})_{k\geq 0}$, if it is defined in discrete-time with step size $h > 0$, or $(\vartheta_t)_{t\geq 0}$, if it is defined in continuous-time. In either case, the order of convergence of the algorithm can be measured by considering a distance, initially denoted as D, and a fixed $\varepsilon > 0$, as

$$k_\varepsilon := \inf \left\{k, D(\mathrm{Law}(\theta_{kh}), \mu) \leq \varepsilon\right\},$$

in the discrete-time case and

$$t_\varepsilon := \inf \left\{t, D(\mathrm{Law}(\vartheta_t), \mu) \leq \varepsilon\right\},$$

in the continuous-time case. In simple words, k_ε is the number of iterations (similarly, t_ε is the time) required to achieve an ε-error in terms of distance D. Clearly, the order of convergence depends on the distance considered.

In our work, we mainly mention the following distances between two probability measures ν and ν' on $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$, where $\mathcal{B}(\mathbb{R}^d)$ is the Borel σ-algebra on $\mathbb{R}^d$ and we assume that $\nu' << \nu$ for the Kullback-Leibler and Rényi divergences.

- Kullback-Leibler divergence (or Relative entropy):

$$\mathrm{KL}(\nu'|\nu) = Ent_\nu\left(\frac{\nu'}{\nu}\right) := \int_{\mathbb{R}^d}\left(\frac{\nu'}{\nu}\right)\log\left(\frac{\nu'}{\nu}\right)\mathrm{d}\nu.$$

  We interchangeably use Kullback-Leibler divergence or relative entropy.

- Total variation distance:

$$\|\nu - \nu'\|_{\mathrm{TV}} := \sup_{A\in\mathcal{B}(\mathbb{R}^d)} |\nu(A) - \nu'(A)|.$$

- p-Wasserstein distance, with $p \geq 1$:

$$\mathcal{W}_p(\nu, \nu') := \inf\left\{(\mathbb{E}[\|\theta - \theta'\|_2^p])^{\frac{1}{p}}, \theta \sim \nu, \theta' \sim \nu'\right\},$$

  where $\|\cdot\|_2$ denotes the Euclidean norm of $\mathbb{R}^d$.

- Rényi divergence, with $p > 0$ and $p \neq 1$:

$$\mathcal{R}_p(\nu'|\nu) := \frac{1}{p-1}\log\left(\left\|\frac{\nu'}{\nu}\right\|^p_{\mathbb{L}^p_\nu(\mathbb{R}^d)}\right),$$

  where $\|f\|^p_{\mathbb{L}^p_\nu(\mathbb{R}^d)} = \int_{\mathbb{R}^d}|f|^p\mathrm{d}\nu$.

The Kullback-Leibler divergence measures the information that is lost when ν' is used to approximate ν. The total variation distance measures the maximum absolute difference between the probabilities assigned to each event by the two probability distributions. Wasserstein distance is also known as the optimal transport cost. Rényi divergence generalizes various notions of entropy. It is important to mention that both divergences are not metric since they do not satisfy the triangular inequality.

There exist several links between these distances, some of them are:

- Pinsker's inequality
$$\|\nu - \nu'\|_{\mathrm{TV}} \leq \sqrt{\frac{1}{2}\mathrm{KL}(\nu'|\nu)}. \tag{5}$$

- If ν satisfies a log-Sobolev inequality with constant C_{LSI} (see (10)), then the Talagrand's inequality holds, that is,
$$\frac{1}{2C_{\mathrm{LSI}}}\mathcal{W}_2^2(\nu,\nu') \leq \mathrm{KL}(\nu'|\nu). \tag{6}$$

- If ν satisfies a Poincaré inequality with constant C_{PI} (see (9)), then the following inequality holds true
$$\max\left\{\mathrm{KL}(\nu'|\nu), \log\left(1 + \frac{1}{2C_{\mathrm{PI}}}\mathcal{W}_2^2(\nu,\nu')\right)\right\} \leq \mathcal{R}_2(\nu'|\nu). \tag{7}$$

The first two points above are well-known results and may be found in [Villani et al., 2009]. On the other hand, the third point above is a consequence of [Liu et al., 2020] and the Poincaré inequality.

The Poincaré and log-Sobolev inequalities are powerful tools of functional analysis which are studied in more detail in Chapter 1.

Convexity and Kurdyka-Łojasiewicz hypothesis

Returning to the sampling problem, we are interested in a probability measure $\mu(\theta) \propto e^{-U(\theta)}$, where $\theta \in \mathbb{R}^d$ and the potential U is a weakly convex function, that is, U is assumed to be convex but not strongly convex. Let us formalize these definitions and provide some examples.

A function $U : \mathbb{R}^d \to \mathbb{R}$ is *convex* if the following condition is satisfied

$$U(\alpha x + (1-\alpha)y) \leq \alpha U(x) + (1-\alpha)U(y), \quad \forall x, y \in \mathbb{R}^d, \quad \alpha \in (0,1).$$

The concept of convexity is closely related to the eigenvalues of the Hessian matrix of U. That is, for any function U twice differentiable, the spectrum of the Hessian matrix of U is denoted as $Sp(\nabla^2 U(\theta))$, and if U is also convex, then

$$0 \leq \underline{\lambda}_{\nabla^2 U}(\theta) := \inf Sp(\nabla^2 U(\theta)), \quad \forall \theta \in \mathbb{R}^d.$$

In simple words, if U is convex and twice differentiable then, for all $\theta \in \mathbb{R}^d$, the eigenvalues of $\nabla^2 U(\theta)$ are non-negative. Using the characterization of a convex

function through its eigenvalues, we can define the concept of strong convexity: a twice differentiable function U is *strongly convex* if there exists a strictly positive constant c, such that

$$0 < c \leq \underline{\lambda}_{\nabla^2 U}(\theta), \quad \forall \theta \in \mathbb{R}^d.$$

Usually, we say that U is c-strongly convex. Informally, a strongly convex function grows at least as fast as a quadratic function. For example, $\theta \to \frac{c}{2}\|\theta\|_2^2$ is c-strongly convex for any $c > 0$.

On the other hand, a function is *weakly convex* if it is convex but not strongly convex.

Example 1. *The following families of functions are weakly convex.*

a) $U(\theta) = (1 + \|\theta\|^2)^\alpha$, $\forall \theta \in \mathbb{R}^d$ *and* $1/2 \leq \alpha < 1$.

b) $W(\theta) = 1 + e^{-\sqrt{L}\theta}$, $\forall \theta \in \mathbb{R}$ *and* $L > 0$.

Let $\mathcal{C}^2(\mathbb{R}^d)$ be the set of real-valued functions defined in $\mathbb{R}^d$ that are twice continuously differentiable. Since we always assume that $U \in \mathcal{C}^2(\mathbb{R}^d)$, the definition of strong convexity is presented for twice differentiable functions. However, the general definition does not require this assumption. For a more in-depth study on convex functions, we refer to [Nesterov, 2013].

As mentioned above, we focus our analysis on weakly convex potentials and parameterize our framework with the help of the Kurdyka-Łojasiewicz (KL) inequality [Kurdyka, 1998, Lojasiewicz, 1963]. This tool is commonly used in optimization to describe the transition between convexity and strong convexity and allows finding explicit bounds.

We now introduce a parametric form of the KL inequalities following [Gadat and Panloup, 2022].

Hypothesis $(\mathcal{H}^{\mathbf{r}}_{\mathbf{KL}}(\mathfrak{c}, L))$**.** *We say that a function* $V : \mathbb{R}^d \to \mathbb{R}$ *satisfies a* $\mathcal{H}^{\mathbf{r}}_{\mathbf{KL}}(\mathfrak{c}, L)$*-condition if:*

a) V *is a* $\mathcal{C}^2(\mathbb{R}^d)$ *function,*

b) V *is convex and* $\min_{\theta \in \mathbb{R}^d} V(\theta) = V(\theta^*) > 0$,

c) ∇V *is* L*-Lipschitz and*

d) there exist some constants $0 \leq r < 1$ *and* $\mathfrak{c} > 0$ *such that*

$$\mathfrak{c} V^{-r}(\theta) \leq \underline{\lambda}_{\nabla^2 V(\theta)}, \quad \forall \theta \in \mathbb{R}^d.$$

In Example 1, the family of functions in part *a* satisfies hypothesis $\mathcal{H}^{\mathbf{r}}_{\mathbf{KL}}(\mathfrak{c}, L)$, while the one in part *b* does not satisfy the last condition. This assumption is the main hypothesis of our work and is studied in detail in Section 1.5.

Finally, we said that a function $e^{-U(\theta)}$ is *log-concave* (*strongly log-concave* or *weakly log-concave*) if U is convex (strongly convex or weakly convex, respectively).

An overview on Markov Diffusion semigroup

The following section has a more theoretical approach and could be skipped by the experienced reader.

In order to study the convergence of some Langevin-type algorithms, we rely on semigroup theory and functional analysis. Hence, we present some concepts that are necessary during the rest of the chapters and that we have mentioned above without defining them formally.

Notations

In this section we employ general notation to describe the concepts introduced, these are some of them:

- $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d), m)$: a probability space.
- $\mathbb{L}^p_m(\mathbb{R}^d)$: the set of real-valued functions defined on $\mathbb{R}^d$ that are measurable with respect to the measure m and has finite p-norm $\|f\|_{\mathbb{L}^p_m(\mathbb{R}^d)} = [\int_{\mathbb{R}^d} |f|^p \mathrm{d}m]^{1/p}$, $1 \leq p < \infty$.
- $\langle \cdot, \cdot \rangle$: the scalar product on $\mathbb{R}^d$.
- $\| \cdot \|_2$: euclidean norm on $\mathbb{R}^d$.

Although our objective is the study of a Markov diffusion semigroup, it is important to keep in mind that it is closely related to the description of the law of a Markov process, as we commented before.

Consider a stochastic process $(X_t)_{t\geq 0}$ defined in the probability space $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d), m)$ with fixed initial point x. We denote by $\mathcal{F}_t = \sigma(X_u, u \leq t)$, $t \geq 0$, the natural filtration of $(X_t)_{t\geq 0}$. Then, we said that $(X_t)_{t\geq 0}$ is a *Markov process* if it satisfies the Markov property, that is, for any $0 \leq s < t$, the law of $X_t|\mathcal{F}_s$ is the law of $X_t|X_s$. Informally, X_t depends on all previous information up to time s only through X_s, where $0 \leq s < t$.

Let us now present some basic definitions and well-known facts and refer to [Bakry et al., 2014] for a more in-depth study.

Definition 1 (Markov semigroup)**.** *Let $(P_t)_{t\geq 0}$ be a family of operators defined on some set of real-valued functions. Then $(P_t)_{t\geq 0}$ is a Markov semigroup if the following conditions hold true:*

i) For every $t \geq 0$, P_t is a linear operator sending bounded functions to bounded measurable functions.

ii) (Initial condition) $P_0 = I_d$.

iii) (Mass conservation) $P_t(\mathbb{1}) = \mathbb{1}$, where $\mathbb{1}$ is the constant function equal to 1.

iv) (Positivity-preserving) If $f \geq 0$ then $P_t f \geq 0$.

v) (Semigroup property) $P_{t+s} = P_t \circ P_s$, for all $t, s \geq 0$.

vi) (Invariant property) There exists a σ-finite measure m on $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$ that is invariant under $(P_t)_{t\geq 0}$, which means that for all bounded, positive and measurable functions f and for every $t \geq 0$,

$$\int_{\mathbb{R}^d} P_t f \mathrm{d}m = \int_{\mathbb{R}^d} f \mathrm{d}m.$$

As a consequence of positivity and mass conservation properties, by Jensen's inequality, we deduce that for any convex function $\phi : \mathbb{R} \to \mathbb{R}$ and any smooth enough function f,

$$P_t\phi(f) \geq \phi(P_t f).$$

In particular, taking $\phi(x) = |x|^p$, $x \in \mathbb{R}$ and $1 \leq p \leq \infty$, then $P_t|f|^p \geq |P_t f|^p$, $t \geq 0$. This property is used in Chapter 1.

Before presenting an example of a Markov semigroup, we establish an important link between Markov process and Markov semigroup. The Markov process $(X_t)_{t\geq 0}$ is *associated with the semigroup* $(P_t)_{t\geq 0}$ if for any function $f \in \mathbb{L}^1_m(\mathbb{R}^d)$ and any $x \in \mathbb{R}^d$,

$$P_t f(x) = \mathbb{E}[f(X_t)|X_0 = x], \quad t > 0.$$

One of the most basic examples is the Ornstein - Uhlenbeck (OU, for short) semigroup.

Example 2 (Ornstein - Uhlenbeck semigroup)**.** *The OU semigroup defined on $\{f : \mathbb{R} \to \mathbb{R}\}$ is*

$$P_t f(x) = \mathbb{E}_{\gamma_1}\left[f\left(e^{-t}x + \sqrt{1-e^{-2t}}X\right)\right], \quad x \in \mathbb{R},$$

where X is a random variable with standard Gaussian distribution and γ_1 is its density function. It is not difficult to verify that $(P_t)_{t\geq 0}$ is a Markov semigroup with invariant measure $m = \gamma_1$. The Markov process $(X_t)_{t\geq 0}$ solution of the stochastic differential equation

$$\mathrm{d}X_t = -X_t \mathrm{d}t + \sqrt{2}\mathrm{d}B_t, \quad t > 0,$$

is called the OU process and is associated with the OU semigroup.

A Markov semigroup is characterized by its infinitesimal generator. This concept is of crucial importance in our study, although in this introductory chapter we avoid going into more technical issues such as its domain of definition.

Definition 2 (Infinitesimal generator)**.** *The infinitesimal generator $\mathcal{L}$ of the Markov semigroup $(P_t)_{t\geq 0}$ with invariant measure m is defined in the domain*

$$\mathcal{D}(\mathcal{L}) = \left\{ f \in \mathbb{L}^2_m(\mathbb{R}^d), \lim_{t\searrow 0} \frac{P_t f - f}{t} \text{ exists and is in } \mathbb{L}^2_m(\mathbb{R}^d) \right\},$$

and for any function $f \in \mathcal{D}(\mathcal{L})$,

$$\mathcal{L}f := \lim_{t\searrow 0} \frac{P_t f - f}{t}.$$

The notation $t \searrow 0$ means that t converges decreasingly towards 0. We then introduce some fundamental properties of a semigroup and its infinitesimal generator that may be obtained using basic tools of differential calculus.

Proposition 1. *Let $f \in \mathcal{D}(\mathcal{L})$.*

a) $P_t f \in \mathcal{D}(\mathcal{L})$ and $\mathcal{L}P_t f = P_t \mathcal{L} f$ for any $t \geq 0$, so that the mapping $t \mapsto P_t f$ is continuous on $[0, \infty)$.

b) The mapping $t \mapsto P_t f$ is differentiable in $[0, \infty)$,

$$\frac{\mathrm{d}}{\mathrm{d}t} P_t f = \mathcal{L} P_t f, \quad t > 0,$$

and then

$$P_t f = f + \int_0^t \mathcal{L} P_s f \mathrm{d}s, \quad t > 0.$$

Let us return to the example of the OU semigroup.

Example 3. *Using Itô's lemma, we can verify that the infinitesimal generator of the OU semigroup is*

$$\mathcal{L} f(x) = f''(x) - x f'(x), \quad \forall f \in \mathcal{D}(\mathcal{L}).$$

The computations made in example 3 may be extended to a diffusion process, but first we formally define what a diffusion process is.

Definition 3 (Diffusion process)**.** *Consider the stochastic differential equation*

$$\begin{aligned} \mathrm{d}X_t &= b(X_t)\mathrm{d}t + \sigma(X_t)\mathrm{d}B_t, \quad t > 0, \\ X_0 &= x, \end{aligned} \tag{8}$$

where $b : \mathbb{R}^d \to \mathbb{R}^d$ and $\sigma : \mathbb{R}^d \to \mathbb{R}^d$ are Lipschitz functions and $(B_t)_{t \geq 0}$ is a d-dimensional Brownian Motion. The solution $(X_t)_{t \geq 0}$ of Equation (8) is a Markov process and is called a diffusion process.

Let $(P_t)_{t \geq 0}$ be the Markov semigroup associated with the diffusion process $(X_t)_{t \geq 0}$ solution of (8). Using the Itô lemma, we get that the infinitesimal generator of $(P_t)_{t \geq 0}$ is

$$\mathcal{L} f = \langle b, \nabla f \rangle + \frac{1}{2} Tr(\sigma^t \nabla^2 f \sigma), \quad \forall f \in \mathcal{D}(\mathcal{L}).$$

A Markov semigroup and its infinitesimal generator are naturally associated with a bilinear form defined in a subspace $\mathcal{A} \subset \mathcal{D}(\mathcal{L})$. As we commented before, we do not focus on rigorously describing the subspace $\mathcal{A}$, we only assume that it is an algebra.

Definition 4 (Carré du champ operator)**.** *The bilinear form*

$$\Gamma_{\mathcal{L}}(f, g) := \frac{1}{2} \left(\mathcal{L}(fg) - f \mathcal{L} g - g \mathcal{L} f \right),$$

defined for all $f, g \in \mathcal{A}$, is called the carré du champ operator of the Markov generator $\mathcal{L}$. We denote $\Gamma_{\mathcal{L}}(f,f) = \Gamma_{\mathcal{L}}(f)$.

In the particular case where $\mathcal{L}f = \langle b, \nabla f\rangle + \Delta f$, that is, $\mathcal{L}$ is the generator of a diffusion process with $\sigma(x) = \sqrt{2}I_d$, for any $x \in \mathbb{R}^d$. Then, for any $f, g \in \mathcal{A}$,

$$\Gamma_{\mathcal{L}}(f,g) = \langle \nabla f, \nabla g\rangle \quad \text{and} \quad \Gamma_{\mathcal{L}}(f) = \|\nabla f\|_2^2.$$

Throughout our work, we focus only on generators such that $\Gamma_{\mathcal{L}}(f) = \|\nabla f\|_2^2$, for any appropriate function f. In this context, we introduce the Dirichlet form associated with the infinitesimal generator $\mathcal{L}$, where m is the invariant distribution and $\Gamma_{\mathcal{L}}$ is the carré du champ operator.

Definition 5 (Dirichlet form)**.** *The Dirichlet form is defined as*

$$\mathcal{E}_m(f) := \int_{\mathbb{R}^d} \|\nabla f\|_2^2 \mathrm{d}m = -\int_{\mathbb{R}^d} f\mathcal{L}f \mathrm{d}m,$$

for any function f such that $\nabla f \in \mathbb{L}^2_m(\mathbb{R}^d)$.

The Dirichlet form appears recurrently in all the following chapters, where it plays an important role for some functional inequalities. In this introduction, we will not go into detail on this subject, since it is studied in Chapter 1.

Poincaré and log-Sobolev inequalities

We briefly present the definitions of these two functional inequalities mentioned in this introductory chapter and refer the reader to Chapter 1, which is entirely dedicated to the study of these and other functional inequalities.

For any function $f \in \mathbb{L}^1_m(\mathbb{R}^d)$, we define its *variance* as:

$$Var_m(f) := \int_{\mathbb{R}^d} (f - m[f])^2 \mathrm{d}m,$$

where $m[f] := \int_{\mathbb{R}^d} f \mathrm{d}m$ and when $f : \mathbb{R}^d \to \mathbb{R}^+$ is a non-negative function such that $\int_{\mathbb{R}^d} f|\log(f)|\mathrm{d}m < \infty$, we define its *entropy* as:

$$Ent_m(f) = \int_{\mathbb{R}^d} f\log(f)\mathrm{d}m - \int_{\mathbb{R}^d} f\mathrm{d}m \log\left(\int_{\mathbb{R}^d} f\mathrm{d}m\right).$$

In this definition, $0\log 0$ is interpreted as 0. The Poincaré and log-Sobolev inequalities upper bound the terms $Var_m(f)$ and $Ent_m(f^2)$ by the Dirichlet form of f, $\mathcal{E}_m(f)$.

Let $\mathcal{H}^1_m(\mathbb{R}^d) = \left\{f : \mathbb{R}^d \to \mathbb{R}; f \in \mathbb{L}^2_m(\mathbb{R}^d), \nabla f \in \left(\mathbb{L}^2_m(\mathbb{R}^d)\right)^d\right\}$, where ∇f is defined in the weak sense.

Definition 6 (Poincaré's inequality (PI))**.** *The probability measure m satisfies a PI if there exists a constant $C_{\mathrm{PI}}(m)$ such that for any $f \in \mathcal{H}^1_m(\mathbb{R}^d)$,*

$$Var_m(f) \le C_{\mathrm{PI}}(m)\mathcal{E}_m(f). \tag{9}$$

Definition 7 (Log-Sobolev inequality (LSI))**.** *The probability measure m satisfies an LSI if there exists a constant $C_{\mathrm{LSI}}(m)$ such that for any $f \in \mathcal{H}^1_m(\mathbb{R}^d)$,*

$$Ent_m(f^2) \leq C_{\mathrm{LSI}}(m)\mathcal{E}_m(f). \tag{10}$$

Some well-known results are as follows: LSI is stronger than a Poincaré inequality, that is, if a probability measure satisfies an LSI, then it also satisfies a PI. If m is log-concave then it satisfies a PI. On the other hand, if m is strongly log-concave then it satisfies an LSI.

State of the art results

Approximating measures have a long history and rely on Markov dynamics. We briefly introduce and describe below some recent results that are based on the over-damped Langevin diffusion. However, there is another approach which is based on the under-damped Langevin diffusion. See, for example, [Cheng et al., 2018], [Dalalyan and Riou-Durand, 2020], [Eberle et al., 2019] and more recently [Camrud et al., 2023].

From now on, we assume that $U : \mathbb{R}^d \to \mathbb{R}$ is a coercive function.

(Over-damped) Langevin Dynamics (LD)

$$\mathrm{d}\vartheta_t = -\nabla U(\vartheta_t)\mathrm{d}t + \sqrt{2}\mathrm{d}B_t. \tag{11}$$

Under mild conditions on the potential function U, [Chiang et al., 1987, Roberts and Tweedie, 1996] proved that the LD defined in (11) converges to the unique stationary distribution $\mu(\theta) \propto \exp(-U(\theta))$, $\theta \in \mathbb{R}^d$. The ergodicity and quantitative mixing properties of LD and many other sampling algorithms are popular subjects of research, initiated in probabilistic works around, roughly speaking, two strategies.

Coupling approaches: The first relies on path-wise considerations and dynamical properties of random dynamical systems and is built with some coupling argument and Lyapunov controls. We refer to the seminal contributions [Meyn and Tweedie, 2012, Khasminskii, 2012], which exploit the Doeblin coupling approach and total variation bounds. Many extensions may be derived from this Lyapunov approach and may lead to Wasserstein or $\mathbb{L}^2_\mu(\mathbb{R}^d)$ upper bounds, we refer to [Cattiaux and Guillin, 2017] and the references therein of the same authors for a description of the link between Lyapunov conditions and ergodicity.

Functional inequality approaches: The second strategy derives from spectral properties of Markov operators and is related to famous functional inequalities (Poincaré and log-Sobolev, among others). The general idea is to differentiate the distance along the time evolution and apply a Gronwall lemma to obtain a quantitative estimate of the long-time evolution of the semigroup. We refer to the seminal contributions of [Holley and Stroock, 1988, Bakry and Émery, 1985], and to [Bakry et al., 2014] for an almost exhaustive survey of all possible inequalities

and consequences on the ergodicity of Markov semigroups. Finally, we emphasize that strong links exist between the spectral and Lyapunov approaches, as pointed out by [Bakry et al., 2007]. If functional inequalities are strongly related to mixing properties, especially in quantitative terms, it is therefore necessary to develop a machinery that is able to assess these inequalities carefully, especially with a specific attention to our statistical setting where the target measure is weakly log-concave, which is a common feature of Bayesian posterior distributions.

Unadjusted Langevin Algorithm (ULA)

The Unadjusted Langevin Algorithm is also known as the Langevin Monte Carlo and refers to the Euler-Maruyama discretization of LD. This algorithm approximates $\mu(\theta) \propto \exp(-U(\theta))$, for $\theta \in \mathbb{R}^d$.

With a step-size $h > 0$, ULA is defined as

$$\theta_{(k+1)h} = \theta_{kh} - h\nabla U(\theta_{kh}) + \sqrt{2h}\xi_{k+1}, \quad k \geq 0, \tag{12}$$

where $(\xi_k)_{k\geq 1}$ are standard Gaussian random variables in $\mathbb{R}^d$, mutually independent and independent of θ_0.

The mixing properties of ULA have been largely investigated during the past decade among the statistical and machine learning communities, strongly motivated by learning methods such as Exponentially Weighted Aggregate introduced by [Dalalyan and Tsybakov, 2012], which involves sampling a non-log-concave and heavy-tailed posterior distribution. Then, several works derive some quantitative estimates in simple or sophisticated frameworks.

Strongly log-concave situations: [Dalalyan, 2017] establishes the cost of ULA to obtain an ε TV bound in terms of d and ρ when the target measure is ρ-strongly log-concave and proposes a penalized version of ULA to circumvent the lack of strong log-concavity when the target distribution is only log-concave. Since this pioneering paper, a huge impressive literature ensued.

Log-concave situations: Other papers relax the strongly convex assumption on U using a modification of the numerical scheme (12), for example: we refer to [Dalalyan et al., 2022] where they studied in depth a penalized ULA in a non-strongly log-concave situation. Among all these papers, the lack of strong log-concavity is first addressed by modifying the initial ULA using a surrogate and asymptotically vanishing penalty. Second, these papers assume that a noiseless gradient of the log-posterior is available at each iteration of the algorithm, which may not be realistic, especially when $U = \frac{1}{n}\sum_{i=1}^n U_i$ with large n. Finally, [Altschuler and Talwar, 2023] provides some upper-bounds on the mixing time of ULA in both *constrained* convex and strongly convex cases using some explicit coupling and projections.

Coupling approaches: Explicit and non-asymptotic bounds have been obtained in different papers, using coupling techniques and Lyapunov functions. Among them, we refer to [Durmus and Moulines, 2017], on the total variation

distance, [Durmus and Moulines, 2019], on the Wasserstein distance, and [Durmus et al., 2019], on the Kullback-Leibler divergence. In particular, the previous paper achieves the best rate known for a strongly log-concave measure.

Functional inequality approaches: On the machine learning side, mixing of ULA has also received an impressive recent amount of interest: [Vempala and Wibisono, 2019] proves the convergence in the Kullback-Leibler divergence assuming that the posterior distribution, defined as e^{-U}, satisfies a log-Sobolev inequality and the Hessian of U is bounded. Later, [Balasubramanian et al., 2022] developed a close analysis using the Fisher information distance and the Poincaré inequality. Metropolis-Adjusted Langevin Algorithm (MALA below), which is close to ULA (with the addition of a Metropolis correction), was studied in [Chewi et al., 2021] using different divergences for the class of smooth and strongly convex potentials. More recently, in a weakly log-concave setting, [Erdogdu and Hosseinzadeh, 2021] studied the convergence in terms of the Kullback-Leibler divergence of ULA using a weak version of log-Sobolev inequality and established a link between the tail assumption in e^{-U} and the inequality mentioned. As emphasized in Section 1.5, the assumptions they made are closely related to ours and we will provide further details later. Using similar ideas, the authors of [Chewi et al., 2022a] assume an α-tail behavior for the target measure and then, with the help of the Latała-Oleszkiewicz or *modified* log-Sobolev inequalities, they guaranteed the convergence of ULA in the Rényi divergence.

In order to summarize some of the above results, Table 1 presents the iteration complexity in different metrics and the main assumptions made in these works. For some metrics, we present the result using its square root, since it is easier to compare it using the inequalities (5), (6), and (7).

Work	Algorithm	Assumptions	Convergence rate	Metric
[Dalalyan, 2017]	ULA	strongly convex	$d\varepsilon^{-2}$	TV
[Dalalyan et al., 2022]	penalized ULA	convex and 2nd moment	$d^2\varepsilon^{-6}$	$\mathcal{W}_2$
[Balasubramanian et al., 2022]	averaged ULA	PI	$d^2\varepsilon^{-2}$	TV
[Durmus et al., 2019]	ULA	strongly convex	$\sqrt{d\varepsilon^{-1}}$	$\sqrt{\mathrm{KL}}$
[Vempala and Wibisono, 2019]	ULA	LSI	$\sqrt{d\varepsilon^{-1}}$	$\sqrt{\mathrm{KL}}$
[Erdogdu and Hosseinzadeh, 2021]	ULA	tail $\sim \lVert\theta\rVert^\alpha$ and β-Hölder gradient	$d^{\frac{1}{\beta}+\frac{1+\beta}{\beta}(\frac{2}{\alpha}-\mathbb{1}_{\{\alpha\neq 1\}})}\varepsilon^{-\frac{1}{\beta}}$	$\sqrt{\mathrm{KL}}$

Work	Algorithm	Assumptions	Convergence rate	Metric
[Chewi et al., 2021]	MALA	strongly convex	$\sqrt{d}$	TV
[Chewi et al., 2022a]	ULA	tail $\sim \|\theta\|^\alpha$ and β-Hölder gradient	$d^{\frac{1}{\alpha}(1+\frac{1}{\beta})-\frac{1}{\beta}}\varepsilon^{-\frac{1}{\beta}}$	$\sqrt{\mathcal{R}_2}$

Table 1: Comparison of convergence rate. In almost all works, it is assumed that ∇U is $L-$Lipschitz, except for those that use β-Hölder.

We point out that using (7), we observe that if the potential U is weakly convex and its gradient is L-Lipschitz, the best convergence rate (in terms of entropy) obtained to our knowledge is

$$d^{\frac{4}{\alpha}-1}\varepsilon^{-1},$$

proved in [Chewi et al., 2022a].

Lower bounds: Recently, [Chewi et al., 2022b] in dimension one and [Chewi et al., 2024] in dimension $d \geq 2$ have investigated the minimum number of queries required for sampling a measure which is strongly log-concave and log-smooth. This type of result would allow us to identify optimal algorithms for sampling; however, it is an axis of study that is just being developed.

Stochastic Gradient Langevin Dynamic (SGLD)

This last framework studies the behavior of ULA when an additional noise term is incorporated in (12), which generates a perturbed discretization:

$$\theta_{(k+1)h} = \theta_{kh} - h\widehat{\nabla U}(\theta_{kh}) + \sqrt{2h}\xi_{k+1}, \tag{13}$$

where $\widehat{\nabla U}(\theta_{kh})$ is an unbiased estimation of $\nabla U(\theta_{kh})$ at iteration k. For example, if $U = \frac{1}{n}\sum_{i=1}^n U_i$, then $\widehat{\nabla U}(\theta_{kh}) = \frac{1}{B}\sum_{i=1}^B \nabla U_{I_i}(\theta_{kh})$ will appear in (13), where $1 \leq B \leq n$ and $I_1, \ldots, I_B \overset{i.i.d.}{\sim}$ uniform$([n])$. The calculation of the full gradient over the entire dataset is replaced by the computation of the gradient over a subset.

Stochastic Gradient Langevin Dynamics (SGLD below) has attracted the interest of several works: [Welling and Teh, 2011] introduced this method and described its efficiency from a numerical point of view in the particular case of Bayesian learning, which is exactly our framework. Some recent advances and related contributions may also be cited: in [Raginsky et al., 2017], the authors derive some non-asymptotic bounds (in terms of 2-Wasserstein distance) assuming a dissipative and smooth potential for optimization purposes.

The contribution of [Xu et al., 2018] is also related to our work: the authors develop a machinery for the study of SGLD based essentially on the Poincaré inequality, but the way the lower bound on the spectral gap involved in the ULA is dealt with appears to be inappropriate. In particular, the diffusion involved in SGLD is used at a very low temperature, proportional to $1/n$, which generates some important troubles in the size of the spectral gap in a non-strongly log-concave framework.

Under assumptions similar to those in [Raginsky et al., 2017], [Zou et al., 2021] proves the convergence of SGLD in total variation distance using an isoperimetric inequality linked to the Cheeger constant. On the other hand, in [Wang et al., 2021], the authors introduce a Laplacian smoothing version of SGLD and establish the convergence of the algorithm in the 2-Wasserstein distance under the assumption that the potential is dissipative, smooth and satisfies a bound variance property. They examine both the convex and non-convex cases; however, the dissipative and smoothness conditions ensure that the target measure satisfies a log-Sobolev inequality.

The most recent work is probably [Das et al., 2023], where the convergence of SGLD is studied in the Kullback-Leibler divergence and in total variation. In both situations, they assumed smoothness of the potential, in addition to a log-Sobolev inequality and a 4th moment growth condition for the convergence in Kullback-Leibler divergence and Poincaré inequality and a 6th moment growth condition in total variation.

Table 2 below shows some of the main results of the SGLD convergence.

Work	Assumptions	Convergence rate	Metric
[Raginsky et al., 2017]	component smooth, dissipative	$\mathrm{poly}(d)\varepsilon^{-4}$	W_2
[Zou et al., 2021]	component smooth, dissipative	$d^4\varepsilon^{-2}$	TV
[Das et al., 2023]	smooth, LSI, and 4th moment	$d^{1.5}\varepsilon^{-2}$	$\sqrt{\mathrm{KL}}$
	smooth, PI, 6th moment	$d^{3.5}\varepsilon^{-2}$	TV

Table 2: Comparison of the convergence rate for SGLD.

Role of functional inequalities and convexity

From previous discussions on the related literature on LD, ULA, and SGLD, it appears that functional inequalities play a key role, even in the log-concave setting, to approximate the measure e^{-U}.

Returning to [Raginsky et al., 2017], the authors identify the important dependency of the spectral gap, denoted by λ_* in their paper, on the temperature level $1/\beta$ they introduced. They derive highly pessimistic bounds in some general situations (see their discussion in Section 4 of [Raginsky et al., 2017]) and conclude emphasizing the urgent need to identify non-trivial scenarios where some better lower bounds of λ_* can be obtained.

Metastability and spectral gap asymptotic:

Indeed, the final remark of Section 4 of [Raginsky et al., 2017] is related to the well-known meta-stability phenomenon: at low temperatures, the mixing rates of many reversible and irreversible Markov semigroups are strongly deteriorated by the low-temperature settings, which are implicitly induced by a Bayesian posterior sampling problem with a large number n of observations. In a regime of variance noise of the order $O(\beta^{-1})$, the first study of the large deviation principle of invariant measures traces back to [Freidlin and Wentzell, 1984], where the authors establish the asymptotic of the spectral gap of the over-damped Langevin diffusion as $\exp(-I\beta)$ (Chapter 6 of [Freidlin and Wentzell, 1984]), where I is an explicit constant that depends on the potential of the Gibbs field. This result has been extended in depth by [Holley and Stroock, 1988], leading to the first precise analyses of the so-called simulated annealing method (see, for example, [Hajek, 1988, Miclo, 1992]). These works, along with more recent contributions involving irreversible dynamical systems in stochastic settings ([Gadat et al., 2013, Gadat et al., 2018]) show that there is almost nothing to expect in meta-stable situations in terms of the asymptotic behavior of the spectral gap and indirectly in terms of the mixing rate.

Weakly convex case and Kurdyka-Łojasiewicz inequality: Hence, the only situation that may lead to reasonable results is an intermediate situation between the (almost) trivial strongly log-concave case and the meta-stable multiwelled case. This is the purpose of the weakly log-concave situation, which is described by the family of Kurdyka-Łojasiewicz inequalities [Kurdyka, 1998, Lojasiewicz, 1963], used in optimization theory [Bolte et al., 2010a, Bolte et al., 2010b] and shown to be efficient for stochastic optimization [Gadat and Panloup, 2022] or for sampling [Gadat et al., 2022]. Later, we will intensively use this parameterization of the "weak convexity" setting and will also relate this assumption to the recent one introduced in [Erdogdu and Hosseinzadeh, 2021]. We also refer to recent contributions [Cattiaux et al., 2022], which derive some functional inequalities within an intermediate framework where the curvature ρ is related to their keystone function α, controlling the constants involved in the functional inequalities studied.

Contributions and structure of the thesis

To conclude this chapter, this section presents and summarizes the four main contributions of the thesis, which is focused on the study of a continuous-time version of the Stochastic Gradient Langevin Dynamics (SGLD) and the discrete-time unadjusted Langevin Algorithm for sampling a probability distribution over $\mathbb{R}^d$ in a weakly log-concave framework. This condition is, as previously mentioned, parameterized using the Kurdyka-Łojasiewicz inequality, which allows us to handle vanishing curvature settings that are far less restrictive compared to the simple strongly log-concave case.

Throughout the manuscript, we frequently use the term relative entropy instead of the Kullback-Leibler divergence. It is important to note that both terms represent the same quantity. In addition, we pay special attention to the dependence of all terms on the constants d (the dimension of the ambient space), n (the number of observations involved in the SGLD), and ε (the entropy error).

Functional inequalities: The first contribution of the thesis falls within the framework of functional analysis and is focused on generalizing the weak log-Sobolev inequality (WLSI, for short), introduced in [Cattiaux et al., 2007], to a space of unbounded functions. The standard WLSI provides an upper bound for the entropy of a function $f \in \mathcal{C}_b^1(\mathbb{R}^d)$ in terms of its Dirichlet form and $Osc(f) = \sup f - \inf f$. The new functional inequality, named the p-weak log-Sobolev inequality (or p-WLSI, for short), is defined in a subspace of $\mathbb{L}_m^p(\mathbb{R}^d)$ with $p > 2$.

Definition 8 (p-weak log-Sobolev inequality)**.** *Let $p > 2$ be a fixed constant. The probability measure m in $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$ satisfies a p-WLSI if a non-increasing function $\varphi_p : (0, \infty) \mapsto \mathbb{R}_+$ exists such that for any $f \in \mathcal{H}_m^1(\mathbb{R}^d) \cap \mathbb{L}_m^p(\mathbb{R}^d)$,*

$$Ent_m(f^2) \leq \varphi_p(s)\mathcal{E}_m(f) + s\|f\|^2_{\mathbb{L}_m^p(\mathbb{R}^d)}, \qquad \forall s > 0.$$

Based on a measure-capacity inequality, we derived a sufficient condition to obtain a p-WLSI and studied its link with the Poincaré, the log-Sobolev, and the standard weak log-Sobolev inequalities. Figure 1 briefly summarizes the links between the functional inequalities that we present and prove in Chapter 1. In general, the opposite implications do not hold.

Although Propositions 1.2.14 and 1.3.7 have been previously established in the literature (see [Bakry et al., 2014] and [Cattiaux et al., 2007]), Corollary 1.4.4 and Proposition 1.4.10 constitute original contributions of this work.

The equivalence in Figure 1 should be understood in the following sense: if a measure m satisfies WLSI, then it satisfies p-WLSI, for any $p > 2$; and, conversely, if a measure m satisfies p-WLSI with a fixed constant $p > 2$, then it satisfies WLSI.

Ergodicity: Strongly related to the previous contribution, we proved that a diffusion semigroup $(P_t)_{t\geq 0}$ converges to its invariant measure μ (previously denoted as m) if this measure satisfies a p-WLSI.

Furthermore, under the $\mathcal{H}^{\mathfrak{r}}_{\mathbf{KL}}(\mathfrak{c}, L)$ hypothesis, we established explicit non-asymptotic bounds for the convergence in entropy. Table 3 summarizes the

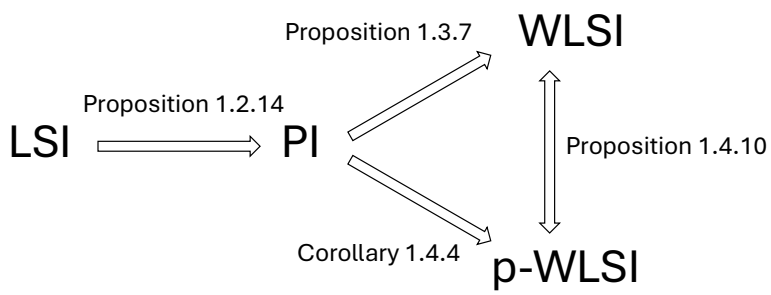


Figure 1: Link between the studied functional inequalities.

results obtained, where the last column shows the time t (up to a multiplicative constant) for which

$$Ent_\mu(P_t f) \leq \varepsilon,$$

where $\varepsilon > 0$ is a sufficiently small constant and f is a positive function chosen with the appropriate smoothness for each scenario.

$\mathcal{H}^{\mathbf{r}}_{\mathbf{KL}}(\mathfrak{c}, L)$	functional inequalities	convergence rate
$(r = 0)$ μ is strongly log-concave	LSI	$\mathfrak{c}^{-1}\left(\log(\varepsilon^{-1}) + \log(Ent_\mu(f))\right)$
$(0 < r < 1)$ μ is weakly log-concave	PI $\implies$ WLSI	$C_{\mathrm{PI}}\left(\log^2(\varepsilon^{-1}) + \log^2(Osc^2(\sqrt{f}))\right)$
	PI $\implies$ p-WLSI	$C_{\mathrm{PI}}\left(\log^2(\varepsilon^{-1}) + \log^2\left(\left\|\sqrt{f}\right\|^2_{\mathbb{L}^p_\mu(\mathbb{R}^d)}\right)\right)$

Table 3: Convergence rate estimates in entropy when an $\mathcal{H}^{\mathbf{r}}_{\mathbf{KL}}(\mathfrak{c}, L)$ hypothesis is satisfied.

In the weakly log-concave situation, μ satisfies a Poincaré inequality, and thus it satisfies both a WLSI and a p-WLSI, with $p > 2$. The terms C_{PI}, $Osc(\sqrt{f})$, and $\left\|\sqrt{f}\right\|^2_{\mathbb{L}^p_\mu(\mathbb{R}^d)}$ depend on the convexity parameter r and on the dimension d, even though this dependence does not appear explicitly. On the other hand, in the strongly log-concave situation, μ satisfies a log-Sobolev inequality with constant $2\mathfrak{c}^{-1}$, and the term $Ent_\mu(f)$ could depend on d.

Comparing both contexts, we observe that in the weakly log-concave case, there is a worsening of convergence with respect to ε and the Poincaré constant, which, unlike $\mathfrak{c}$, depends on d.

The following contributions are strongly based on the functional inequalities studied under the hypothesis $\mathcal{H}^{\mathbf{r}}_{\mathbf{KL}}(\mathfrak{c}, L)$.

Continuous-time Stochastic Gradient Langevin Dynamics: The third part of the thesis focuses on investigating a continuous-time version of the Stochastic Gradient Langevin Dynamics, which was introduced in [Welling and Teh, 2011]. Our algorithm incorporates a stochastic sampling step inside the traditional over-damped Langevin diffusion.

Let $n \geq 1$ and $\mathbf{X}_1$, $\mathbf{X}_2$, ..., $\mathbf{X}_n$ be n i.i.d. observations distributed according to p_{θ^*}. Given a prior distribution π_0 on $\mathbb{R}^d$, the posterior distribution μ_n is then defined as:

$$\mu_n(\theta) \propto \pi_0(\theta) \times \prod_{i=1}^{n} p_\theta(\mathbf{X}_i) \propto \exp(-U_{\nu_n}(\theta)), \quad \theta \in \mathbb{R}^d.$$

The continuous-time SGLD replaces the role of the function U_{ν_n} (which depends on all the observations $\mathbf{X}_1, \mathbf{X}_2, \ldots, \mathbf{X}_n$) in the over-damped Langevin Dynamics by U_{X_t}, where X_t is a jump process that takes the value of a single unique observation chosen uniformly from $\mathbf{X}_1, \mathbf{X}_2, \ldots, \mathbf{X}_n$ during exponential time intervals.

Formally, the continuous-time SGLD is the solution $(\theta_t, X_t)_{t\geq 0}$ of the stochastic differential equation

$$\mathrm{d}\theta_t = -\nabla_\theta U_{X_t}(\theta_t)\mathrm{d}t + \sqrt{2}\mathrm{d}B_t, \quad t > 0,$$

where $(B_t)_{t\geq 0}$ is a multivariate standard Brownian Motion. When n is large, this process can be seen as an online algorithm, in which the observations $\mathbf{X}_1$, $\mathbf{X}_2$, ..., $\mathbf{X}_n$ become available one at a time.

We derive our analysis on the $\mathcal{H}^{\mathbf{r}}_{\mathbf{KL}}(\mathfrak{c}, L)$ framework and establish an explicit bound on the final horizon of simulation to obtain an ε approximation (in terms of entropy).

Before presenting this result, let us recall and specify some quantities: n is the number of observations, d is the dimension of the ambient space where the parameter of interest resides, r, $\mathfrak{c}$ and L are fixed constants associated with hypothesis $\mathcal{H}^{\mathbf{r}}_{\mathbf{KL}}(\mathfrak{c}, L)$, and $1/\alpha_n$ is the mean time in which X_t has a jump. We define J_t as the relative entropy of the distribution of θ_t with respect to μ_n.

Theorem. *If one of the following conditions is satisfied, then $J_t \leq \varepsilon$.*

a) $r = 0$, $\alpha_n = n$ *and* $t \gtrsim_{uc} n^{-1}\left[\log(\varepsilon^{-1}) + \log(d) + \log(n)\right]$.

b) $0 < r < 1$, $n \geq 2$, $t \gtrsim_{uc} \alpha_n^{-2}\left(d\log^2(n)\right)^{-(1+r)^2}$, *and*

$$t \gtrsim_{uc} \left(d\log^2(n)\right)^{(1+r)^2}\left[\log^2(\varepsilon^{-1}) + n^2\left(d\log^2(n)\right)^{2(1+r)}\right].$$

The notation $a \gtrsim_{uc} b$ ($a \lesssim_{uc} b$) means that there exists a positive constant c independent of n and d such that $a \geq cb$ ($a \leq cb$). If a constant is independent of d and n, we say that it is a *universal constant*.

In order to understand this result, Table 4 shows a comparison between the convergence rate of the continuous-time process SGLD and the over-damped

Langevin Dynamics associated with μ_n, which is described with the diffusion

$$\mathrm{d}\vartheta_t = -\nabla U_{\nu_n}(\vartheta_t)\mathrm{d}t + \sqrt{2}\mathrm{d}B_t, \quad t > 0.$$

Abusing the notation, we refer to continuous-time SGLD as SGLD and over-damped Langevin Dynamics as LD.

$\mathcal{H}^{\mathbf{r}}_{\mathbf{KL}}(\mathfrak{c}, L)$	algorithm	N° of obs.	convergence rate
$(r = 0)$ μ is strongly log-concave	LD	n	$n^{-1}\left[\log(\varepsilon^{-1}) + \log(d)\right]$
	SGLD	1	$n^{-1}\left[\log(\varepsilon^{-1}) + \log(d) + \log(n)\right]$
$(0 < r < 1)$ μ is weakly log-concave	LD	n	$\left(d\log^{2\beta}(n)\right)^{(1+r)^2}\left[\log^2(\varepsilon^{-1}) + n^2\left(d\log^{2\beta}(n)\right)^{2(1+r)}\right]$
	SGLD	1	$\left(d\log^{2\beta}(n)\right)^{(1+r)^2}\left[\log^2(\varepsilon^{-1}) + n^2\left(d\log^{2\beta}(n)\right)^{2(1+r)}\right]$

Table 4: Convergence rate to obtain an ε approximation (in terms of entropy). Where $\alpha_n = n$ if $r = 0$ and $\alpha_n > \left(d\log^2(n)\right)^{-(1+r)^2}$ if $0 < r < 1$ for the continuous-time SGLD.

In the strongly log-concave situation, the order of convergence of LD is faster than that of SGLD, although, in the weakly log-concave situation, the orders are equal. However, LD requires having all observations from the initial instant, and the computation of ∇U_{ν_n} could be complex when n is large. On the other hand, as we mentioned above, SGLD can be seen as an online algorithm, which only requires having a new observation available with an exponential clock. This arises from the fact that the observations are i.i.d. . Thus, SGLD could be considered as a good alternative, especially when all the observations are not available at the initial time.

We illustrate our result through two examples. The first one has a pedagogical character: the potential function is quadratic, allowing for some terms to be computed explicitly. We also observe that the simulations agree with our choice of parameter α_n. The second example is based on a Bayesian logistic regression problem, for which we establish a posterior distribution whose potential satisfies the hypothesis $\mathcal{H}^{\mathbf{r}}_{\mathbf{KL}}(\mathfrak{c}, L)$. In this context, it is not possible to find an explicit solution for the SGLD. Hence, we rely on its Euler-Maruyama approximation with a small step size h, to graphically show the decay of the error of the Césaro average under the influence of the dimension d and the parameter r. When t is large, we observe that the curves are ordered according to the value of r, with the curve for $r = 0$ being the one with the least error since it corresponds to the strongly log-concave situation.

Discrete-time Unadjusted Langevin Algorithm: The fourth contribution of the thesis is related to the unadjusted Langevin Algorithm (ULA) defined in (12). We investigate a potential U that satisfies a $\mathcal{H}^{\mathbf{r}}_{\mathbf{KL}}(\mathfrak{c}, L)$-condition with fixed constants

r, $\mathfrak{c}$ and L, and such that

$$\|\arg\min U\|_2 \lesssim_{uc} \sqrt{d} \quad \text{and} \quad \min_{\theta\in\mathbb{R}^d} U(\theta) \lesssim_{uc} d.$$

We assume that there exist two positive constants A_d (which could depend on d) and $0 < b < 1$ such that the density function m_k of θ_{kh} is uniformly bounded in k and h as follows:

$$m_k(\theta) \leq A_d e^{-bU(\theta)}, \quad \theta \in \mathbb{R}^d, \quad k \geq 0.$$

Furthermore, $\theta_0 \sim \mathcal{N}(0_d, \sigma_0^2 I_d)$, where $\sigma_0 > 0$ is conveniently chosen.

In the strongly log-concave case, for example, when $U(\theta) = \frac{1}{2}\|\theta\|_2^2$ and $h \leq \frac{1}{2}\left(1 - \frac{1}{\sigma_0^2}\right)$, this bound is immediately obtained where $A_d = (2\pi)^{-\frac{d}{2}}$ and $b = [2(\sigma_0^2 + \frac{4}{3})]^{-1}$. In the weakly log-concave context, this bound is much more complex to verify.

In [Konakov and Mammen, 2000], the authors investigate local bounds for the transition probabilities of diffusion processes, and as a result, a similar behavior for m_k. However, this approach assumes that U is a bounded function, which does not apply in our context. Since this is not the primary focus of our study, we will adopt this assumption and leave a more detailed exploration of this direction for future work.

Using the result above, we apply a p-WLSI to the function $f = \sqrt{m_k/\mu}$, for which there exists a value of $p > 2$ such that the p-norm of f is uniformly bounded in k and h. We then prove explicit bounds in k, the number of iterations and h, the step size of the algorithm, to achieve an ε error in entropy. The following theorem summarizes the results we obtain in Chapter 3, where J_k is the relative entropy of m_k with respect to μ, for any $k \geq 0$.

Theorem. *If one of the following conditions is satisfied, then $J_k \leq \varepsilon$.*

a) $r = 0$, $h \lesssim_{uc} \varepsilon d^{-1}$ *and* $k \gtrsim_{uc} \varepsilon^{-1} d \log(d/\varepsilon)$.

b) $0 < r < 1$, $h \lesssim_{uc} \varepsilon(d^{1+(1+r)^2} \log(A_d/\varepsilon))^{-1}$, *and*

$$k \gtrsim_{uc} \varepsilon^{-1} d^{1+2(1+r)^2} \log^2(A_d/\varepsilon) \log(d/\varepsilon).$$

The result of part *a* recovers the best bound shown in [Vempala and Wibisono, 2019] under an LSI inequality. However, the result of part *b* does not reach the best existing bound. It is important to note that, to make an adequate comparison, for instance, with the works of [Erdogdu and Hosseinzadeh, 2021] and [Chewi et al., 2022a], their hypotheses need to be adjusted to align with our context. In several of their assumptions, they treat certain terms as constants, whereas we propose more general hypotheses in which these constants can depend on d. This is discussed in detail in Chapter 3. Returning to the comparison, it appears that by using the *modified* log-Sobolev inequality instead of the p-WLSI, the multiplicative terms lead to a better dependence on dimension d. Another possible reason is that we use the Bobkov bound for the Poincaré constant C_{PI} associated with μ, which we bound by $C_{\mathrm{PI}} \lesssim_{uc} d^{(1+r)^2}$. A better bound on this constant would improve this result. This avenue of research still requires further development.

In Chapter 4, we study a problem in the Bayesian logistic regression framework, where we apply ULA using real datasets. Furthermore, we present a simulation study in which we estimate the entropy to verify its decay as $t \to \infty$, as proven in the main theorem.

Structure of the thesis

The thesis consists of five chapters.

Chapter 1 focuses on functional analysis and the Kurdyka-Łojasiewicz hypothesis. In the first sections, we introduce and study the Poincaré, log-Sobolev, and weak log-Sobolev inequalities. We present some well-known techniques and results that form the basis for developing the main contribution of the chapter: a generalization of the weak log-Sobolev inequality to a space of unbounded functions, called the p-WLSI. In addition, we prove that a diffusion semigroup converges to its invariant measure if that measure satisfies a p-WLSI. The final section of the chapter is dedicated to the study of the above inequalities under the Kurdyka-Łojasiewicz hypothesis. To do so, we first present some basic consequences of this hypothesis and establish explicit non-asymptotic bounds on the convergence in entropy.

In Chapter 2, we investigate a continuous-time version of the Stochastic Gradient Langevin Dynamics that incorporates a stochastic sampling step inside the traditional over-damped Langevin diffusion. We pay specific attention to the computational cost in terms of n (the number of observations that produces the posterior distribution) and d (the dimension of the ambient space where the parameter of interest is living). We derive our analysis from the $\mathcal{H}^{\mathbf{r}}_{\mathbf{KL}}(\mathfrak{c}, L)$ framework and establish an explicit bound on the final horizon of simulation to obtain an ε approximation (in terms of entropy).

We study the unadjusted Langevin Algorithm in Chapter 3. This discrete-time algorithm is commonly used in machine learning to sample a given distribution $\mu(\theta) \propto e^{-U(\theta)}$, with $\theta \in \mathbb{R}^d$. We investigate a potential function U which satisfies a $\mathcal{H}^{\mathbf{r}}_{\mathbf{KL}}(\mathfrak{c}, L)$-condition and prove explicit bounds on the number of iterations and the step size of the algorithm to achieve an ε error in entropy.

The main objective of Chapter 4 is to present an algorithm, some illustrative examples, and a simulation study for SGLD and ULA. We discuss the choice of the parameter α_n associated with the SGLD. We study a Bayesian logistic regression problem where the potential of the posterior distribution satisfies a $\mathcal{H}^{\mathbf{r}}_{\mathbf{KL}}(\mathfrak{c}, L)$-condition and we use two data sets taken from the UCI repository. We also verify the decay of the entropy when $t \to \infty$ and observe the influence of the parameter r involved in the hypothesis $\mathcal{H}^{\mathbf{r}}_{\mathbf{KL}}(\mathfrak{c}, L)$.

Chapter 5 is a very short chapter that focuses on presenting some future perspectives: extensions and generalizations of this work.

Chapter 1

Functional inequalities and Kurdyka-Łojasiewicz hypothesis

1.1 Introduction

In this chapter, we introduce and study some functional inequalities that link the entropy of a function and its Dirichlet form, in particular, under a Kurdyka-Łojasiewicz hypothesis. The chapter is divided into four parts.

Section 1.2 presents Poincaré (PI) and log-Sobolev (LSI) inequalities. It also describes the standard techniques to quantify the behavior of the variance and the relative entropy along a diffusion semigroup using these inequalities, respectively. Section 1.3 is devoted to the study of the weak log-Sobolev inequality (WLSI), as introduced in [Cattiaux et al., 2007], for a set of bounded functions. It also develops some basic results that will be necessary later, for example, the link between Poincaré's inequality and WLSI. Both sections contain well-known techniques and results and are the basis for developing the main contributions of this chapter, which appear in sections 1.4 and 1.5.

Section 1.4 generalizes WLSI to a set of unbounded functions, in particular for those with a finite p-norm, $p > 2$. This new functional inequality is introduced as the p-weak log-Sobolev inequality (p-WLSI). An objective of this chapter is to prove that if a measure verifies a Poincaré inequality, then it satisfies a p-WLSI, for any $p > 2$. This section summarizes all the links that exist between the inequalities that will be studied throughout the chapter.

All the above inequalities are defined in a probability space, so the last section focuses on probability measures that satisfy a Kurdyka-Łojasiewicz hypothesis. This condition was introduced in [Gadat et al., 2022] and describes log-concave measures with a behavior between the Gaussian and Laplace situation. Section 1.5 presents quantitative estimates for the constants and functions involved in the functional inequalities studied. These results are the cornerstone for the development of subsequent chapters.

1.2 Poincaré and log-Sobolev inequalities

These inequalities are defined in a Sobolev space and link the norm of a function or a related quantity to the norm of its derivatives. This family of inequalities has a wide range of applications in partial differential equations, functional analysis, theory of Sobolev spaces, probability theory and statistics, etc. Particularly in these last two areas, they are powerful tools used to derive concentration inequalities and the convergence of Markov processes. See [Ledoux, 2001] and [Bakry et al., 2014] for a more in-depth study.

Poincaré's inequality

We recall some definitions and well-known facts. Let $\mathcal{B}(\mathbb{R}^d)$ be the Borel σ-algebra on $\mathbb{R}^d$ and m a probability measure defined on $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$. Consider a function $f : \mathbb{R}^d \to \mathbb{R}$. For any $1 \leq p < \infty$, we define $\mathbb{L}^p_m(\mathbb{R}^d)$: the space of functions with finite p-norm

$$\|f\|_{\mathbb{L}^p_m(\mathbb{R}^d)} := \left[\int_{\mathbb{R}^d} |f|^p \mathrm{d}m\right]^{1/p},$$

whenever this quantity is finite. Likewise, $\mathbb{L}^\infty_m(\mathbb{R}^d)$ is the space of functions with finite ∞-norm,

$$\|f\|_{\mathbb{L}^\infty_m(\mathbb{R}^d)} := \inf\{K : m(|f| > K) = 0\}.$$

We say that a function is *bounded* if its ∞-norm is finite. Let

$$\mathcal{H}^1_m(\mathbb{R}^d) = \left\{f : \mathbb{R}^d \to \mathbb{R}; f \in \mathbb{L}^2_m(\mathbb{R}^d), \nabla f \in \left(\mathbb{L}^2_m(\mathbb{R}^d)\right)^d\right\},$$

where ∇f is defined in the weak sense. $\mathcal{C}^1_b(\mathbb{R}^d)$ is the set of bounded and once differentiable functions on $\mathbb{R}^d$. For any function $f \in \mathbb{L}^1_m(\mathbb{R}^d)$, we define its variance as:

$$Var_m(f) := \int_{\mathbb{R}^d} (f - m[f])^2 \mathrm{d}m,$$

where $m[f] = \int_{\mathbb{R}^d} f \mathrm{d}m$ and when $f \in \mathcal{H}^1_m(\mathbb{R}^d)$, the Dirichlet form of f is defined as:

$$\mathcal{E}_m(f) := \int_{\mathbb{R}^d} \|\nabla f\|_2^2 \mathrm{d}m.$$

In general, the Dirichlet form is associated with an infinitesimal generator $\mathcal{L}$ as in Definition 5. However, in this chapter, we avoid the definition of the infinitesimal generator $\mathcal{L}$ and just consider that $\Gamma_{\mathcal{L}}(f, g) = \langle \nabla f, \nabla g \rangle$ and that m is the invariant distribution associated with $\mathcal{L}$.

Poincaré's inequality links the variance of f to its Dirichlet form.

Definition 1.2.1 (Poincaré's inequality)**.** *The probability measure m satisfies a Poincaré inequality if there exists a constant $C_{\mathrm{PI}}(m)$ such that for any $f \in \mathcal{H}^1_m(\mathbb{R}^d)$,*

$$Var_m(f) \leq C_{\mathrm{PI}}(m)\mathcal{E}_m(f).$$

The optimal constant $C_{\mathrm{PI}}(m)$ is referred to as the Poincaré constant.

Example 1.2.2. *Let be $\sigma > 0$ a fixed constant and γ_{σ^2} the density function of a centered Gaussian probability measure in $\mathbb{R}^d$ with covariance matrix $\sigma^2 I_d$, where I_d is the identity matrix of size d, that is*

$$\gamma_{\sigma^2}(\theta) = (2\pi\sigma^2)^{-d/2} e^{-\frac{\|\theta\|_2^2}{2\sigma^2}}, \quad \theta \in \mathbb{R}^d.$$

Then γ_{σ^2} satisfies a Poincaré inequality with constant $C_{\mathrm{PI}}(\gamma_{\sigma^2}) = \sigma^2$.

An important property of log-concave measures is that they satisfy the Poincaré inequality. We state this result in the following and refer to Section 4.6.2 of [Bakry et al., 2014] for further details.

Proposition 1.2.3. *Assume that m is a log-concave measure, then it satisfies a Poincaré inequality.*

Let us briefly recall some basic definitions of probability. If $X = (X_1, X_2, \ldots, X_d)$ is a random variable (or random vector) in $\mathbb{R}^d$ and m is the density function of X, then for any Borel function g defined in $\mathbb{R}^d$

$$\mathbb{E}_m(g(X)) := \int_{\mathbb{R}^d} g(x) \mathrm{d}m(x)$$

whenever this quantity exists. If $m \in \mathbb{L}^1_m(\mathbb{R}^d)$, then we define the variance of the probability measure m as

$$Var_m(X) := \mathbb{E}_m \left(\|X - \mathbb{E}_m(X)\|_2^2 \right).$$

Moreover, if $m \in \mathbb{L}^2_m(\mathbb{R}^d)$, then the covariance matrix of m is defined as

$$Cov_m(X) := (Cov_m(X_i, X_j))_{1 \le i,j \le d},$$

where $Cov_m(X_i, X_j) = E_m\left((X_i - \mathbb{E}_m(X_i))(X_j - \mathbb{E}_m(X_j))\right)$, for all i and j.

Since we focus our study on log-concave measures, we need to specify how to compute or bound the Poincaré constant. That is, in fact, the aim of the next remark.

Remark 1.2.4. *If a probability measure m is p-strongly log-concave, then the Poincaré constant is $C_{\mathrm{PI}}(m) = 1/p$. In the weakly log-concave case the following bound on the Poincaré constant may be found in Theorem 1.2 of [Bobkov, 1999]:*

$$C_{\mathrm{PI}}(m) \le 4K^2 Var_m(X), \tag{1.1}$$

where K is a universal constant and X denotes a random variable with density function m.

Throughout all chapters of this work, we will bound the Poincaré constant using the above remark whenever the measure m is weakly log-concave. However, it is also of interest to present the conjecture of Kannan, Lovász and Simonovits also known as KLS conjecture, see [Kannan et al., 1995].

Conjecture 1.2.5 (KLS conjecture)**.** *Assume that m is a log-concave probability measure with covariance matrix I_d. Then, there exists a universal constant K' such that*

$$C_{\mathrm{PI}}(m) \leq K'.$$

In the general case where the covariance matrix $Cov_m(X)$ is not the identity, using the stability of the Poincaré constant with respect to Lipschitz transport (see Theorem 1.7 of [Milman, 2018]), the KLS conjecture gives that for any log-concave measure m in $\mathbb{R}^d$,

$$C_{\mathrm{PI}}(m) \leq K'\overline{\lambda}_{Cov_m(X)},$$

where $\overline{\lambda}_{Cov_m(X)}$ denotes the largest eigenvalue of $Cov_m(X)$. This value may depend on d.

Example 1.2.6. *a) Let $\sigma > 0$ be a fixed constant and $m = \gamma_{\sigma^2}$ the density function of a centered Gaussian random variable in $\mathbb{R}^d$ with covariance matrix $\sigma^2 I_d$. Then*

$$\overline{\lambda}_{Cov_{\gamma_{\sigma^2}}(X)} = \sigma^2.$$

From Example 1.2.2, we already know that $C_{\mathrm{PI}}(\gamma_{\sigma^2}) = \sigma^2$, so the KLS conjecture holds for $K' = 1$.

b) For an arbitrary density function m, the Cauchy-Schwarz inequality implies that for any $1 \leq i, j \leq d$,

$$Cov_m(X_i, X_j) \leq \sqrt{Var_m(X_i)Var_m(X_j)} \leq \max_{1\leq k\leq d} Var_m(X_k).$$

Let u_j be the j-th component of $u \in \mathbb{R}^d$, for $1 \leq j \leq d$. The largest eigenvalue $\overline{\lambda}_{Cov_m(X)}$ is upper bounded as follows

$$\begin{aligned}
\overline{\lambda}_{Cov_m(X)} &\leq \sup_{u\in\mathbb{R}^d, \|u\|^2=1} u^T Cov_m(X) u \\
&= \sup_{u\in\mathbb{R}^d, \|u\|^2=1} \sum_{i,j=1}^{d} u_i Cov_m(X_i, X_j) u_j \\
&\leq \max_{1\leq k\leq d} Var_m(X_k) \sup_{u\in\mathbb{R}^d, \|u\|^2=1} \sum_{i,j=1}^{d} u_i u_j \\
&= d \max_{1\leq k\leq d} Var_m(X_k).
\end{aligned}$$

This bound is reached, for example, by a Gaussian probability measure centered with a covariance matrix $Cov_m(X)$ such that $Cov_m(X_i, X_j) = 1$, for all i, j.

In a general case, the KLS conjecture implies that the Poincaré constant is bounded by

$$C_{\mathrm{PI}}(m) \leq K' d \max_{1\leq k\leq d} Var_m(X_k).$$

We will return to this conjecture during the last section of this chapter; let us just present the most recent advance in the KLS conjecture, deduced from Theorem 1.2 in [Klartag, 2023].

Proposition 1.2.7. *For any $d \geq 2$ and any log-concave measure m in $\mathbb{R}^d$ with a covariance matrix $Cov_m(X)$, there exists a universal constant K' such that*

$$C_{\mathrm{PI}}(m) \leq K' \log(d) \overline{\lambda}_{Cov_m(X)},$$

where $\overline{\lambda}_{Cov_m(X)}$ denotes the largest eigenvalue of $Cov_m(X)$.

The following result is obtained as a consequence of Proposition 4.4.2 in [Bakry et al., 2014] and shows that if the Poincaré inequality is satisfied, then the tail of the measure is exponentially small. This gives a very useful necessary condition for this functional inequality to hold.

Proposition 1.2.8. *Assume that m satisfies a Poincaré inequality with constant $C_{\mathrm{PI}}(m)$. Then there exist two positive constants c and C such that for any R large enough,*

$$m(\|\theta\|_2 > R) \leq C e^{-\frac{cR}{\sqrt{C_{\mathrm{PI}}(m)}}}.$$

Example 1.2.9. *The following two families of probability measures do not satisfy the Poincaré inequality:*

a) Let $\alpha > 0$,

$$m_\alpha(\theta) = \frac{\alpha}{2}(1 + |\theta|)^{-1-\alpha}, \quad \theta \in \mathbb{R}.$$

b) Let $0 < \alpha < 1$,

$$m^\alpha(\theta) \propto e^{-|\theta|^\alpha}, \quad \theta \in \mathbb{R}.$$

Variance decay of a diffusion semigroup

Let $(P_t)_{t\geq 0}$ be a diffusion semigroup with invariant measure m and infinitesimal generator $\mathcal{L}$. The following result, taken from Theorem 4.2.5 of [Bakry et al., 2014], shows that the Poincaré inequality induces an exponential convergence in $\mathbb{L}^2_m(\mathbb{R}^d)$ of the semigroup $(P_t)_{t\geq 0}$ to the equilibrium.

Proposition 1.2.10. *Assume that m satisfies the Poincaré inequality with constant $C_{\mathrm{PI}}(m)$. Then, for any function $f \in \mathbb{L}^2_m(\mathbb{R}^d)$ and every $t \geq 0$,*

$$Var_m(P_t f) \leq e^{-\frac{2t}{C_{\mathrm{PI}}(m)}} Var_m(f).$$

Proof. For simplicity, consider $m[f] = 0$. Using the invariant property of the semigroup, then $m[P_t f] = m[f] = 0$ and

$$Var_m(P_t f) = \int_{\mathbb{R}^d} (P_t f)^2 \mathrm{d}m.$$

By integration by parts,

$$\frac{\mathrm{d}}{\mathrm{d}t} Var_m(P_t f) = 2 \int_{\mathbb{R}^d} P_t f \mathcal{L} P_t f \mathrm{d}m = -2\mathcal{E}_m(P_t f).$$

Therefore, the Poincaré inequality translates into the differential inequality

$$\frac{\mathrm{d}}{\mathrm{d}t} Var_m(P_t f) \leq -\frac{2}{C_{\mathrm{PI}}(m)} Var_m(P_t f).$$

The conclusion follows by using Gronwall's lemma. □

In fact, this condition is also necessary, although we will not prove this fact.

Example 1.2.11. *The OU semigroup has an invariant distribution $m = \gamma_1$, then it satisfies a Poincaré inequality of constant $C_{\mathrm{PI}}(\gamma_1) = 1$. Using Proposition 1.2.10, we deduce that*

$$Var_{\gamma_1}(P_t f) \leq e^{-2t} Var_{\gamma_1}(f), \tag{1.2}$$

for any $f \in \mathbb{L}^2_\mu(\mathbb{R}^d)$.

Log-Sobolev inequality

The log-Sobolev inequality (LSI) is one of the most studied functional inequality for semigroups. Let us briefly introduce some definitions and basic properties. For any non-negative function $f : \mathbb{R}^d \to \mathbb{R}^+$ such that $\int_{\mathbb{R}^d} f|\log(f)|\mathrm{d}m < \infty$, we define the entropy of f as:

$$Ent_m(f) := \int_{\mathbb{R}^d} f \log(f) \mathrm{d}m - \int_{\mathbb{R}^d} f \mathrm{d}m \log\left(\int_{\mathbb{R}^d} f \mathrm{d}m\right).$$

In this definition, $0 \log 0$ is interpreted as 0. Using Jensen's inequality applied to the strictly convex function $x \mapsto x \log(x)$, we could verify that $Ent_m(f) \geq 0$ for any f and $Ent_m(f) = 0$ if and only if f is a constant function m-almost surely. Moreover, for any $c > 0$ and any non-negative function f with finite entropy,

$$Ent_m(cf) = cEnt_m(f).$$

This property is called homogeneity of the entropy.

For future use, we establish the variational definition of the entropy

$$Ent_m(f) = \sup\left\{\int_{\mathbb{R}^d} fg \mathrm{d}m : \int_{\mathbb{R}^d} e^g \mathrm{d}m \leq 1\right\}. \tag{1.3}$$

An immediate consequence is the subadditivity of the entropy: for any two non-negative functions f_1 and f_2 with finite entropy,

$$Ent_m(f_1 + f_2) \leq Ent_m(f_1) + Ent_m(f_2).$$

Let us recall Rothau's lemma.

Lemma 1.2.12 (Rothau's lemma)**.** *Let $f : \mathbb{R}^d \to \mathbb{R}$ be a function such that $\int_{\mathbb{R}^d} f^2 \log(1 + f^2)\mathrm{d}m < \infty$. Then, for any constant $a \in \mathbb{R}$,*

$$Ent_m((f + a)^2) \leq Ent_m(f^2) + 2m[f^2].$$

A consequence of this lemma is that

$$Ent_m(f^2) \leq Ent_m(\tilde{f}^2) + 2Var_m(f), \tag{1.4}$$

where $\tilde{f} = f - m[f]$.

Let us now define the LSI, which links the entropy of a function f^2 and the Dirichlet form of f.

Definition 1.2.13 (Log-Sobolev inequality)**.** *The probability measure m satisfies an LSI if there exists a constant $C_{\mathrm{LSI}}(m)$ such that for any $f \in \mathcal{H}^1_m(\mathbb{R}^d)$,*

$$Ent_m(f^2) \leq C_{\mathrm{LSI}}(m)\mathcal{E}_m(f).$$

Using the homogeneity of the entropy, we could study $Ent_m(f^2)$ assuming that $m[f^2] = 1$. In this case $\mathrm{d}\nu = f^2\mathrm{d}m$ is a probability measure and the quantity

$$H(\nu|m) := \int_{\mathbb{R}^d} \log(\nu/m)\mathrm{d}\nu = Ent_m(\nu/m),$$

is known as the relative entropy of ν with respect to m, while

$$I(\nu|m) := 4\int_{\mathbb{R}^d} \left\|\nabla\left(\sqrt{\nu/m}\right)\right\|_2^2 \mathrm{d}m = 4\mathcal{E}_m\left(\sqrt{\nu/m}\right)$$

is called Fisher information of ν with respect to m. LSI compares the entropy to the information,

$$H(\nu|m) \leq \frac{C_{\mathrm{LSI}}(m)}{4} I(\nu|m).$$

These two terms belong to the context of information theory, where LSI was first considered.

As expected, an LSI is stronger than a Poincaré inequality. The following proposition shows this result, which was taken from Section 5.1.2 in [Bakry et al., 2014].

Proposition 1.2.14. *Assume that m satisfies an LSI with constant $C_{\mathrm{LSI}}(m)$. Then m satisfies a Poincaré inequality with constant $C_{\mathrm{PI}}(m) = C_{\mathrm{LSI}}(m)/2$.*

The following proposition may be found in [Bakry and Émery, 1985].

Proposition 1.2.15. *If m is a p-strongly log-concave measure, then an LSI is verified, and the log-Sobolev constant $C_{\mathrm{LSI}}(m)$ is $2/p$, which is independent of the dimension d.*

The following example is an immediate application of Proposition 1.2.15 to the Gaussian distribution.

Example 1.2.16. *Let $\sigma > 0$ be a fixed constant and γ_{σ^2} the density function of a centered Gaussian probability measure in $\mathbb{R}^d$, as in Example 1.2.2. Then γ_{σ^2} satisfies an LSI with constant $C_{\mathrm{LSI}}(\gamma_{\sigma^2}) = 2\sigma^2$.*

Entropy decay of a diffusion semigroup

An LSI induces the convergence of the diffusion semigroup $(P_t)_{t\geq 0}$ towards its invariant distribution m. To verify this statement, let us present the link between the Dirichlet form and the entropy along the semigroup. So, once again, we consider the infinitesimal generator $\mathcal{L}$. The following results are taken from Section 5.2.1 of [Bakry et al., 2014].

Proposition 1.2.17 (De Bruijn's identity)**.** *For any strictly positive function f with finite entropy and $t > 0$,*

$$\frac{\mathrm{d}}{\mathrm{d}t} Ent_m\left(P_t f\right) = -4\mathcal{E}_m\left(\sqrt{P_t f}\right).$$

Proof. Using integration by parts and the identity $\langle \nabla g, \nabla \log(g)\rangle = -4\|\nabla(\sqrt{g})\|_2^2$, for any strictly positive and differentiable function g, we then get

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}t} Ent_m\left(P_t f\right) &= \int_{\mathbb{R}^d} \left(1 + \log(P_t f)\right) \mathcal{L} P_t f \mathrm{d}m \\
&= -\int_{\mathbb{R}^d} \Gamma_{\mathcal{L}}\left(P_t f, 1 + \log(P_t f)\right) \mathrm{d}m \\
&= -4\int_{\mathbb{R}^d} \left\|\nabla\left(\sqrt{P_t f}\right)\right\|_2^2 \mathrm{d}m \\
&= -4\mathcal{E}_m\left(\sqrt{P_t f}\right).
\end{aligned}$$

□

Proposition 1.2.18. *Assume that m satisfies an LSI with constant $C_{\mathrm{LSI}}(m)$. Then, for any positive function f with finite entropy and for any $t > 0$,*

$$Ent_m(P_t f) \leq e^{-\frac{4t}{C_{\mathrm{LSI}}(m)}} Ent_m(f).$$

Proof. As an immediate consequence of De Bruijn's identity and LSI,

$$\frac{\mathrm{d}}{\mathrm{d}t} Ent_m\left(P_t f\right) \leq -\frac{4}{C_{\mathrm{LSI}}(m)} Ent_m\left(P_t f\right).$$

The conclusion follows by using Gronwall's lemma. □

Let us return to the example of the OU semigroup.

Example 1.2.19. *Since $m = \gamma_1$ and $C_{\mathrm{LSI}}(\gamma_1) = 2$, then*

$$Ent_{\gamma_1}(P_t f) \leq e^{-2t} Ent_{\gamma_1}(f). \tag{1.5}$$

Comparing inequalities (1.2) *and* (1.5) *is not easy, but we can verify that one difference is the class of admissible functions f for which they are well defined. For example, taking $f(x) = e^{\frac{cx^2}{2}}$, $x \in \mathbb{R}$ and $c \in (0,1)$, then $Var_{\gamma_1}(f)$ does not exists if $c > 1/2$, while $Ent_{\gamma_1}(f) = \frac{1}{2}\left(\frac{1}{c} - 1 + \log(c)\right)$ for any $c \in (0,1)$.*

1.3 Weak log-Sobolev inequality

A sufficient condition to guarantee LSI is that the measure m is strongly log-concave. However, it is not clear that this result holds in general for weakly log-concave measures. Hence, modified versions of LSI arise.

This section studies the weak log-Sobolev inequality (WLSI), which, as its name indicates, is a weak version of LSI and is defined for bounded functions. This functional inequality was introduced in [Cattiaux et al., 2007] and we refer to this work for a more in-depth study of the results that will appear in this section.

Definition 1.3.1 (Weak log-Sobolev inequality)**.** *The probability measure m satisfies a WLSI if a non-increasing function $\varphi : (0,\infty) \mapsto \mathbb{R}_+$ exists such that for any $f \in \mathcal{C}_b^1(\mathbb{R}^d)$ and any $s > 0$,*

$$Ent_m(f^2) \leq \varphi(s)\mathcal{E}_m(f) + s\, Osc^2(f),$$

where $Osc(f) = \sup f - \inf f$.

As the following remark shows, the above definition is only necessary for small values of s.

Remark 1.3.2. *Let $f \in \mathcal{C}_b^1(\mathbb{R}^d)$. Using Rothau's lemma as in* (1.4)*, we get that:*

$$Ent_m(f^2) \leq Ent_m(\tilde{f}^2) + 2Var_m(f),$$

where $\tilde{f} = f - m[f]$. Moreover, Popoviciu's inequality establishes that

$$Var_m(f) \leq \frac{1}{4} Osc^2(f).$$

We also note that $Osc(f) = 1$ implies $\sup \tilde{f}^2 \leq 1$ and then $Ent_m(\tilde{f}^2) \leq \frac{1}{e}$. Hence, by homogeneity of the entropy:

$$Ent_m(f^2) \leq \left(\frac{1}{e} + \frac{1}{2}\right) Osc^2(f).$$

In conclusion, for $s \geq \frac{1}{e} + \frac{1}{2}$, we could take $\varphi(s)$ as a constant (in particular $\varphi(s) = 0$) in the definition of WLSI.

In order to produce a criterion to satisfy a WLSI, we need to introduce a family of functional inequalities named measure - capacity inequalities.

A measure - capacity condition

The measure - capacity inequalities are a class of inequalities introduced by [Barthe and Roberto, 2003] and involve the capacity of measurable sets, which we define below. They are commonly used to characterize some functional inequalities, such as Poincaré and LSI. For a more in-depth study, we refer to Chapter 8 of [Bakry et al., 2014].

Let us define the capacity of a measurable set. Let A and Ω be two sets in $\mathcal{B}(\mathbb{R}^d)$ such that $A \subset \Omega$, the capacity $Cap_m(A, \Omega)$ is defined as

$$Cap_m(A, \Omega) := \inf \left\{ \mathcal{E}_m(f), \mathbb{1}_A \leq f \leq \mathbb{1}_\Omega \right\},$$

where f is a Lipschitz function in $\mathbb{R}^d$. If $m(A) \leq \frac{1}{2}$, then we denote

$$Cap_m(A) := \inf \left\{ Cap_m(A, \Omega), A \subset \Omega, m(\Omega) \leq \frac{1}{2} \right\}.$$

Definition 1.3.3 (Measure - capacity inequality)**.** *The probability measure m satisfies a measure - capacity inequality if there exists an increasing function $\Psi : [0, \frac{1}{2}] \to (0, \infty)$ such that*

$$Cap_m(A) \geq \Psi(m(A)),$$

for any $A \in \mathcal{B}(\mathbb{R}^d)$ such that $m(A) \leq \frac{1}{2}$.

The following result, linking the Poincaré inequality with a measure - capacity inequality, is taken from Proposition 8.3.1 of [Bakry et al., 2014].

Proposition 1.3.4. *Assume that the probability measure m satisfies a Poincaré inequality with constant $C_{\mathrm{PI}}(m) > 0$, then m verifies a measure - capacity inequality with function $\Psi(x) = \frac{x}{2C_{\mathrm{PI}}(m)}$, for any $x \in [0, \frac{1}{2}]$. That is,*

$$Cap_m(A) \geq \frac{m(A)}{2C_{\mathrm{PI}}(m)},$$

for any $A \in \mathcal{B}(\mathbb{R}^d)$ such that $m(A) \leq \frac{1}{2}$.

Proof. Let be $A, \Omega \in \mathcal{B}(\mathbb{R}^d)$ such that $A \subset \Omega$ and $m(\Omega) \leq \frac{1}{2}$. A function f as in the definition of $Cap_m(A, \Omega)$ satisfies the requirement that its support is contained in Ω. Then using the Cauchy - Schwarz inequality:

$$\left(\int_{\mathbb{R}^d} f \mathrm{d}m \right)^2 \leq m(\Omega) \int_{\mathbb{R}^d} f^2 \mathrm{d}m,$$

Then

$$Var_m(f) = \int_{\mathbb{R}^d} f^2 \mathrm{d}m - \left(\int_{\mathbb{R}^d} f \mathrm{d}m \right)^2 \geq (1 - m(\Omega)) \int_{\mathbb{R}^d} f^2 \mathrm{d}m \geq \frac{m(A)}{2},$$

where in the last inequality we used $\mathbb{1}_A \leq f^2$ and $m(\Omega) \leq \frac{1}{2}$. Poincaré's inequality implies that,

$$\mathcal{E}_m(f) \geq \frac{Var_m(f)}{C_{\mathrm{PI}}(m)} \geq \frac{m(A)}{2C_{\mathrm{PI}}(m)}.$$

We conclude the result by taking the infimum over f. □

For future use, let us state two propositions taken from [Cattiaux et al., 2007] that characterize WLSI through measure - capacity inequalities.

Proposition 1.3.5. *Assume that m satisfies a WLSI with function φ, then for any $A \in \mathcal{B}(\mathbb{R}^d)$ such that $0 < m(A) \leq \frac{1}{2}$,*

$$\frac{m(A)\log\left(1+\frac{1}{2m(A)}\right)-s}{\varphi(s)} \leq Cap_m(A), \quad \forall s > 0. \tag{1.6}$$

The second proposition shows a sufficient condition to verify a WLSI. Condition (1.7) within is, in fact, a measure - capacity inequality.

Proposition 1.3.6. *Let $\phi : (0,\infty) \to \mathbb{R}^+$ be a non-increasing function such that for any $A \in \mathcal{B}(\mathbb{R}^d)$ with $0 < m(A) \leq \frac{1}{2}$,*

$$\frac{m(A)\log\left(1+\frac{e^2}{m(A)}\right)-s}{\phi(s)} \leq Cap_m(A), \quad \forall s > 0. \tag{1.7}$$

Then m satisfies a WLSI with the function $\varphi(s) = 16\phi(3s/14)$ for $s > 0$.

The proof of Propositions 1.3.5 and 1.3.6 may be found in [Cattiaux et al., 2007]. In Section 1.4, we prove a generalization of Proposition 1.3.6 using the same technique.

From Poincaré's inequality to WLSI

An important result is that if a measure satisfies a Poincaré inequality, then it also satisfies a WLSI. Although this result is part of Proposition 3.1 in [Cattiaux et al., 2007], we rewrite its proof to explicitly find a function φ as required in the WLSI definition.

Proposition 1.3.7. *Assume that the probability measure m satisfies a Poincaré inequality with constant $C_{\mathrm{PI}}(m)$. Then,*

a) There exists a non-increasing and positive function ϕ such that (1.7) holds. Moreover, the function could be defined as

$$\phi(s) = \mathfrak{a} C_{\mathrm{PI}}(m)\log(1/s), \quad 0 < s \leq e^{-1},$$

where $\mathfrak{a} > 0$ is a universal constant.

b) Consequently, a WLSI is verified with function

$$\varphi(s) = \mathfrak{b} C_{\mathrm{PI}}(m)\log(1/s), \quad 0 < s \leq e^{-1},$$

where $\mathfrak{b} > 0$ is a universal constant.

Using Remark 1.3.2, it suffices to define φ when s is small, hence why in part b of Proposition 1.3.7 we define $\varphi(s)$ only for $0 < s \leq e^{-1}$, although a general definition may be found in its proof.

Proof. a). Let $A \in \mathcal{B}(\mathbb{R}^d)$ with $0 < m(A) \leq \frac{1}{2}$. From Proposition 1.3.4, if m satisfies a Poincaré inequality with constant $C_{\mathrm{PI}}(m)$, then:

$$Cap_m(A) \geq \frac{m(A)}{2C_{\mathrm{PI}}(m)}.$$

Using the inequality above, a positive function ϕ verifies (1.7) if, for any $s > 0$,

$$2C_{\mathrm{PI}}(m)\left[\log\left(1+\frac{e^2}{m(A)}\right)-\frac{s}{m(A)}\right] \leq \phi(s).$$

Let us fix $s > 0$ small and define the function $g_s(x) = \log\left(1+\frac{e^2}{x}\right) - \frac{s}{x}$, for $0 < x \leq \frac{1}{2}$. Using that g_s reaches its maximum at

$$x_{\max} = \begin{cases} \frac{se^2}{e^2-s}, & 0 < s \leq \frac{e^2}{2e^2+1} \\ \frac{1}{2}, & s > \frac{e^2}{2e^2+1} \end{cases},$$

then $g_s(x) \leq g_s(x_{\max})$ where

$$g_s(x_{\max}) = \begin{cases} \log\left(\frac{1}{s}\right)+\frac{s}{e^2}+1, & 0 < s \leq \frac{e^2}{2e^2+1} \\ \log(1+2e^2)-2s, & s > \frac{e^2}{2e^2+1} \end{cases}.$$

We define $\phi(s) = 2C_{\mathrm{PI}}(m)g_s(x_{\max})$ if $s > e^{-1}$, while we choose

$$\phi(s) = \mathfrak{a}C_{\mathrm{PI}}(m)\log(1/s) \geq 2C_{\mathrm{PI}}(m)g_s(x_{\max}),$$

for any $0 < s \leq e^{-1}$, where $\mathfrak{a} = 2 + e^{-3}$.

b). From Proposition 1.3.6, the proof of the second statement is reduced to verify some simplifications and upper bounds. In general, we could define $\varphi(s) = 16\phi(\frac{3s}{14})$, where ϕ is the function defined in part *a*. However, to simplify its expression we consider φ as:

$$\varphi(s) = \mathfrak{b}C_{\mathrm{PI}}(m)\log(1/s), \quad 0 < s \leq e^{-1},$$

where $\mathfrak{b} = 16\mathfrak{a}(1+\log(14/3))$ is a universal constant. □

Entropy decay of a diffusion semigroup

Let us prove that a WLSI guarantees the convergence in entropy of the semigroup $(P_t)_{t\geq 0}$ with invariant measure m. Let f be a bounded density function. The following result is taken from [Cattiaux et al., 2007] and connects the decay of the entropy with $\|f\|_{\mathbb{L}^\infty_m(\mathbb{R}^d)}$.

Proposition 1.3.8. *Assume that m satisfies a WLSI with function φ. Then, for any positive function $f \in \mathcal{C}^1_b(\mathbb{R}^d)$ and t large enough,*

$$Ent_m(P_t f) \leq \left(\frac{1}{2}+\frac{1}{e}+\varepsilon\right)\xi_\varepsilon(t)Osc^2\left(\sqrt{f}\right),$$

where ε is a small constant and $\xi_\varepsilon(t)$ is given by

$$t = \xi_\varepsilon^{-1}(s) = -\frac{1}{4}\varphi(s)\log(s/\varepsilon),$$

for s small enough.

Proof. De Bruijn's identity and WLSI imply that, for any $s > 0$,

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}t}Ent_m\left(P_t f\right) &= -4\mathcal{E}_m\left(\sqrt{P_t f}\right)\\ &\leq -\frac{4}{\varphi(s)}Ent_m(P_t f) + \frac{4s}{\varphi(s)}Osc^2\left(\sqrt{P_t f}\right).\end{aligned}$$

Using Gronwall's lemma,

$$Ent_m\left(P_t f\right) \leq \inf_{s>0}\left\{s\sup_{u\in[0,t]}Osc^2\left(\sqrt{P_u f}\right) + e^{-\frac{4t}{\varphi(s)}}Ent_m\left(f\right)\right\}.$$

By definition, we already know that for any $x \in \mathbb{R}^d$,

$$P_t f(x) = \mathbb{E}[f(X_t)|X_0 = x], \quad t > 0.$$

Then, $P_t f$ is bounded, $\inf f \leq P_t f(x) \leq \sup f$, for any $x \in \mathbb{R}^d$ and

$$Osc\left(\sqrt{P_t f}\right) \leq Osc\left(\sqrt{f}\right), \quad t > 0.$$

From Remark 1.3.2, we recall that $Ent_m\left(f\right) \leq \left(\frac{1}{2} + \frac{1}{e}\right)Osc^2\left(\sqrt{f}\right)$. Then

$$Ent_m\left(P_t f\right) \leq \inf_{s>0}\left(s + e^{-\frac{4t}{\varphi(s)}}\left(\frac{1}{2} + \frac{1}{e}\right)\right)Osc^2\left(\sqrt{f}\right).$$

Finally, we choose $s = s_t$ so that $s = \varepsilon e^{-\frac{4t}{\varphi(s)}}$. □

The above result is not explicit in terms of t. Then, let us study the particular case in which the Poincaré inequality is satisfied and therefore, thanks to Proposition 1.3.7 we find that a WLSI is also satisfied with a simpler function φ.

Corollary 1.3.9. *Assume that m satisfies the Poincaré inequality with constant $C_{\mathrm{PI}}(m)$. Then, for any positive function $f \in \mathcal{C}_b^1(\mathbb{R}^d)$ and t large enough,*

$$Ent_m(P_t f) \leq \left(\frac{1}{2} + \frac{1}{e} + \varepsilon\right)Osc^2\left(\sqrt{f}\right)e^{-\sqrt{\frac{4t}{\mathfrak{b}C_{\mathrm{PI}}(m)}}},$$

where ε is a small constant and $\mathfrak{b}$ is a universal constant.

Proof. From Proposition 1.3.7, we deduce that m satisfies a WLSI with

$$\varphi(s) = \mathfrak{b}C_{\mathrm{PI}}(m)\log(1/s), \quad 0 < s < e^{-1},$$

where $\mathfrak{b}$ is a positive constant. Let us choose $s = s_t = e^{-A\sqrt{t}}$, for $A > 0$, in the proof of Proposition 1.3.8. Then, if $t > A^{-2}$,

$$\varphi(s_t) = A\mathfrak{b}C_{\mathrm{PI}}(m)\sqrt{t}$$

and

$$\begin{aligned} Ent_m\left(P_t f\right) &\leq \left(e^{-A\sqrt{t}} + e^{-\frac{4\sqrt{t}}{A\mathfrak{b}C_{\mathrm{PI}}(m)}}\left(\frac{1}{2}+\frac{1}{e}\right)\right) Osc^2\left(\sqrt{f}\right) \\ &\leq \left(\frac{3}{2}+\frac{1}{e}\right) Osc^2\left(\sqrt{f}\right) e^{-\sqrt{\frac{4t}{\mathfrak{b}C_{\mathrm{PI}}(m)}}}, \end{aligned}$$

where we have chosen $A = \frac{2}{\sqrt{\mathfrak{b}C_{\mathrm{PI}}(m)}}$. □

However, it is in our best interest to consider the general case. Let us study an example where Poincaré's inequality does not hold and for which we obtain a polynomial convergence of the entropy.

Example 1.3.10. *As we proved in Example 1.2.9, $m_\alpha(\theta) = \frac{\alpha}{2}(1+|\theta|)^{-1-\alpha}$, for $\theta \in \mathbb{R}$ and $\alpha > 0$ does not satisfy Poincaré's inequality. However, it satisfies a WLSI with function*

$$\varphi(s) = Cs^{-2/\alpha}\log^{1+2/\alpha}(1/s),$$

where $C > 0$ is a universal constant. This example was taken from [Cattiaux et al., 2007], but we will choose the value of s_t differently to simplify some computations.

We define $s = s_t = t^{-\frac{\alpha}{2(2+\alpha)}}$ when t is large, then $\varphi(s_t) = Ct^{\frac{1}{2+\alpha}}\log^{1+\frac{2}{\alpha}}(t^{\frac{\alpha}{2(2+\alpha)}})$ and

$$\begin{aligned} Ent_{m_\alpha}(P_t f) &\lesssim_{uc} \left(s_t + e^{-\frac{4t}{\varphi(s_t)}}\right) Osc^2\left(\sqrt{f}\right) \\ &\lesssim_{uc} t^{-\frac{\alpha}{2(2+\alpha)}} Osc^2\left(\sqrt{f}\right). \end{aligned}$$

1.4 Weak log-Sobolev inequality for unbounded functions

According to the definition of WLSI, the function f has to be bounded, which is, in general, a very strong condition. We now introduce a new functional inequality, which we named p - weak log-Sobolev inequality (p-WLSI) where $p > 2$ is a constant. As its name indicates, this inequality is a WLSI-type inequality and is suitable for a family of unbounded functions, particularly the functions in $\mathbb{L}^p_m(\mathbb{R}^d)$.

Definition 1.4.1 (p-weak log-Sobolev inequality)**.** *Let $p > 2$ be a fixed constant. The probability measure m in $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$ satisfies a p-WLSI if there exists a non-increasing function $\varphi_p : (0,\infty) \mapsto \mathbb{R}_+$ such that for any $f \in \mathcal{H}^1_m(\mathbb{R}^d) \cap \mathbb{L}^p_m(\mathbb{R}^d)$ and any $s > 0$,*

$$Ent_m(f^2) \leq \varphi_p(s)\mathcal{E}_m(f) + s\|f\|^2_{\mathbb{L}^p_m(\mathbb{R}^d)}.$$

The main difference between the WLSI and the p-WLSI is the space of functions on which we define each inequality. WLSI is valid on $\mathbb{L}^{\infty}_m(\mathbb{R}^d)$, so the inequality contains a quantity related to the ∞-norm, while the p-WLSI is valid for functions in $\mathbb{L}^p_m(\mathbb{R}^d)$, so the p-norm appears in the definition.

It is important to mention that this inequality generalizes the inequality studied in Section 3.4 in [Cattiaux et al., 2007] where, instead of $\|f\|^2_{\mathbb{L}^p_m(\mathbb{R}^d)}$, they use $Var_m(f^2)$, which would correspond to $p = 4$.

Remark 1.4.2. *Let us check that, similarly to Remark 1.3.2, the inequality above should be considered only for values of s below a fixed constant (in general such a constant depends on p).*

Let $f \in \mathcal{H}^1_m(\mathbb{R}^d) \cap \mathbb{L}^p_m(\mathbb{R}^d)$ and H be a positive function such that $\int_{\mathbb{R}^d} e^H \mathrm{d}m \leq 1$. Then

$$\int_{\mathbb{R}^d} f^2 H \mathrm{d}m \leq \left(\int_{\mathbb{R}^d} \left(f^2\right)^{p/2} \mathrm{d}m\right)^{\frac{2}{p}} \left(\int_{\mathbb{R}^d} H^q \mathrm{d}m\right)^{\frac{1}{q}},$$

where we used Holder's inequality with constants $\frac{p}{2} > 1$ and $q = \frac{p}{p-2} > 1$. The first integral on the right-hand side is $\|f\|^2_{\mathbb{L}^p_m(\mathbb{R}^d)}$, so let us study the second integral.

The inequality $\left(1 + \frac{x}{y}\right)^y \leq e^x$ for any $x, y > 0$, implies that

$$\int_{\mathbb{R}^d} H^q \mathrm{d}m \leq q^q \int_{\mathbb{R}^d} (1 + H/q)^q \mathrm{d}m \leq q^q \int_{\mathbb{R}^d} e^H \mathrm{d}m \leq q^q.$$

Thus, for any positive function H such that $\int_{\mathbb{R}^d} e^H \mathrm{d}m \leq 1$, we proved that

$$\int_{\mathbb{R}^d} f^2 H \mathrm{d}m \leq q \|f\|^2_{\mathbb{L}^p_m(\mathbb{R}^d)}.$$

Taking supremum over the functions H and using the variational definition of the entropy in (1.3)*, we conclude that*

$$Ent(f^2) \leq q \|f\|^2_{\mathbb{L}^p_m(\mathbb{R}^d)}. \tag{1.8}$$

The value of q depends on the value of p, so if p is large, q would be close to 1 (although always greater than 1). The function φ_p in the definition of p-WLSI should always be considered as 0 for values of $s > q = \frac{p}{p-2}$.

A condition to get WLSI for unbounded functions

We present one of the main results of this chapter, which is a generalization of Proposition 1.3.6 taken from [Cattiaux et al., 2007]. We assume that m satisfies the measure - capacity inequality 1.7 stated in 1.3.6 and then we prove that it also satisfies the WLSI for unbounded functions for any $p > 2$.

Theorem 1.4.3. *Let $\phi : (0, \infty) \to \mathbb{R}^+$ be a non-increasing function that satisfies the measure - capacity inequality* (1.7)*. That is, for any $A \in \mathcal{B}(\mathbb{R}^d)$ with $0 < m(A) \leq \frac{1}{2}$,*

$$\frac{m(A) \log\left(1 + \frac{e^2}{m(A)}\right) - s}{\phi(s)} \leq Cap_m(A), \quad \forall s > 0.$$

Then, for any $p > 2$ the probability measure m satisfies a p-WLSI with the function $\varphi_p(s) = 16\phi\left((s/a_p)^{\frac{p+6}{p-2}}\right)$ where a_p is a positive constant that depends on p.

The proof of the above theorem is postponed in order to recall some basic results on entropy and to show how to avoid the $Osc(f)$ term that appears in the definition of WLSI.

The following corollary is an immediate consequence of the part a of Proposition 1.3.7 and Theorem 1.4.3.

Corollary 1.4.4. *Assume that m satisfies the Poincaré inequality with constant $C_{\mathrm{PI}}(m)$. Then m satisfies a p-WLSI, for any $p > 2$. Moreover, the function φ_p could be defined as*

$$\varphi_p(s) = b_p C_{\mathrm{PI}}(m) \log\left(a_p/s\right), \quad 0 < s \leq s_p,$$

where a_p is the constant obtained in Theorem 1.4.3, and b_p and s_p are positive constants depending of p.

Proof. We focus only on finding the values of the constants b_p and s_p. From part a in Proposition 1.3.7, if m satisfies a Poincaré inequality with constant $C_{\mathrm{PI}}(m)$, then

$$\phi(s) = \mathfrak{a} C_{\mathrm{PI}}(m) \log\left(1/s\right), \quad 0 < s \leq e^{-1},$$

satisfies the inequality (1.7). Theorem 1.4.3 implies that m satisfies a p-WLSI for any $p > 2$ with the function

$$\varphi_p(s) = b_p C_{\mathrm{PI}}(m) \log\left(a_p/s\right), \quad 0 < s \leq s_p,$$

where $b_p = \frac{16\mathfrak{a}(p+6)}{p-2}$ and $s_p = \min\left\{\frac{p}{p-2}, a_p e^{-\frac{p-2}{p+6}}\right\}$. □

If m is a log-concave measure, then it satisfies the Poincaré inequality and we could apply Corollary 1.4.4 to deduce that m also satisfies a p-WLSI. So, let us study an example where the measure is not log-concave and does not satisfy Poincaré's inequality.

Example 1.4.5. *Let $\alpha \in (0,1)$ and $m^\alpha(\theta) \propto e^{-|\theta|^\alpha}$, for $\theta \in \mathbb{R}$. As we proved in Example 1.2.9, this probability measure does not satisfy Poincaré's inequality. However, in [Cattiaux et al., 2007] it was shown that m^α satisfies a WLSI with function*

$$\varphi(s) = C(\log(1/s))^{\frac{2-\alpha}{\alpha}}, \quad s > 0,$$

for some $C > 0$. Using Proposition 1.3.5, we deduce that φ verifies the measure - capacity (1.6). Taking into account that there exists a constant $c \in (0,1)$ such that

$$c \log\left(1 + \frac{e^2}{x}\right) \leq \log\left(1 + \frac{1}{2x}\right), \quad \forall x \in (0, 1/2],$$

then, for any $A \in \mathcal{B}(\mathbb{R}^d)$ *with* $0 < m^\alpha(A) < 1/2$,

$$\frac{m^\alpha(A)\log\left(1+\frac{e^2}{m^\alpha(A)}\right)-s}{c^{-1}\varphi(cs)} \leq Cap_{m^\alpha}(A), \quad \forall s>0.$$

Theorem 1.4.3 concludes that, for any $p>2$, m^α *satisfies* p*-WLSI with function* $\varphi_p = 16c^{-1}\varphi\left(c\,(s/a_p)^{\frac{p+6}{p-2}}\right)$, *where* a_p *is a positive constant that depends on* p.

Proof of the main result

Before presenting the proof of Theorem 1.4.3, let us recall some results on the entropy. We introduce as well a preliminary proposition which will allow us to dispense of the boundedness of f.

A review of some entropy results

Let us present two lemmas taken from [Barthe and Roberto, 2003] that, although simple, are very important to prove Theorem 1.4.3. Because of its importance, from now on, we denote

$$\mathcal{H} = \left\{H : \mathbb{R}^d \to \mathbb{R}_+, \int_{\mathbb{R}^d} e^H \mathrm{d}m \leq 1+e^2\right\}. \tag{1.9}$$

Lemma 1.4.6. *Let* f *be a non-negative function with finite entropy. Then,*

$$Ent_m(f) + 2m[f] \leq \sup_{H\in\mathcal{H}} \int_{\mathbb{R}^d} fH\mathrm{d}m.$$

Lemma 1.4.7 is a simplified version of Lemma 6 from [Barthe and Roberto, 2003]: we simply consider the case where m is a probability measure and the supremum on the left-hand side is taken only over $\mathcal{H}$.

Lemma 1.4.7. *Let* m *be a probability measure in* $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$. *Let* $A \in \mathcal{B}(\mathbb{R}^d)$ *be a subset with* $m(A) > 0$. *Then,*

$$\sup_{H\in\mathcal{H}} \int_A H\mathrm{d}m = m(A)\log\left(1+\frac{e^2}{m(A)}\right).$$

The proofs of Lemmas 1.4.6 and the general version of 1.4.7 may be found in Section 2 of [Barthe and Roberto, 2003].

A technical result to avoid $Osc(f)$

We recall that for a function $f \in \mathcal{C}_b^1(\mathbb{R}^d)$, we define

$$Osc(f) = \sup f - \inf f.$$

To replace the role of the oscillation in the proof of Proposition 2.2 in [Cattiaux et al., 2007], we start by showing how they used the boundedness of f.

For any function $f \in \mathcal{H}^1_m(\mathbb{R}^d)$, let M be the median of f with respect to the probability measure m, that is, $m(f > M) = m(f < M) = 1/2$. Consider a subset $\Omega \subset \{f > M\}$ and an arbitrary non-negative function $H \in \mathcal{H}$. In [Cattiaux et al., 2007], they use the following bound:

$$\int_\Omega (f - M)^2 H \mathrm{d}m \leq Osc^2(f) \int_\Omega H \mathrm{d}m,$$

since the function f is assumed to be bounded.

Now, instead of bounding $\int_\Omega (f - M)^2 H \mathrm{d}m$, let us just consider

$$\int_\Omega f^2 H \mathrm{d}m,$$

since $0 \leq f - M \leq f$ on Ω and then $\int_\Omega (f - M)^2 H \mathrm{d}m \leq \int_\Omega f^2 H \mathrm{d}m$. The following result is a key point to prove Theorem 1.4.3.

Proposition 1.4.8. *Assume that m is a probability measure in $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$. Let $p > 2$ and $f \in \mathbb{L}^p_m(\mathbb{R}^d)$. Then, for any $H \in \mathcal{H}$,*

$$\int_\Omega f^2 H \mathrm{d}m \leq C_q \, \|f\|^2_{\mathbb{L}^p_m(\mathbb{R}^d)} \left(\int_\Omega H \mathrm{d}m \right)^{1/2q},$$

where $q = \frac{p}{p-2}$ and C_q is positive constant depending on q.

Proof. We use Holder's inequality with constants $\frac{p}{2} > 1$ and $q = \frac{p}{p-2} > 1$ and then Cauchy-Schwarz inequality to get that

$$\begin{aligned} \int_\Omega f^2 H \mathrm{d}m &\leq \left[\int_\Omega |f|^p \mathrm{d}m \right]^{\frac{2}{p}} \left[\int_\Omega H^q \mathrm{d}m \right]^{\frac{1}{q}} \\ &\leq \|f\|^2_{\mathbb{L}^p_m(\mathbb{R}^d)} \left[\int_\Omega H \mathrm{d}m \right]^{\frac{1}{2q}} \left[\int_{\mathbb{R}^d} H^{2q-1} \mathrm{d}m \right]^{\frac{1}{2q}}. \end{aligned} \tag{1.10}$$

Let's study the last term in (1.10). Inequality $\left(1 + \frac{x}{y}\right)^y \leq e^x$ for any $x, y > 0$, implies that

$$\int_{\mathbb{R}^d} H^{2q-1} \mathrm{d}m \leq (2q-1)^{2q-1} \int_{\mathbb{R}^d} e^H \mathrm{d}m \leq (2q-1)^{2q-1}(1 + e^2), \tag{1.11}$$

where we used hypothesis $\int_{\mathbb{R}^d} e^H \mathrm{d}m \leq 1 + e^2$. The statement is obtained immediately by replacing (1.11) into (1.10) and taking $C_q = (2q-1)^{1-1/2q}(1 + e^2)^{1/2q}$. □

Proof of Theorem 1.4.3

The principal difference between the proofs of Proposition 1.4.3 and Proposition 1.3.6 (which is stated in [Cattiaux et al., 2007]) is based on the Proposition 1.4.8. The general idea remains similar; however, we rewrote the entire first half of the proof differently.

Let $p > 2$ be fixed and $f \in \mathcal{H}^1_m(\mathbb{R}^d) \cap \mathbb{L}^p_m(\mathbb{R}^d)$. $M > 0$ denotes the median of f with respect to the measure m and set $\Omega_+ = \{f > M\}$, $\Omega_- = \{f < M\}$,

$$F_+ = (f - M)\mathbb{1}_{\Omega_+} \quad \text{and} \quad F_- = (f - M)\mathbb{1}_{\Omega_-}.$$

Using Rothau's lemma and the subadditivity of the entropy, we get that

$$\begin{aligned} Ent_m(f^2) &\leq Ent_m((f-M)^2) + 2m[(f-M)^2] \\ &\leq Ent_m(F_+^2) + Ent_m(F_-^2) + 2m[F_+^2] + 2m[F_-^2]. \end{aligned}$$

Lemma 1.4.6 implies that

$$Ent_m(f^2) \leq \sup_{H \in \mathcal{H}} \int F_+^2 H \mathrm{d}m + \sup_{H \in \mathcal{H}} \int F_-^2 H \mathrm{d}m. \tag{1.12}$$

Let's study each term separately.

- First term of (1.12). Let be $c > 0$ and $\rho \in (0,1)$. We introduce the sequence of measurable subsets

  $$\Omega_i = \left\{F_+^2 \geq c\rho^i\right\}, \quad i = 0, 1, 2, \ldots,$$

  which is increasing and $\bigcup_{i \geq 0} \Omega_i = \Omega_+$ so that, for any function $H \geq 0$,

  $$\int F_+^2 H \mathrm{d}m = \int_{\Omega_0} F_+^2 H \mathrm{d}m + \sum_{i \geq 1} \int_{\Omega_i \setminus \Omega_{i-1}} F_+^2 H \mathrm{d}m,$$

  and then

  $$\sup_{H \in \mathcal{H}} \int F_+^2 H \mathrm{d}m \leq \sup_{H \in \mathcal{H}} \int_{\Omega_0} F_+^2 H \mathrm{d}m + \sup_{H \in \mathcal{H}} \sum_{i \geq 1} \int_{\Omega_i \setminus \Omega_{i-1}} F_+^2 H \mathrm{d}m. \tag{1.13}$$

  - First term of (1.13). On Ω_0, $F_+^2 \leq f^2$, then there exist two constants $q = \frac{p}{p-2}$ and C_q (which depends on q) such that

    $$\begin{aligned} \sup_{H \in \mathcal{H}} \int_{\Omega_0} F_+^2 H \mathrm{d}m &\leq \sup_{H \in \mathcal{H}} \int_{\Omega_0} f^2 H \mathrm{d}m \\ &\leq C_q \|f\|^2_{\mathbb{L}^p_m(\mathbb{R}^d)} \sup_{H \in \mathcal{H}} \left(\int_{\Omega_0} H \mathrm{d}m\right)^{\frac{1}{2q}} \\ &= C_q \|f\|^2_{\mathbb{L}^p_m(\mathbb{R}^d)} \left[m(\Omega_0) \log\left(1 + \frac{e^2}{m(\Omega_0)}\right)\right]^{\frac{1}{2q}} \end{aligned}$$

    where we used Proposition 1.4.8 in the second line and Lemma 1.4.7 in the last line.

    Using that $\Omega_0 \subset \{f^2 \geq c\}$ and $c > 0$, we verify that

    $$m(\Omega_0) = \int_{\Omega_0} \mathrm{d}m \leq \frac{1}{c^{p/2}} \int_{\Omega_0} |f|^p \mathrm{d}m \leq \left(\frac{\|f\|^2_{\mathbb{L}^p_m(\mathbb{R}^d)}}{c}\right)^{p/2}.$$

Moreover, the inequality $\log(1+x) \leq \sqrt{x}$, for all $x \geq 0$, guarantees that

$$m(\Omega_0)\log\left(1+\frac{e^2}{m(\Omega_0)}\right) \leq e\sqrt{m(\Omega_0)} \leq e\left(\frac{\|f\|^2_{\mathbb{L}^p_m(\mathbb{R}^d)}}{c}\right)^{p/4}$$

then

$$\sup_{H\in\mathcal{H}}\int_{\Omega_0} F_+^2 H\mathrm{d}m \leq eC_q\|f\|^{2+p/4q}_{\mathbb{L}^p_m(\mathbb{R}^d)}c^{-\frac{p}{8q}}. \tag{1.14}$$

– Second term of (1.13). Due to the fact that $c\rho^i \leq F_+^2 \leq c\rho^{i-1}$ on $\Omega_i\backslash\Omega_{i-1}$, for all $i \geq 1$, we have that

$$\int_{\Omega_i\backslash\Omega_{i-1}} F_+^2 H\mathrm{d}m \leq c\rho^{i-1}\int_{\Omega_i\backslash\Omega_{i-1}} H\mathrm{d}m.$$

Using again Lemma 1.4.7, for any $i \geq 1$, we obtain that

$$\begin{aligned}\sup_{H\in\mathcal{H}}\int_{\Omega_i\backslash\Omega_{i-1}} F_+^2 H\mathrm{d}m &\leq c\rho^{i-1}m(\Omega_i\setminus\Omega_{i-1})\log\left(1+\frac{e^2}{m(\Omega_i\setminus\Omega_{i-1})}\right)\\ &\leq sc\rho^{i-1}+c\rho^{i-1}\phi(s)Cap_m(\Omega_i\setminus\Omega_{i-1}),\end{aligned}$$

where we used the measure - capacity.

Let set for any $i \geq 1$,

$$g_i = \begin{cases} 1 & \text{on } \Omega_i \\ \frac{F_+-\sqrt{c\rho^{i+1}}}{\sqrt{c\rho^i}-\sqrt{c\rho^{i+1}}} & \text{on } \Omega_{i+1}\setminus\Omega_i \\ 0 & \text{on } \Omega_{i+1}^c \end{cases},$$

so that we have $\mathbb{1}_{\Omega_i} \leq g_i \leq \mathbb{1}_{\Omega_+}$, recall that $m(\Omega_+) = 1/2$. This implies, using the definition of $Cap_m(\Omega_i\setminus\Omega_{i-1})$, that

$$c\rho^{i-1}Cap_m(\Omega_i\setminus\Omega_{i-1}) \leq \frac{1}{\rho(1-\sqrt{\rho})^2}\int_{\Omega_{i+1}\backslash\Omega_i}\|\nabla F_+\|_2^2\mathrm{d}m,$$

then

$$\sup_{H\in\mathcal{H}}\int_{\Omega_i\backslash\Omega_{i-1}} F_+^2 H\mathrm{d}m \leq sc\rho^{i-1}+\frac{\phi(s)}{\rho(1-\sqrt{\rho})^2}\int_{\Omega_{i+1}\backslash\Omega_i}\|\nabla F_+\|_2^2\mathrm{d}m. \tag{1.15}$$

We replace (1.14) and (1.15) in (1.13), then

$$\begin{aligned}\sup_{H\in\mathcal{H}}\int F_+^2 H\mathrm{d}m &\leq \sup_{H\in\mathcal{H}}\int_{\Omega_0} F_+^2 H\mathrm{d}m + \sup_{H\in\mathcal{H}}\sum_{i\geq 1}\int_{\Omega_i\backslash\Omega_{i-1}} F_+^2 H\mathrm{d}m\\ &\leq eC_q\|f\|^{2+p/4q}_{\mathbb{L}^p_m(\mathbb{R}^d)}c^{-\frac{p}{8q}}+\frac{sc}{1-\rho}+\frac{\phi(s)}{\rho(1-\sqrt{\rho})^2}\int\|\nabla F_+\|_2^2\mathrm{d}m.\end{aligned}$$

The optimal values of c and ρ are $\|f\|^2_{\mathbb{L}^p_m(\mathbb{R}^d)}\, s^{-\frac{8q}{p+8q}}$ and $1/4$ respectively, so we get

$$\sup_{H\in\mathcal{H}} \int F_+^2 H \mathrm{d}m \leq \left(eC_q + \frac{4}{3}\right) \|f\|^2_{\mathbb{L}^p_m(\mathbb{R}^d)}\, s^{\frac{p}{p+8q}} + 16\phi(s) \int_{\Omega_+} \|\nabla f\|_2^2 \mathrm{d}m.$$

- Second term of (1.12). Since $0 \leq f \leq M$ on Ω_-, the second term of (1.12) will be treated similarly but changing the following bounds:

$$\sup_{H\in\mathcal{H}} \int_{\Omega^0} F_-^2 H \mathrm{d}m \leq M^2 \sup_{H\in\mathcal{H}} \int_{\Omega^0} H \mathrm{d}m = M^2 m\left(\Omega^0\right) \log\left(1 + \frac{e^2}{m\left(\Omega^0\right)}\right),$$

where $\Omega^0 = \left\{F_-^2 \geq c\right\}$, and then we verify that

$$m(\Omega^0) = \int_{\Omega^0} \mathrm{d}m \leq \frac{1}{c^{p/4q}} \int_{\Omega^0} (F_-^2)^{p/4q} \mathrm{d}m \leq \left(\frac{M^2}{c}\right)^{p/4q}.$$

Once again, the inequality $\log(1+x) \leq \sqrt{x}$, for all $x \geq 0$, guarantees that

$$\sup_{H\in\mathcal{H}} \int_{\Omega^0} F_-^2 H \mathrm{d}m \leq eM^2 \sqrt{m\left(\Omega_0\right)} \leq eM^{2+p/4q} c^{-\frac{p}{8q}}.$$

Then, after optimizing the value of c as $M^2 s^{-\frac{8q}{p+8q}}$, we obtain that for any $s > 0$,

$$\sup_{H\in\mathcal{H}} \int F_-^2 H \mathrm{d}m \leq \left(e + \frac{4}{3}\right) M^2 s^{\frac{p}{p+8q}} + 16\phi(s) \int_{\Omega_-} \|\nabla f\|_2^2 \mathrm{d}m.$$

Using the inequality

$$\|f\|_{\mathbb{L}^p_m(\mathbb{R}^d)} \geq \left(\int_{f>M} |f|^p \mathrm{d}m\right)^{\frac{1}{p}} \geq 2^{-\frac{1}{p}} M,$$

we deduce that $M^2 \leq 2\|f\|_{\mathbb{L}^p_m(\mathbb{R}^d)}$, since $p > 2$, and

$$\sup_{H\in\mathcal{H}} \int F_-^2 H \mathrm{d}m \leq 2\left(e + \frac{4}{3}\right) s^{\frac{p}{p+8q}} \|f\|^2_{\mathbb{L}^p_m(\mathbb{R}^d)} + 16\phi(s) \int_{\Omega_-} \|\nabla f\|_2^2 \mathrm{d}m.$$

We obtain that

$$Ent_m(f^2) \leq a_p s^{\frac{p}{p+8q}} \|f\|^2_{\mathbb{L}^p_m(\mathbb{R}^d)} + 16\phi(s) \int_{\mathbb{R}^d} \|\nabla f\|_2^2 \mathrm{d}m,$$

where $a_p = e(2+C_q)+4$, and we recall that C_q is the constant defined in Proposition 1.4.8. Taking $u = a_p s^{\frac{p}{p+8q}} = a_p s^{\frac{p-2}{p+6}}$ and $\varphi_p(u) = 16\phi\left(\left(\frac{u}{a_p}\right)^{\frac{p+6}{p-2}}\right)$, we finally get that

$$Ent_m(f^2) \leq u\|f\|^2_{\mathbb{L}^p_m(\mathbb{R}^d)} + \varphi_p(u) \int_{\mathbb{R}^d} \|\nabla f\|_2^2 \mathrm{d}m, \quad \forall u > 0.$$

Entropy decay of a diffusion semigroup

If a p-WLSI is verified, then we can obtain a result similar to that of Proposition 1.3.8, where again, the convergence of the entropy is not explicit as a function of t. The proof of the first part of the said result is analogous to the proof of Proposition 1.3.8, so we omit it. The only differences reside on the consideration $\epsilon = 1$, which is done for simplicity and the replacement of $\frac{1}{2} + \frac{1}{e}$ by $q = \frac{p}{p-2}$, according to Remark 1.4.2.

The proof of the second part of the statement is based on the fact that if a measure verifies Poincaré's inequality, then it also verifies p-WLSI with the particular function φ_p.

Theorem 1.4.9. *Assume that $p > 2$ and m satisfies a p-WLSI with function φ_p. Then, for any positive function f such that $\sqrt{f} \in \mathbb{L}^p_m(\mathbb{R}^d)$ and any t large enough,*

$$Ent_m(P_t f) \leq (1+q)\,\xi(t) \left\| \sqrt{f} \right\|^2_{\mathbb{L}^p_m(\mathbb{R}^d)},$$

where $q = \frac{p}{p-2}$ and $\xi(t)$ is given by $t = \xi^{-1}(s) = -\frac{1}{4}\varphi(s)\log(s)$, for s small enough. Furthermore, if m also satisfies a Poincaré inequality with constant $C_{\mathrm{PI}}(m)$, then,

$$Ent_m(P_t f) \leq (a_p + q) e^{-\sqrt{\frac{4t}{b_p C_{\mathrm{PI}}(m)}}} \left\| \sqrt{f} \right\|^2_{\mathbb{L}^p_m(\mathbb{R}^d)},$$

where a_p and b_p are two positive constant depending on p.

Proof. As we mentioned earlier, we prove only the second part of the statement. From Corollary 1.4.4, we deduce that m satisfies a p-WLSI with

$$\varphi_p(s) = b_p C_{\mathrm{PI}}(m) \log(a_p/s), \quad 0 < s < s_p,$$

where a_p, b_p and s_p are positive constants. Then, De Bruijn's identity and p-WLSI imply that, for any $s > 0$,

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}t} Ent_m\left(P_t f\right) &= -4\mathcal{E}_m\left(\sqrt{P_t f}\right) \\ &\leq -\frac{4}{\varphi_p(s)} Ent_m(P_t f) + \frac{4s}{\varphi_p(s)} \left\| \sqrt{P_t f} \right\|^2_{\mathbb{L}^p_m(\mathbb{R}^d)}. \end{aligned}$$

Using Gronwall's lemma, we get

$$Ent_m\left(P_t f\right) \leq \inf_{s>0} \left\{ s \sup_{u\in[0,t]} \left\| \sqrt{P_u f} \right\|^2_{\mathbb{L}^p_m(\mathbb{R}^d)} + e^{-\frac{4t}{\varphi_p(s)}} Ent_m\left(f\right) \right\}.$$

We choose $s = s_t = a_p e^{-A\sqrt{t}}$ when t is large and $A > 0$ such that $s_t < s_p$ then

$$\varphi_p(s_t) = A b_p C_{\mathrm{PI}}(m) \sqrt{t}$$

and

$$\begin{aligned} Ent_m\left(P_t f\right) &\leq a_p e^{-A\sqrt{t}} \sup_{u\in[0,t]} \left\|\sqrt{P_u f}\right\|^2_{\mathbb{L}^p_m(\mathbb{R}^d)} + e^{-\frac{4\sqrt{t}}{Ab_p C_{\mathrm{PI}}(m)}} Ent_m\left(f\right) \\ &\leq e^{-\sqrt{\frac{4t}{b_p C_{\mathrm{PI}}(m)}}} \left(a_p \sup_{u\in[0,t]} \left\|\sqrt{P_u f}\right\|^2_{\mathbb{L}^p_m(\mathbb{R}^d)} + Ent_m\left(f\right)\right), \end{aligned}$$

where we chose $A = \frac{2}{\sqrt{b_p C_{\mathrm{PI}}(m)}}$.

Since $p > 2$ and $x \mapsto x^{p/2}$ is convex, using Jensen's inequality and that m is the invariant measure of $(P_t)_{t\geq 0}$, then, for any $t > 0$,

$$\begin{aligned} \left\|\sqrt{P_t f}\right\|^{2p}_{\mathbb{L}^p_m(\mathbb{R}^d)} &= \left(\int_{\mathbb{R}^d} (P_t f)^{p/2} \mathrm{d}m\right)^2 \\ &\leq \left(\int_{\mathbb{R}^d} P_t f^{p/2} \mathrm{d}m\right)^2 \\ &= \left(\int_{\mathbb{R}^d} f^{p/2} \mathrm{d}m\right)^2 \\ &= \left\|\sqrt{f}\right\|^{2p}_{\mathbb{L}^p_m(\mathbb{R}^d)}. \end{aligned}$$

We conclude by using inequality (1.8) in Remark 1.4.2. □

Summary of the links between functional inequalities

Although it is not part of the main objective of this chapter, it is not difficult to prove that m satisfies a WLSI if and only if it satisfies a p-WLSI, $p > 2$. As mentioned above, the main difference lies in the set of functions on which they are defined.

Proposition 1.4.10. *The following conditions are satisfied.*

a) If m satisfies a WLSI, then it satisfies a p-WLSI for any $p > 2$.

b) If there exists $p > 2$ such that m satisfies p-WLSI, then it satisfies WLSI.

Proof. *a)* The proof is based on Example 1.4.5. If m satisfies a WLSI with function φ, then, using Proposition 1.3.5, we deduce that φ verifies the measure - capacity (1.6). Taking into account that

$$c \log\left(1 + \frac{e^2}{x}\right) \leq \log\left(1 + \frac{1}{2x}\right), \quad \forall x \in (0, 1/2], \tag{1.16}$$

where $c = \frac{\log(2)}{\log(1+2e^2)}$, then, for any $A \in \mathcal{B}(\mathbb{R}^d)$ with $0 < m(A) < 1/2$,

$$\frac{m(A)\log\left(1 + \frac{e^2}{m(A)}\right) - s}{c^{-1}\varphi(cs)} \leq Cap_m(A), \quad \forall s > 0.$$

Theorem 1.4.3 concludes that for any $p > 2$, m satisfies a p-WLSI with function $\varphi_p = 16c^{-1}\varphi\left(c\,(s/a_p)^{\frac{p+6}{p-2}}\right)$, where a_p is a positive constant that depends on p.

b) This second part is similar to the proof of Proposition 1.3.5 that appears in [Cattiaux et al., 2007]. Let $A, \Omega \in \mathcal{B}(\mathbb{R}^d)$ be two subsets with $A \subset \Omega$ and $m(\Omega) \leq 1/2$. Consider a local Lipschitz function f with $\mathbb{1}_A \leq f \leq \mathbb{1}_\Omega$. The variational definition of the entropy implies

$$Ent_m(f^2) \geq \int_\Omega f^2 g \mathrm{d}m,$$

for all g such that $\int_{\mathbb{R}^d} e^g \mathrm{d}m \leq 1$. We apply this inequality with

$$g = \begin{cases} \log\left(1 + \frac{1}{2m(A)}\right), & \text{on } A \\ 0, & \text{on } \Omega \setminus A \\ -\infty, & \text{on } \Omega^c \end{cases},$$

which satisfies $\int_{\mathbb{R}^d} e^g \mathrm{d}m = m(\Omega) + 1/2 \leq 1$. It yields

$$Ent_m(f^2) \geq m(A) \log\left(1 + \frac{1}{2m(A)}\right).$$

Therefore, by the p-WLSI with $p > 2$ and the definition of the capacity, we obtain

$$m(A) \log\left(1 + \frac{1}{2m(A)}\right) \leq \varphi_p(s) Cap_m(A, \Omega) + \frac{s}{4},$$

where we used $\|f\|_{\mathbb{L}^p_m(\mathbb{R}^d)} \leq m(\Omega) \leq 1/2$. Using (1.16) and then taking the infimum over Ω, we get

$$\frac{m(A) \log\left(1 + \frac{e^2}{m(A)}\right) - s}{c^{-1}\varphi_p(4cs)} \leq Cap_m(A), \quad \forall s > 0.$$

Using Proposition 1.3.6, we conclude that m satisfies a WLSI with function $\varphi(s) = 16c^{-1}\varphi_p(6cs/7)$.

□

Figure 1.4.1 briefly summarizes the links, between functional inequalities, that we have shown and proved during this chapter. In general, the opposite implications do not hold. Although Propositions 1.2.14 and 1.3.7 have been previously established in the literature (see [Bakry et al., 2014] and [Cattiaux et al., 2007]), Corollary 1.4.4 and Proposition 1.4.10 constitute original contributions of this work.

Chapters 2 and 3 make repeated use of the Poincaré inequality to derive WLSI or p-WLSI. The exact choice depends specifically on the space over which f is defined. The goal of the following chapters will be to use LSI and its weak versions to deduce the entropy decay of a diffusion semigroup $(P_t)_{t\geq 0}$ with presumably invariant measure m.

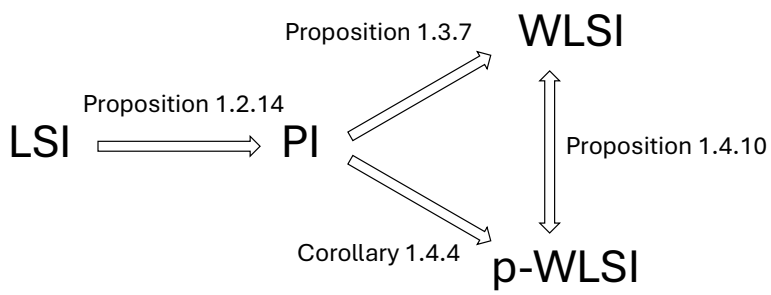


Figure 1.4.1: Link between the studied functional inequalities. The equivalence should be understood in the sense of Proposition 1.4.10.

1.5 Functional inequalities under $\mathcal{H}^{\mathbf{r}}_{\mathbf{KL}}(\mathfrak{c}, L)$ hypothesis

This section studies the above functional inequalities under a Kurdyka-Łojasiewicz hypothesis, that is, we consider the probability measure

$$\mu(\theta) = \frac{e^{-U(\theta)}}{\mathcal{Z}}, \quad \theta \in \mathbb{R}^d,$$

where $\mathcal{Z}$ is the normalizing constant and U satisfies a $\mathcal{H}^{\mathbf{r}}_{\mathbf{KL}}(\mathfrak{c}, L)$-condition, which is defined below. As we will discuss later, a value of $r = 0$ means that μ is strongly log-concave, and then an LSI is verified. Using this functional inequality, we deduce the convergence of the diffusion semigroup $(P_t)_{t\geq 0}$ towards its invariant distribution μ. However, if $0 < r < 1$, we are in the case where μ is weakly log-concave. We then choose to study two modifications of LSI (WLSI for bounded and unbounded functions) to prove the convergence of the semigroup. These results will be summarized at the end of the section.

It is worth mentioning that we pay special attention to the dependence of all constants involved in functional inequalities in terms of d.

Kurdyka-Łojasiewicz hypothesis

The hypothesis $\mathcal{H}^{\mathbf{r}}_{\mathbf{KL}}(\mathfrak{c}, L)$ was introduced in Chapter as a tool to parameterize the lack of strong convexity and is built using the Kurdyka-Łojasiewicz inequality [Kurdyka, 1998, Lojasiewicz, 1963]. As was mentioned before, this hypothesis is commonly used in optimization to describe the transition between convexity and strong convexity and allows for finding bounds that are more explicit.

We return to its definition here to comment on some basic ideas and consequences. We recall that $\mathcal{C}^2(\mathbb{R}^d)$ is the set of twice continuously differentiable functions. For any $\mathcal{C}^2(\mathbb{R}^d)$-function V, the spectrum of the Hessian matrix of V is

denoted as $Sp(\nabla^2 V(\theta))$. Furthermore, if V is convex, we denote

$$\underline{\lambda}_{\nabla^2 V}(\theta) = \inf Sp(\nabla^2 V(\theta)).$$

Hypothesis 1.5.1 ($\mathcal{H}^{\mathbf{r}}_{\mathbf{KL}}(\mathfrak{c}, L)$)**.** *We say that a function $V : \mathbb{R}^d \to \mathbb{R}$ satisfies a $\mathcal{H}^{\mathbf{r}}_{\mathbf{KL}}(\mathfrak{c}, L)$-condition if:*

a) V is a $\mathcal{C}^2(\mathbb{R}^d)$ function,

b) V is convex and $\min_{\theta\in\mathbb{R}^d} V(\theta) = V(\theta^*) > 0$,

c) ∇V is L-Lipschitz and

d) there exist some constants $0 \le r < 1$ and $\mathfrak{c} > 0$ such that

$$\mathfrak{c} V^{-r}(\theta) \le \underline{\lambda}_{\nabla^2 V(\theta)}, \quad \forall \theta \in \mathbb{R}^d. \tag{1.17}$$

Let us briefly discuss this assumption.

- In [Gadat et al., 2022], a slightly different parameterization is used with the introduction of another exponent $0 \le q \le r$ related to $\overline{\lambda}_{\nabla^2 V(\theta)} = \sup Sp(\nabla^2 V(\theta))$, $\theta \in \mathbb{R}^d$. The authors also assume the upper bound $\overline{\lambda}_{\nabla^2 V(\theta)} \le \tilde{\mathfrak{c}} V^{-q}(\theta)$. When $r = q$, they recover a global standard KL inequality (see [Gadat and Panloup, 2022, Bolte et al., 2010b]). Here, we have chosen to simplify this assumption and use a rough upper bound on the eigenvalues of the Hessian matrix given by the Lipschitz constant L, *i.e.* in the last inequality we simply use $\tilde{\mathfrak{c}} = L$ and $q = 0$.

- The case $r = 0$ is, of course, associated with the strongly convex situation where the curvature of the function is uniformly lower bounded by $\mathfrak{c}$, and the case $r = 1$ would correspond to the limiting Laplace situation.

- We shall observe that if $V(\theta) = (1+\|\theta\|_2^2)^\alpha$ with $\alpha \in [1/2, 1]$, then V satisfies a $\mathcal{H}^{\mathbf{r}}_{\mathbf{KL}}(\mathfrak{c}, L)$-condition with $r = \frac{1-\alpha}{\alpha}$ (equivalently $\alpha = \frac{1}{1+r}$) and $\mathfrak{c} = 2\alpha(1-2(1-\alpha))$, see Remark 7 of [Gadat et al., 2022] for further details. In particular, the larger α, the smaller r, which translates into a better curvature of the potential function.

- The assumption $\mathcal{H}^{\mathbf{r}}_{\mathbf{KL}}(\mathfrak{c}, L)$ in our work is closely related to that of [Erdogdu and Hosseinzadeh, 2021]. They assume that the potential function U is degenerately convex at infinity, which means that there exists a function $\widetilde{U}$ such that for a constant $\epsilon \ge 0$, $\|U - \widetilde{U}\|_\infty \le \epsilon$ where $\underline{\lambda}_{\nabla^2 \widetilde{U}}(\theta) \ge \kappa(1+\frac{1}{4}\|\theta\|_2^2)^{-\tau/2}$ for some $\kappa > 0$ and $\tau \ge 0$. This parameterization is similar to the one considered in (1.17), however, we bound the eigenvalues by a power of the same function U, while they use a power of $\|\theta\|_2^2$. The example above which satisfies a $\mathcal{H}^{\mathbf{r}}_{\mathbf{KL}}(\mathfrak{c}, L)$-condition with $r = \frac{1-\alpha}{\alpha}$ is also degenerated at infinity taking $\widetilde{U}(\theta) = U(\theta) = (1 + \|\theta\|_2^2)^\alpha$ and $\tau = 2(1-\alpha) = \frac{2r}{1+r}$.

  We will establish that an important consequence of $\mathcal{H}^{\mathbf{r}}_{\mathbf{KL}}(\mathfrak{c}, L)$ is that the invariant distribution verifies a p-WLSI, which will be useful for evaluating the

ergodic behavior in Chapter 3. In [Erdogdu and Hosseinzadeh, 2021], even though not exactly equivalent, the strategy is rather close and the authors establish a *modified* log-Sobolev inequality with the help of a perturbation strategy (see in particular Appendix A of [Erdogdu and Hosseinzadeh, 2021]).

Growth properties under the Kurdyka-Łojasiewicz hypothesis

We recall here some important consequences of the $\mathcal{H}^{\mathbf{r}}_{\mathbf{KL}}(\mathfrak{c}, L)$ inequality that imply several relationships between the function and the norm of its gradient. The proof of these inequalities may be found in Lemma 15 of [Gadat et al., 2022] (we correct a small mistake that appears there by including a factor 2 in our work).

Proposition 1.5.2. *Assume that a function V satisfies $\mathcal{H}^{\mathbf{r}}_{\mathbf{KL}}(\mathfrak{c}, L)$, then for all $\theta \in \mathbb{R}^d$,*

$$\frac{2\mathfrak{c}}{1-r}\left[V^{1-r}(\theta) - \min V^{1-r}\right] \leq \|\nabla V(\theta)\|_2^2 \leq 2L\left[V(\theta) - \min V\right] \leq 2LV(\theta).$$

It is furthermore possible to assess a minimal and maximal growth property of any function that satisfies $\mathcal{H}^{\mathbf{r}}_{\mathbf{KL}}(\mathfrak{c}, L)$, which is necessarily lower and upper bounded by a positive power of the distance to its minimizer.

Proposition 1.5.3. *Assume that a function V satisfies $\mathcal{H}^{\mathbf{r}}_{\mathbf{KL}}(\mathfrak{c}, L)$, then for all $\theta \in \mathbb{R}^d$,*

$$V^{1+r}(\theta) \geq V^{1+r}(\theta) - \min(V)^{1+r} \geq \frac{(1+r)\mathfrak{c}}{2}\|\theta - \arg\min V\|_2^2$$

and

$$V(\theta) - \min(V) \leq \frac{L}{2}\|\theta - \arg\min V\|_2^2.$$

Bound on the normalizing constant

We state the following result, which upper bounds the normalizing constant

$$\mathcal{Z} = \int_{\mathbb{R}^d} e^{-U(\theta)} \mathrm{d}\theta$$

of μ by using Proposition 1.5.3.

Proposition 1.5.4. *Assume that U satisfies a $\mathcal{H}^{\mathbf{r}}_{\mathbf{KL}}(\mathfrak{c}, L)$-condition. Then the normalizing constant of μ verifies the following inequality*

$$\mathcal{Z} \leq 2\left(2\pi/\mathfrak{c}\right)^{\frac{d}{2}} d^{\frac{dr}{2}}.$$

Proof. Using Proposition 1.5.3, we get

$$U(\theta) \geq c_r \|\theta - \theta^*\|_2^{\frac{2}{1+r}}, \tag{1.18}$$

where $c_r = \left(\frac{(1+r)\mathfrak{c}}{2}\right)^{\frac{1}{1+r}}$ and $\theta^* = \arg\min U$. Then,

$$\mathcal{Z} = \int_{\mathbb{R}^d} e^{-U(\theta)} \mathrm{d}\theta \leq \int_{\mathbb{R}^d} e^{-c_r \|\theta-\theta^*\|_2^{\frac{2}{1+r}}} \mathrm{d}\theta.$$

Using the well known equality

$$\int_{\mathbb{R}^d} e^{-a\|\theta\|_2^\ell} \mathrm{d}\theta = \frac{\pi^{d/2}\Gamma(d/\ell+1)}{a^{d/\ell}\Gamma(d/2+1)}, \quad \forall a>0, \quad \forall \ell>0, \tag{1.19}$$

we then deduce that

$$\mathcal{Z} \leq \left(\frac{2\pi}{(1+r)\mathfrak{c}}\right)^{\frac{d}{2}} \frac{\Gamma\left(d(1+r)/2+1\right)}{\Gamma\left(d/2+1\right)} \leq 2\left(\frac{2\pi}{\mathfrak{c}}\right)^{\frac{d}{2}} d^{\frac{dr}{2}},$$

where we used standard relations on the Gamma function. □

Functional inequalities and entropy decay under KL hypothesis

As we discussed before, assuming the hypothesis $\mathcal{H}^{\mathbf{r}}_{\mathbf{KL}}(\mathfrak{c}, L)$, if $r=0$, the function U is strongly convex and if $0<r<1$ then it is weakly convex. This leads to the need to study each case separately, since the functional inequality verified by μ depends on whether it is strongly log-concave or not.

From now on, we will assume that $(P_t)_{t\geq 0}$ is a diffusion semigroup with invariant distribution μ. The function f will be assumed to satisfy the conditions for each scenario.

Strongly log-concave case ($r=0$)

The probability measure μ is $\mathfrak{c}$-strongly log-concave, so a simple way to derive the decay in entropy of $(P_t)_{t\geq 0}$ is through LSI. The following result is an immediate consequence of Propositions 1.2.15 and 1.2.18, therefore, we omit the proof.

Theorem 1.5.5. *Assume that U satisfies a $\mathcal{H}^{\mathbf{r}}_{\mathbf{KL}}(\mathfrak{c}, L)$-condition with $r=0$. Then,*

a) μ satisfies an LSI with constant $C_{\mathrm{LSI}} = 2/\mathfrak{c}$, which is independent of d.

b) for any function f with finite entropy and for any $t>0$,

$$Ent_\mu(P_t f) \leq e^{-2\mathfrak{c}t} Ent_\mu(f).$$

In Sections 2.4 and 3.3 we will use this theorem to derive quantitative estimates of the order of convergence of two different stochastic processes when the (presumably) invariant measure is strongly log-concave. The constants have been explicitly kept throughout the chapter in order to be able to track the dependence on the dimension d. As we have seen in part b, the only term that could depend on d is $Ent_\mu(f)$.

Weakly log-concave case $(0 < r < 1)$

In order to study the decay of the entropy, we link the entropy and the Dirichlet form through WLSI and p-WLSI. It is important to recall that WLSI is valid when f is a bounded function, while p-WLSI holds for any function $f \in \mathbb{L}^p_m(\mathbb{R}^d)$, $p > 2$.

Theorem 1.5.6. *Assume that U satisfies a $\mathcal{H}^{\mathfrak{r}}_{\mathbf{KL}}(\mathfrak{c}, L)$-condition with $0 < r < 1$. Then,*

a) μ satisfies a Poincaré inequality with constant $C_{\mathrm{PI}}(\mu)$.

b) There exists a positive constant $\mathfrak{b}$ such that μ satisfies a WLSI with the function

$$\varphi(s) = \mathfrak{b} C_{\mathrm{PI}}(\mu) \log(1/s), \quad 0 < s \leq e^{-1}.$$

Moreover, for any positive function $f \in \mathcal{C}^1_b(\mathbb{R}^d)$ and t large enough,

$$Ent_\mu(P_t f) \leq (3/2 + e^{-1}) e^{-\sqrt{\frac{4t}{\mathfrak{b} C_{\mathrm{PI}}(\mu)}}} Osc^2\left(\sqrt{f}\right).$$

c) For any $p > 2$, there exist three positive constants a_p, b_p and s_p such that μ satisfies a p-WLSI with function

$$\varphi_p(s) = b_p C_{\mathrm{PI}}(\mu) \log(a_p/s), \quad 0 < s \leq s_p.$$

Moreover, for any function f such that $\sqrt{f} \in \mathbb{L}^p_\mu(\mathbb{R}^d)$ and any t large enough,

$$Ent_\mu(P_t f) \leq (a_p + q) e^{-\sqrt{\frac{4t}{b_p C_{\mathrm{PI}}(\mu)}}} \left\|\sqrt{f}\right\|^2_{\mathbb{L}^p_\mu(\mathbb{R}^d)},$$

where $q = \frac{p}{p-2}$.

We omit the proof of the previous theorem, which is a consequence of several results: Proposition 1.2.3, part b of Proposition 1.3.7, Corollaries 1.3.9 and 1.4.4 and Theorem 1.4.9.

Let us discuss the influence of r, $\mathfrak{c}$ and d on the entropy decay of Theorem 1.5.6. It is important to recall that $\mathfrak{b}$ is a universal constant, that is, it does not depend on the dimension d nor the $\mathcal{H}^{\mathfrak{r}}_{\mathbf{KL}}(\mathfrak{c}, L)$ parameters r and $\mathfrak{c}$. On the other hand, in general, the choice of the constant p in part c could depend on r, $\mathfrak{c}$ and even on d. As a consequence, this dependency extends to a_p, b_p, and s_p. Although in Chapter 3, we assume a condition that allows us to choose a value of p independent of all the previous parameters.

To our knowledge, the estimates of the terms associated with $\sqrt{f}$ ($Osc(\sqrt{f})$ in part b and $\left\|\sqrt{f}\right\|^2_{\mathbb{L}^p_\mu(\mathbb{R}^d)}$ in part c) will depend on r, $\mathfrak{c}$ and d for almost any admissible choice of f, since f is a density function in $\mathbb{L}^2_\mu(\mathbb{R}^d)$. Therefore, these terms must be treated carefully when finding quantitative estimates of $Ent_\mu(P_t f)$.

The Poincaré constant is, without question, one of the key points in our study. As was discussed in Remark 1.2.4, we could use the Bobkov bound to get

$$C_{\mathrm{PI}}(\mu) \leq 4K^2 Var_\mu(\theta),$$

where K is a universal constant and, abusing notation, θ is a random variable with distribution μ. We will use the Bobkov bound in Sections 2.B and 3.C. However, it is clear that $Var_\mu(\theta)$ will depend on r, $\mathfrak{c}$ and d. This bound could introduce a stronger dependency on the parameters, but the KLS conjecture does not seem to be of much help in a $\mathcal{H}^{\mathbf{r}}_{\mathbf{KL}}(\mathfrak{c}, L)$ setting, since bounding the variance of a component of θ does not yield better results than using the Bobkov bound.

Parts b and c of Theorem 1.5.6 will be used in Sections 2.4 and 3.4 respectively to then derive quantitative estimates of the order of convergence of two stochastic processes when the (presumably) invariant measure is weakly log-concave.

Conclusions

Consider a diffusion semigroup $(P_t)_{t\geq 0}$ with invariant distribution μ, a positive and small constant ε, and a function f chosen with the appropriate smoothness for each scenario. Then, the results obtained in this section are summarized in Table 1.5.1 where the last column shows the time t for which

$$Ent_\mu(P_t f) \leq \varepsilon.$$

$\mathcal{H}^{\mathbf{r}}_{\mathbf{KL}}(\mathfrak{c}, L)$	functional inequalities	Convergence rate
$(r = 0)$ μ is strongly log-concave	LSI	$\mathfrak{c}^{-1}\left(\log(\varepsilon^{-1}) + \log(Ent_\mu(f))\right)$
$(0 < r < 1)$ μ is weakly log-concave	PI $\implies$ WLSI	$C_{\mathrm{PI}}\left(\log^2(\varepsilon^{-1}) + \log^2(Osc^2(\sqrt{f}))\right)$
	PI $\implies$ p-WLSI	$C_{\mathrm{PI}}\left(\log^2(\varepsilon^{-1}) + \log^2\left(\left\|\sqrt{f}\right\|^2_{\mathbb{L}^p_\mu(\mathbb{R}^d)}\right)\right)$

Table 1.5.1: Convergence rate estimates in entropy when an $\mathcal{H}^{\mathbf{r}}_{\mathbf{KL}}(\mathfrak{c}, L)$ hypothesis is satisfied. PI denotes the Poincaré inequality.

As we discussed above, in the weakly log-concave situation, the terms C_{PI}, $Osc(\sqrt{f})$ and $\left\|\sqrt{f}\right\|^2_{\mathbb{L}^p_\mu(\mathbb{R}^d)}$ depend on the convexity parameter r and on the dimension d, even if it does not appear explicitly.

It is important to keep in mind that, in Table 1.5.1, $(P_t)_{t\geq 0}$ is a diffusion semigroup with an invariant measure μ. In Chapters 2 and 3, we study two processes with unknown invariant measure. Then, our objective is to prove that, when t is large, $P_t f$ is close to μ in entropy.

Chapter 2

Stochastic Gradient Langevin Dynamics

2.1 Introduction

In recent years, a huge amount of methods have been developed in machine learning to handle large scale massive datasets with a large number n of observations $(X_1, \ldots, X_n)$ embedded in a high-dimensional space $\mathbb{R}^d$. These methods generally involve either optimization of a data-dependent function (for frequentist learning) or sampling a data-dependent measure (for Bayesian learning with posterior distributions). In both approaches, a bottleneck lies in the size of n and d, which usually generates numerical difficulties for the use of standard algorithms.

In this chapter, we are interested in the approximation of a posterior distribution following a Bayesian point of view with a statistical model described by a collection of densities $(p_\theta)_{\theta \in \mathbb{R}^d}$ on $\mathcal{X}$, where the parameter of interest θ belongs to $\mathbb{R}^d$ and where the $(X_i)_{1 \leq i \leq n}$ are assumed to be i.i.d. observations in $\mathcal{X}$ distributed according to $p_{\theta^\star}$. A standard Bayesian approach consists in defining a prior distribution π_0 on $\mathbb{R}^d$ and then sample the posterior distribution denoted by μ_n (which will be proportional to $\exp(-U_{\nu_n})$ below) using a numerical probabilistic approximation with the help of a Langevin Dynamics (LD for short and also known as over-damped Langevin diffusion):

$$\mathrm{d}\theta_t = -\nabla_\theta U_{\nu_n}(\theta_t)\mathrm{d}t + \sqrt{2}\mathrm{d}B_t.$$

In this chapter, we focus on an adaptation of the Stochastic Gradient Langevin Dynamics (SGLD) algorithm proposed in [Welling and Teh, 2011], that exploits some old ideas of stochastic algorithms introduced in [Robbins and Monro, 1951]: instead of using the previous equation, the authors propose a modification of the diffusion that generates a noisy drift in the SGLD due to a sampling strategy among the set of observations $(X_i)_{1 \leq i \leq n}$.

This process will be studied in the Kurdyka-Łojasiewicz framework, which is parameterized by a real value r, which varies between 0 (strongly log-concave case) and 1 (limiting Laplace asymptotic tail). An important advantage of considering SGLD is that it is not necessary to have all the observations available at the initial instant, but only the total number of observations that will be used and modify the potential every exponential time.

Structure of the chapter: The rest of the introductory section consists of the definition of the algorithm, the way we evaluate the quality of our result with an entropy criterion, and the main assumptions. We state our main result in Section 2.2 and compare our algorithm with LD. In order to prove the main result, we first present in Section 2.3, the classical tools related to Markov semigroups, and we establish an inequality for the entropy that depends on the Dirichlet form. Section 2.4 is dedicated to establish a functional inequality that links entropy and Dirichlet form, allowing us, in Section 2.5, to prove the main result. The technical results may be found in the appendix.

Continuous time evolution

In the following, we briefly define the continuous-time SGLD algorithm for Bayesian learning, for which a discretized form has been introduced in [Welling and Teh, 2011]. For this purpose, we consider a statistical model that is built with the help of a function $(x, \theta) \mapsto p_\theta(x)$, where $\theta \in \mathbb{R}^d$ encodes the parameter of the statistical model and x the observation in a space denoted by $\mathcal{X}$. We then assume that we have n i.i.d. observations denoted by $(\mathbf{X}_1, \ldots, \mathbf{X}_n)$ distributed according to p_{θ^*}. Given a prior distribution π_0 on $\mathbb{R}^d$, the posterior distribution μ_n is then defined as:

$$\mu_n(\theta) \propto \pi_0(\theta) \times \prod_{i=1}^{n} p_\theta(\mathbf{X}_i).$$

We introduce the log-parameterization that leads to the Gibbs form:

$$U_x(\theta) = -[\log \pi_0(\theta) + n \log p_\theta(x)],$$

and we then observe that:

$$\mu_n(\theta) \propto \exp\left(-\frac{1}{n}\sum_{i=1}^{n} U_{\mathbf{X}_i}(\theta)\right) = \exp\left(-U_{\nu_n}(\theta)\right),$$

where ν_n refers to the empirical distribution and U_{ν_n}, the average value of $U_X(\theta)$ when $X \sim \nu_n$:

$$\nu_n(x) = \frac{1}{n}\sum_{i=1}^{n} \delta_{\mathbf{X}_i}(x) \quad \text{and} \quad U_{\nu_n}(\theta) = \mathbb{E}_{X\sim\nu_n}[U_X(\theta)].$$

The standard over-damped Langevin Dynamics approach relies on the ergodic behavior of the stochastic differential equation:

$$\mathrm{d}\vartheta_t = -\nabla U_{\nu_n}(\vartheta_t)\mathrm{d}t + \sqrt{2}\mathrm{d}B_t, \tag{2.1}$$

that possesses under some mild assumptions a unique invariant distribution μ_n.

The SGLD algorithm takes advantage of both sampling with a stochastic differential equation and homogenization of the drift that may be written as an expectation in X that is uniformly sampled over the set of observations according to ν_n. The leading idea is to replace the expectation in U_{ν_n} that depends on the overall

set of observations $(\mathbf{X}_1, \ldots, \mathbf{X}_n)$ by a single unique observation that is randomized uniformly throughout the evolution of the stochastic differential equation, and modified according to a Markov exponential clock. That being said, we can write an explicit formal definition of the algorithm as follows. We define $\left(T_j^{(n)}\right)_{j\geq 1}$ an infinite sequence of exponential random variables of mean α_n^{-1} that will be fixed later on, $\tau_0^{(n)} = 0$ and $\tau_j^{(n)} = \sum_{k=1}^{j} T_k^{(n)}$, for any $j \geq 1$.

We also consider a sequence $\left(V_j^{(n)}\right)_{j\geq 1}$ of i.i.d. random variables uniformly distributed in $[n] = \{1, 2, \ldots, n\}$. We then define the process $(X_t)_{t\geq 0}$ as a jump process that takes its values in $\{\mathbf{X}_1, \ldots, \mathbf{X}_n\}$ such that:

$$X_t = \mathbf{X}_{V_j^{(n)}}, \qquad \text{if } \tau_{j-1}^{(n)} \leq t < \tau_j^{(n)}, \quad j \geq 1. \tag{2.2}$$

Informally, $(X_t)_{t\geq 0}$ should be understood as follows: the process takes the value of one observation uniformly chosen from the n observations $\mathbf{X}_1, \ldots, \mathbf{X}_n$ during exponential times with intensity α_n. The stochastic Langevin over-damped diffusion we consider is then given by the joint evolution $(\theta_t, X_t)_{t\geq 0}$ and that is defined by:

$$\mathrm{d}\theta_t = -\nabla_\theta U_{X_t}(\theta_t)\mathrm{d}t + \sqrt{2}\mathrm{d}B_t, \quad t > 0, \tag{2.3}$$

where $(B_t)_{t\geq 0}$ is a multivariate standard Brownian Motion.

For the completeness of the chapter, we present the continuous-time SGLD algorithm in Section 4.2 in Chapter 4, which is built using (2.3).

Entropic divergence

To assess the long-time behavior of the continuous-time SGLD, we introduce several notations related to the pair $(\theta_t, X_t)_{t\geq 0}$. Below, we denote by λ_d the Lebesgue measure over $\mathbb{R}^d$ and by ν_c the counting measure over $\mathcal{X}$. The semigroup induced by $\mathcal{L}$, defined in Equation (2.7) and being elliptic on the θ coordinate, trivially irreducible and finitely supported in the coordinate x, makes the law of (θ_t, X_t) absolutely continuous with respect to the measure $\lambda_d \times \nu_c$ as soon as $t > 0$.

We introduce the notation of m_t to refer to the joint density of (θ_t, X_t) at time t with respect to $\lambda_d \times \nu_c$. Meanwhile, n_t denotes the marginal distribution of θ_t and $m_t(\cdot|\theta)$ the conditional distribution of X_t given $\theta_t = \theta$. That is,

$$\mathrm{Law}(\theta_t, X_t) = m_t, \quad n_t(\theta) = \sum_{i=1}^{n} m_t(\theta, \mathbf{X}_i), \quad m_t(x|\theta) = \frac{m_t(\theta, x)}{n_t(\theta)},$$

for $\theta \in \mathbb{R}^d$ and $x \in \{\mathbf{X}_1, \ldots, \mathbf{X}_n\}$.

To show that the algorithm recovers the correct asymptotic behavior, *i.e.* that $n_t(\theta) \longrightarrow \mu_n$ when $t \longrightarrow \infty$, we consider the relative entropy (or Kullback-Leibler divergence) of n_t with respect to μ_n that is well defined thanks to the ellipticity, and given by:

$$J_t = Ent_{\mu_n}\left(\frac{n_t}{\mu_n}\right) = \int_{\mathbb{R}^d} \log\left(\frac{n_t(\theta)}{\mu_n(\theta)}\right) \mathrm{d}n_t(\theta). \tag{2.4}$$

For any $t > 0$, J_t measures a divergence between the instantaneous law of the process at time t and the (presumably) invariant distribution μ_n of the process (θ_t, X_t). It would also be possible to measure this difference between the two distributions in terms of the $\mathbb{L}^2_\mu(\mathbb{R}^d)$ or the χ square distance and to produce a theoretical analysis with the help of functional analysis, but it would rely on stronger assumptions on the function U_{ν_n}.

Main assumptions

Weak convexity

We will study a continuous-time version of SGLD in a weakly convex framework, that is, when U_{ν_n} is assumed to be convex but not necessarily strongly convex. SGLD has recently received an important interest in the machine learning community and has been studied in various situations where functional inequalities are involved. We refer to [Xu et al., 2018, Das et al., 2023] (the uniform Poincaré inequality) and to [Raginsky et al., 2017, Wang et al., 2021, Das et al., 2023] (log-Sobolev inequality, LSI for short), where the functional inequalities play a crucial role to analyze the behavior of the process. In [Raginsky et al., 2017, Xu et al., 2018, Wang et al., 2021], the authors develop a 2-Wasserstein analysis of the algorithm that assumes the dissipativity of the potential, which is a restrictive assumption. On the other hand, in [Das et al., 2023], they study the convergence in the Kullback-Leibler divergence and in total variation. In both analysis, they assume smoothness of the potential, in addition to an LSI and a 4th moment growth condition for the convergence in entropy and Poincaré inequality and a 6th moment growth condition in total variation.

Importantly, the Poincaré inequality and LSI are not so innocent since they generally require convexity (see e.g. [Bobkov, 1999, Bakry et al., 2014]) to be reasonably dimension dependent and even strong convexity to be dimension free. Otherwise, the constants involved in these functional inequalities are exponentially degraded by the "temperature" ($1/n$ in our case) and the dimension (d for us) as indicated in [Holley and Stroock, 1988].

We have chosen to parameterize this lack of strong convexity with the help of the Kurdyka-Łojasiewicz inequality [Kurdyka, 1998, Lojasiewicz, 1963], which is a standard tool in optimization to describe the transition between convexity and strong convexity and makes the bounds more explicit. This assumption allows us to observe how the entropy evolves according to the key exponent involved in the KL inequality. In particular, it makes possible to understand the influence of the lack of strong convexity that is more or less hidden in the uniform Poincaré or LSI.

For completeness, we recall the definition of hypothesis $\mathcal{H}^{\mathbf{r}}_{\mathbf{KL}}(\mathfrak{c}, L)$. This hypothesis and its consequences were discussed in Section 1.5. For any $\mathcal{C}^2(\mathbb{R}^d)$ function V, the spectrum of the Hessian matrix of V is denoted as $Sp(\nabla^2 V(\theta))$ and, if V is convex, then

$$\underline{\lambda}_{\nabla^2 V}(\theta) = \inf Sp(\nabla^2 V(\theta)).$$

Hypothesis $(\mathcal{H}^{\mathbf{r}}_{\mathbf{KL}}(\mathfrak{c}, L))$**.** *We say that a function $V : \mathbb{R}^d \to \mathbb{R}$ satisfies a $\mathcal{H}^{\mathbf{r}}_{\mathbf{KL}}(\mathfrak{c}, L)$-condition if:*

a) V is a $\mathcal{C}^2(\mathbb{R}^d)$ function,

b) V is convex and $\min_{\theta\in\mathbb{R}^d} V(\theta) = V(\theta^*) > 0$,

c) ∇V is L-Lipschitz and

d) there exist some constants $0 \le r < 1$ and $\mathfrak{c} > 0$ such that

$$\mathfrak{c}V^{-r}(\theta) \le \underline{\lambda}_{\nabla^2 V(\theta)}, \quad \forall \theta \in \mathbb{R}^d. \tag{2.5}$$

Throughout this chapter, we will assume that there exist three fixed constants $\mathfrak{c} > 0$, $L > 0$, and $0 \le r < 1$ such that for any x: $\theta \longmapsto -\log p_\theta(x)$ satisfies $\mathcal{H}^{\mathbf{r}}_{\mathbf{KL}}(\mathfrak{c}, L)$. Let us just keep in mind the following remark.

Remark 2.1.1. *If inequality* (2.5) *holds for a constant $\mathfrak{c}$, then it holds for all positive values less than $\mathfrak{c}$. For that reason, we assume that $\mathfrak{c} \le (8L/(1+r))^{1+r}$, which will be used in Proposition 2.E.1.*

Others assumptions

We state below an important consequence of a "population" satisfying the assumption $\mathcal{H}^{\mathbf{r}}_{\mathbf{KL}}(\mathfrak{c}, L)$, but before, let us state some mild assumptions on π_0.

Hypothesis 2.1.2 ($\mathcal{H}_{\pi_0}(\ell_0)$)**.** *$\pi_0$ is a $\mathcal{C}^2(\mathbb{R}^d)$ log-concave function such that $\min_{\theta\in\mathbb{R}^d} -\log \pi_0(\theta) > 0$ and $\theta \mapsto \nabla \log \pi_0(\theta)$ is ℓ_0-Lipschitz.*

Since the prior distribution is chosen by the user, our $\mathcal{H}_{\pi_0}(\ell_0)$ hypothesis is not restrictive, and some typical examples satisfy these conditions, such as Gaussian, Weibull, and Gamma, both with shape parameter larger than 1, Gumbel, among others.

The following proposition shows the consequence on the function U_{ν_n} of assuming the hypothesis $\mathcal{H}_{\pi_0}(\ell_0)$ and $\mathcal{H}^{\mathbf{r}}_{\mathbf{KL}}(\mathfrak{c}, L)$ on $\theta \mapsto -\log p_\theta(x)$, for any x. The proof of Proposition 2.1.3 may be found in Section 2.A.

Proposition 2.1.3. *We assume $\mathcal{H}_{\pi_0}(\ell_0)$ and that $\theta \longmapsto -\log p_\theta(x)$ satisfies $\mathcal{H}^{\mathbf{r}}_{\mathbf{KL}}(\mathfrak{c}, L)$, for any x. Then U_{ν_n} satisfies $\mathcal{H}^{\mathbf{r}}_{\mathbf{KL}}(\mathfrak{c}n^{1+r}, nL + \ell_0)$, and in particular, for any $\mathbf{X}_i$, $U_{\mathbf{X}_i}$ satisfies $\mathcal{H}^{\mathbf{r}}_{\mathbf{KL}}(\mathfrak{c}n^{1+r}, nL + \ell_0)$.*

We recall the notation $a \lesssim_{uc} b$ ($a \gtrsim_{uc} b$) which means $a \le cb$ ($a \ge cb$) where c is a universal constant *i.e.* a positive constant independent of n and d.

We assume that the minimizers and the minimums of the functions $U_{\mathbf{X}_i}$ are contained in balls of radius depending on n and d and $n > 1$.

Hypothesis 2.1.4 ($\mathcal{H}^{\beta}_{\min}$)**.** *There exists a constant $\beta \ge 0$ such that*

$$\max_i \|\arg\min U_{\mathbf{X}_i}\|_2 \lesssim_{uc} \sqrt{d}\log^{\beta}(n) \quad \text{and} \quad \max_i \min_{\theta\in\mathbb{R}^d} U_{\mathbf{X}_i}(\theta) \lesssim_{uc} d\log^{2\beta}(n).$$

The assumption $\mathcal{H}^{\beta}_{\min}$ is not restrictive. In dimension $d = 1$, consider a concentrated i.i.d. sample $(X_i)_{1\leq i\leq n}$ with a suitable sub-Gaussian like behavior for which the Laplace transform of $|\arg\min U_{X_i}|$, for any i, is upper bounded as:

$$\mathbb{E}[\exp(\lambda|\arg\min U_{X_i}|)] \leq \exp(\sigma^2\lambda^k), \quad \forall\lambda > 0,$$

where $k \geq 1$ and $\sigma > 0$ are fixed constants. The Markov inequality and the upper bound above imply that, in this case for any $\lambda > 0$ and $s > 0$:

$$\begin{aligned}\mathbb{P}\left(\max_i |\arg\min U_{X_i}|) > s\right) &\leq e^{-s\lambda}\mathbb{E}\left(\max_i \exp(\lambda|\arg\min U_{X_i}|)\right)\\ &\leq e^{-s\lambda}\mathbb{E}\left(\sum_{i=1}^n \exp(\lambda|\arg\min U_{X_i}|)\right)\\ &\leq n\exp(-s\lambda + \sigma^2\lambda^k).\end{aligned}$$

If we choose $\lambda = \sigma^{-\frac{2}{k}}\log^{\frac{1}{k}}(n/\delta)$ and $s = 2\sigma^{\frac{2}{k}}\log^{\frac{k-1}{k}}(n/\delta)$ where δ is a small positive constant, then:

$$\mathbb{P}\left(\max_i |\arg\min U_{\mathbf{X}_i}| > 2\sigma^{\frac{2}{k}}\log^{\frac{k-1}{k}}(n/\delta)\right) \leq \delta.$$

Therefore, the value of β involved in $\mathcal{H}^{\beta}_{\min}$ is given by $\beta = \frac{k-1}{k}$. We recover in particular the situation where $\beta = 1/2$ when $k = 2$. For larger dimensions, the result can be extended using $\|x\|_2^2 \leq d\,\max_{1\leq j\leq d}(x^j)^2$, where x^j is the j-th component of x.

We should keep in mind from this last discussion that even if $\mathcal{H}^{\beta}_{\min}$ is stated (and makes sense) for any value of $\beta \geq 0$, it holds in general for $0 \leq \beta \leq 1$. If we replace $\|\arg\min U_{\mathbf{X}_i}\|_2^2$ by $\min U_{\mathbf{X}_i}$ in the previous example, we obtain an analogous behavior of $\max_i \min U_{\mathbf{X}_i}$.

Hypotheses $\mathcal{H}^{\beta}_{\min}$ and $\mathcal{H}_{\pi_0}(\ell_0)$ lead to an almost similar behavior of the minimizer and the minimum of U_{ν_n}. Details appear in Proposition 2.A.1.

2.2 Long-time entropy convergence

We introduce for any time $t \geq 0$, the density function of θ_t with respect to μ_n, which is given by:

$$h_t(\theta) = \frac{n_t(\theta)}{\mu_n(\theta)}.$$

We will choose the initial distribution n_0 such that h_0 is bounded, then h_t will also be bounded for any $t > 0$. This property is important to prove the main theorem of the chapter and it will be studied in Section 2.3. Let us denote by 0_d and I_d the null vector of $\mathbb{R}^d$ and the identity matrix of size d, respectively.

Hypothesis 2.2.1 ($\mathcal{H}_{\mathbf{n_0}}(L, \ell_0)$)**.** *A positive constant σ^2 exists such that n_0 is the density function of a $\mathcal{N}(0_d, \sigma^2 I_d)$ random variable. Moreover, there exist two*

universal constants c_1 and c_2 such that $0 < c_1 \leq c_2 < 1$ and

$$\frac{c_1}{nL + \ell_0} \leq \sigma^2 \leq \frac{c_2}{nL + \ell_0}.$$

This hypothesis guarantees the boundedness of h_0 and the initial entropy, in particular we will prove that

$$J_0 = \int_{\mathbb{R}^d} \log\left(h_0(\theta)\right) \mathrm{d}n_0(\theta) \lesssim_{uc} n(d\log^{2\beta}(n))^{1+r} + rd\log\left(d/n\right).$$

We refer to Proposition 2.3.2 in Section 2.3 for further details.

The next result assesses a mixing property in terms of decrease in entropy and therefore states the convergence of n_t towards the correct measure μ_n.

Theorem 2.2.2. *Assume $n \geq 2$, $\mathcal{H}_{\pi_0}(\ell_0)$, $\mathcal{H}^{\beta}_{\min}$, $\mathcal{H}_{\mathbf{n_0}}(L, \ell_0)$ and that each $\theta \mapsto -\log p_\theta(\mathbf{X}_i)$ satisfies $\mathcal{H}^{\mathbf{r}}_{\mathbf{KL}}(\mathfrak{c}, L)$.*

a) If $r = 0$, then μ_n satisfies an LSI with constant $C_{\mathrm{LSI}} = \frac{2}{\mathfrak{c}n}$. Moreover, for any $\varepsilon > 0$, if $\alpha_n = n$ and

$$t \gtrsim_{uc} n^{-1}\left[\log(\varepsilon^{-1}) + \log(d) + \log(n)\right],$$

then $J_t \leq \varepsilon$.

b) If $0 < r < 1$, then μ_n satisfies a Poincaré inequality with constant $C_{\mathrm{PI}}(\mu_n)$, indistinctly denoted as C_{PI}. For any $t > 0$,

$$J_t \lesssim_{uc} \left(J_0 + O_{n,d} + c_{n,d}C_{\mathrm{PI}}\right) e^{-\sqrt{\frac{3t}{16\mathfrak{b}C_{\mathrm{PI}}}}},$$

where $c_{n,d} = n^4\left(d\log^{2\beta}(n)\right)^{1+r}$, $O_{n,d} = \left(\frac{C_1 d}{n}\right)^{\frac{dr}{2}} \exp\left(C_2 n\left(d\log^{2\beta}(n)\right)^{1+r}\right)$ and $\mathfrak{b}$, C_1 and C_2 are universal positive constants.

Moreover, for any $\varepsilon > 0$ and $\alpha_n > 0$ such that $t \gtrsim_{uc} \alpha_n^{-2}\left(d\log^{2\beta}(n)\right)^{-(1+r)^2}$ and

$$t \gtrsim_{uc} \left(d\log^{2\beta}(n)\right)^{(1+r)^2}\left[\log^2(\varepsilon^{-1}) + n^2\left(d\log^{2\beta}(n)\right)^{2(1+r)}\right],$$

then $J_t \leq \varepsilon$.

If we denote t_ε the smallest value such that $J_{t_\varepsilon} \leq \varepsilon$, then our choice of $\alpha_n = n$ in the strongly convex case guarantees that the mean number of jumps $\alpha_n t_\varepsilon$ of the process $(X_t)_{0 \leq t \leq t_\varepsilon}$ is the minimum possible.

It is important to note that in the weakly log-concave case, the bound used for the Poincaré constant C_{PI} adds noise in terms of dimension d and possibly in n. Using C_{PI} instead of its bound, then $J_t \leq \varepsilon$ if $t \gtrsim_{uc} \alpha_n^{-2} C_{\mathrm{PI}}^{-1}$ and:

$$t \gtrsim_{uc} C_{\mathrm{PI}}\left[\log^2(\varepsilon^{-1}) + n^2\left(d\log^{2\beta}(n)\right)^{2(1+r)}\right].$$

The term $n\left(d\log^{2\beta}(n)\right)^{1+r}$ appears due to the assumption $\mathcal{H}^{\beta}_{\min}$, in a particular situation in which $\max_i \|\arg\min U_{\mathbf{X}_i}\|_2 \lesssim_{uc} 1$ and $\max_i \ \min_{\theta\in\mathbb{R}^d} \ U_{\mathbf{X}_i}(\theta) \lesssim_{uc} 1$, then instead of $n\left(d\log^{2\beta}(n)\right)^{1+r}$ we would have $\log(d)+n$.

Comparison with over-damped Langevin Dynamics

Let us return to the definition of the algorithm. The continuous-time SGLD is defined by

$$\mathrm{d}\theta_t = -\nabla_\theta U_{X_t}(\theta_t)\mathrm{d}t + \sqrt{2}\mathrm{d}B_t, \quad t>0,$$

where $(X_t)_{t\geq 0}$ is a jump process that takes the value of one observation from $\mathbf{X}_1,\mathbf{X}_2,\ldots,\mathbf{X}_n$ during exponential times. We consider that n is large and instead of choosing an observation uniformly from $\mathbf{X}_1,\mathbf{X}_2,\ldots,\mathbf{X}_n$ following an exponential clock, we could simply assume that observations become available at exponential time intervals. That is, at the initial instant an observation $\mathbf{X}_1$ becomes available, after an exponential time interval $\mathbf{X}_1$ is replaced by a new available observation $\mathbf{X}_2$, and so on, up to $\mathbf{X}_n$.

We recall that $\tau_j^{(n)}$ is the time at which X_t takes a new value for $j\geq 1$ and $\tau_0^{(n)}=0$. We can then define a new jump process Y_t as

$$Y_t = \mathbf{X}_j, \qquad \text{if } \tau_{j-1}^{(n)} \leq t < \tau_j^{(n)}, \quad 1\leq j\leq n, \tag{2.6}$$

where $0\leq t<\tau_n^{(n)}$. The process Y_t is well defined in the time interval $[0,\tau_n^{(n)})$ and using (2.2), we deduce

$$\forall t\in[0,\tau_n^{(n)}), \quad X_t = Y_t \quad \text{in law.}$$

We can then interpret the continuous-time process SGLD as an online algorithm, in which we only need to know the total number of observations that will be available.

Having specified this characteristic, let us compare the continuous-time process SGLD with the over-damped Langevin Dynamics (LD). Since the continuous-time process SGLD has not been studied before, we decided to compare our result to that obtained if we simply used all the observations $\mathbf{X}_1,\mathbf{X}_2,\ldots,\mathbf{X}_n$ to define the standard over-damped Langevin Dynamics as in Equation (2.1). For this reason, we assume hypotheses $\mathcal{H}_{\pi_0}(\ell_0)$ and $\mathcal{H}^{\beta}_{\min}$. As in $\mathcal{H}_{\mathbf{n_0}}(L,\ell_0)$, the initial distribution of the LD is assumed to be Gaussian with the same covariance matrix. In addition, each $\theta\mapsto -\log p_\theta(\mathbf{X}_i)$ satisfies $\mathcal{H}^{\mathbf{r}}_{\mathbf{KL}}(\mathfrak{c},L)$. Let us separately discuss the cases $r=0$ and $0<r<1$.

In the strongly log-concave situation ($r=0$), we use Theorem 1.5.5 to get that if $t\gtrsim_{uc} n^{-1}\left[\log(\varepsilon^{-1})+\log(d)\right]$ then the relative entropy between the law of the LD at time t and its invariant distribution μ_n is smaller than ε. This conclusion appears in Table 2.2.1 below.

We observe that the order of convergence of LD is faster than that of SGLD, where an additional $\log(n)/n$ term appears. However, LD requires having all observations from the initial instant and the computation of ∇U_{ν_n} could be complex when n is large. On the other hand, applying SGLD only requires having a new

process	N° of observations	convergence rate
LD	n	$n^{-1}\left[\log(\varepsilon^{-1})+\log(d)\right]$
SGLD	1	$n^{-1}\left[\log(\varepsilon^{-1})+\log(d)+\log(n)\right]$

Table 2.2.1: Convergence rate in strongly log-concave case.

observation available with an exponential clock of intensity α_n, this comes from the fact that the observations are i.i.d. .

Similarly, in the weakly log-concave situation ($0 < r < 1$), using Theorem 1.5.6 we get the same order of convergence as for SGLD. We arrive at the same conclusion: SGLD could be considered as a good alternative when all the observations are not available at the initial time; see Table 2.2.2.

process	N° of observations	convergence rate
LD	n	$\left(d\log^{2\beta}(n)\right)^{(1+r)^2}\left[\log^2(\varepsilon^{-1})+n^2\left(d\log^{2\beta}(n)\right)^{2(1+r)}\right]$
SGLD	1	$\left(d\log^{2\beta}(n)\right)^{(1+r)^2}\left[\log^2(\varepsilon^{-1})+n^2\left(d\log^{2\beta}(n)\right)^{2(1+r)}\right]$

Table 2.2.2: Convergence rate in a $\mathcal{H}^{\mathbf{r}}_{\mathbf{KL}}(\mathfrak{c}, L)$ context with $0 < r < 1$.

It would be interesting to compare the discrete-time version of SGLD with the Unadjusted Langevin algorithm (ULA, which is the Euler approximation of LD) in the context $\mathcal{H}^{\mathbf{r}}_{\mathbf{KL}}(\mathfrak{c}, L)$. This issue remains a topic for future work to be developed.

2.3 Markov tools and evolution of the entropy

It is straightforward to verify that the joint evolution of $(\theta_t, X_t)_{t\geq 0}$ exists and is weakly unique (in law) with the help of the Martingale problem (MP below). For this purpose, we preliminary define the operator $\mathcal{L}$ that acts on any function $f \in \mathcal{C}^2(\mathbb{R}^d \times \mathcal{X})$ as:

$$\mathcal{L}f(\theta,x) = \underbrace{-\langle \nabla_\theta U_x(\theta), \nabla_\theta f(\theta,x)\rangle + \Delta_\theta f(\theta,x)}_{\mathcal{L}_1 f(\theta,x)} + \underbrace{\frac{\alpha_n}{n}\sum_{i=1}^{n}[f(\theta,\mathbf{X}_i) - f(\theta,x)]}_{\mathcal{L}_2 f(\theta,x)}, \quad (2.7)$$

for all $(\theta, x) \in \mathbb{R}^d \times \mathcal{X}$.

The operator $\mathcal{L}$ is divided into two terms, $\mathcal{L}_1$ acts on the component θ and is associated to the diffusion part, while $\mathcal{L}_2$ is the jump operator that acts on the component x.

Due to the finite number of observations $(\mathbf{X}_1, \ldots, \mathbf{X}_n)$, we can apply the results of Sections 4 and 5 of Chapter 4 of [Ethier and Kurtz, 2009] and deduce the following result:

Proposition 2.3.1. *Assume that for any $x \in \mathcal{X}$, U_x is $\mathcal{C}^2(\mathbb{R}^d)$ and $\nabla_\theta U_x$ is L_x-Lipschitz, then for any initial distribution ν in $\mathbb{R}^d \times \mathcal{X}$, the martingale problem $(\mathcal{L}, \nu)$ is well posed. The associated (weakly) unique process $(\theta_t, X_t)_{t\geq 0}$ is a Feller Markov process associated to the semigroup $\mathcal{L}$. In particular, the θ component verifies the stochastic differential equation* (2.3).

If we denote by $\mathcal{L}^\star$ the adjoint operator of $\mathcal{L}$ in $\mathbb{L}^2_{\lambda_d \times \nu_c}(\mathbb{R}^d \times \mathcal{X})$, the Kolmogorov forward equation (or Fokker-Plank equation) yields:

$$\partial_t\{m_t(\theta, x)\} = \mathcal{L}^\star m_t(\theta, x), \tag{2.8}$$

in the weak sense. However, using Remark 3.19 of [Kusuoka and Stroock, 1984], we could verify that the first and second derivatives of m_t with respect to θ are bounded, therefore m_t is differentiable in time and Equation (2.8) is satisfied in the strong sense.

We now introduce two operators, which will be the keystone of our work. The first one describes the infinitesimal action on the θ coordinate under the average effect of X_t at time t that applies for any function $f \in \mathcal{C}^2(\mathbb{R}^d)$ as:

$$\mathcal{G}_t f(\theta) = -\sum_{i=1}^n \langle \nabla_\theta f(\theta), \nabla_\theta U_{\mathbf{X}_i}(\theta)\rangle m_t(\mathbf{X}_i|\theta) + \Delta_\theta f(\theta). \tag{2.9}$$

The second is very close to the first, except that the average effect of X_t is replaced by the ideal target distribution ν_n. It leads to the definition: for any function $f \in \mathcal{C}^2(\mathbb{R}^d)$,

$$\begin{aligned}\mathcal{G} f(\theta) &= -\sum_{i=1}^n \langle \nabla_\theta f(\theta), \nabla_\theta U_{\mathbf{X}_i}(\theta)\rangle \nu_n(\mathbf{X}_i) + \Delta_\theta f(\theta) \\ &= -\langle \nabla_\theta f(\theta), \nabla_\theta U_{\nu_n}(\theta)\rangle + \Delta_\theta f(\theta). \end{aligned} \tag{2.10}$$

Study of J_0 and $\partial_t\{J_t\}$

Our starting point is to establish a differential inequality for J_t. But first, we state the following proposition, in which it is proved that the initial entropy J_0 is bounded.

Proposition 2.3.2. *Assume $\mathcal{H}^\beta_{\min}$,$\mathcal{H}_{\mathbf{n_0}}(L, \ell_0)$, $\mathcal{H}_{\pi_0}(\ell_0)$ and that, for any x, $\theta \mapsto -\log p_\theta(x)$ satisfies $\mathcal{H}^{\mathbf{r}}_{\mathbf{KL}}(\mathfrak{c}, L)$, then:*

a) Two positive constants C_1 and C_2 exist, which are independent from n and d and such that:

$$\|h_0\|_{\mathbb{L}^\infty_m(\mathbb{R}^d)} \lesssim_{uc} (C_1 d/n)^{\frac{dr}{2}} \exp\left(C_2 n (d \log^{2\beta}(n))^{1+r}\right).$$

b) As a consequence:

$$J_0 = \int_{\mathbb{R}^d} \log\left(h_0(\theta)\right) \mathrm{d}n_0(\theta) \lesssim_{uc} n(d\log^{2\beta}(n))^{1+r} + rd\log\left(d/n\right).$$

The proof of Proposition 2.3.2 may be found in Section 2.B. Let us now define the Dirichlet form of $\sqrt{n_t(\theta)/\mu_n(\theta)}$ (proportional to the Fisher information of n_t with respect to μ_n) as:

$$\mathcal{E}_t = \int_{\mathbb{R}^d} \left\| \nabla \left(\sqrt{\frac{n_t(\theta)}{\mu_n(\theta)}} \right) \right\|_2^2 \mathrm{d}\mu_n(\theta) = \frac{1}{4} \int_{\mathbb{R}^d} \left\| \nabla \log \left(\frac{n_t(\theta)}{\mu_n(\theta)} \right) \right\|_2^2 \mathrm{d}n_t(\theta). \tag{2.11}$$

The following proposition shows the link between $\partial_t\{J_t\}$ and $\mathcal{E}_t$.

Proposition 2.3.3. *Assume $\mathcal{H}^{\beta}_{\min}$, $\mathcal{H}_{\pi_0}(\ell_0)$ and for each $\mathbf{X}_i$, $\theta \to -\log p_\theta(\mathbf{X}_i)$ satisfies $\mathcal{H}^{\mathfrak{r}}_{\mathbf{KL}}(\mathfrak{c}, L)$. Then, for any $t > 0$,*

$$\partial_t\{J_t\} \leq -3\mathcal{E}_t + c_{n,d} e^{-\frac{2\alpha_n}{3} t},$$

where $c_{n,d} \lesssim_{uc} n^4 \left(d\log^{2\beta}(n)\right)^{1+r}$.

Proof of Proposition 2.3.3. The existence of $\partial_t\{J_t\}$ and the equalities:

$$\begin{aligned}\partial_t\{J_t\} &= \int_{\mathbb{R}^d} \left(1 + \log\left(\frac{n_t(\theta)}{\mu_n(\theta)}\right)\right) \partial_t\{n_t(\theta)\}\mathrm{d}\theta \\ &= \int_{\mathbb{R}^d} \mathcal{G}_t \log\left(\frac{n_t(\theta)}{\mu_n(\theta)}\right) \mathrm{d}n_t(\theta).\end{aligned}$$

are justified in Section 2.C. Then, we are led to split $\partial_t\{J_t\}$ into two terms:

$$\begin{aligned}\partial_t\{J_t\} &= \int_{\mathbb{R}^d} \mathcal{G}_t \log\left(\frac{n_t(\theta)}{\mu_n(\theta)}\right) \mathrm{d}n_t(\theta), \\ &= \underbrace{\int_{\mathbb{R}^d} \mathcal{G} \log\left(\frac{n_t(\theta)}{\mu_n(\theta)}\right) \mathrm{d}n_t(\theta)}_{J_{1,t}} + \underbrace{\int_{\mathbb{R}^d} (\mathcal{G}_t - \mathcal{G}) \log\left(\frac{n_t(\theta)}{\mu_n(\theta)}\right) \mathrm{d}n_t(\theta)}_{J_{2,t}}.\end{aligned} \tag{2.12}$$

We study each term separately.

- Study of $J_{1,t}$. Since $\mathcal{G}$ is a diffusion operator and μ_n is the invariant measure associated to $\mathcal{G}$, then we can use De Bruijn's identity (see Proposition 1.2.17):

$$\int_{\mathbb{R}^d} \mathcal{G} \log\left(\frac{n_t(\theta)}{\mu_n(\theta)}\right) \mathrm{d}n_t(\theta) = \int_{\mathbb{R}^d} \frac{n_t(\theta)}{\mu_n(\theta)} \mathcal{G} \log\left(\frac{n_t(\theta)}{\mu_n(\theta)}\right) \mathrm{d}\mu_n(\theta) = -4\mathcal{E}_t. \tag{2.13}$$

- Study of $J_{2,t}$. We use the difference between $\mathcal{G}$ and $\mathcal{G}_t$, for any twice

differentiable function f:

$$\begin{aligned}(\mathcal{G}_t - \mathcal{G})\, f(\theta) &= -\sum_{i=1}^{n} \langle \nabla_\theta f(\theta), \nabla_\theta U_{\mathbf{X}_i}(\theta)\rangle \left[m_t(\mathbf{X}_i|\theta) - \nu_n(\mathbf{X}_i)\right] \\ &= -\frac{1}{n}\sum_{i=1}^{n} \langle \nabla_\theta f(\theta), \nabla_\theta U_{\mathbf{X}_i}(\theta)\rangle \left[\frac{m_t(\mathbf{X}_i|\theta)}{\nu_n(\mathbf{X}_i)} - 1\right],\end{aligned}$$

where $\nu_n(\mathbf{X}_i) = \frac{1}{n}$, for any i. Then, the term $J_{2,t}$ may be computed as follows:

$$\begin{aligned}|J_{2,t}| &= \left|\int_{\mathbb{R}^d} (\mathcal{G}_t - \mathcal{G}) \log\left(\frac{n_t(\theta)}{\mu_n(\theta)}\right) \mathrm{d}n_t(\theta)\right| \\ &= \frac{1}{n}\left|\int_{\mathbb{R}^d} \sum_{i=1}^{n} \langle \nabla_\theta \log\left(\frac{n_t(\theta)}{\mu_n(\theta)}\right), \nabla_\theta U_{\mathbf{X}_i}(\theta)\rangle \left[\frac{m_t(\mathbf{X}_i|\theta)}{\nu_n(\mathbf{X}_i)} - 1\right] \mathrm{d}n_t(\theta)\right|.\end{aligned}$$

Using the Cauchy-Schwarz inequality with respect to the measure $\nu_n(\mathbf{X}_i) \times \mathrm{d}n_t(\theta)$ in the first line and $2ab \leq a^2 + b^2$, for any a and b, in the second line, we obtain that:

$$\begin{aligned}|J_{2,t}| &\leq 2\mathcal{E}_t^{\frac{1}{2}} \left(\frac{1}{n}\int_{\mathbb{R}^d} \sum_{i=1}^{n} \|\nabla_\theta U_{\mathbf{X}_i}(\theta)\|_2^2 \left[\frac{m_t(\mathbf{X}_i|\theta)}{\nu_n(\mathbf{X}_i)} - 1\right]^2 \mathrm{d}n_t(\theta)\right)^{\frac{1}{2}} \\ &\leq \mathcal{E}_t + \frac{1}{n}\int_{\mathbb{R}^d} \sum_{i=1}^{n} \|\nabla_\theta U_{\mathbf{X}_i}(\theta)\|_2^2 \left[\frac{m_t(\mathbf{X}_i|\theta)}{\nu_n(\mathbf{X}_i)} - 1\right]^2 \mathrm{d}n_t(\theta).\end{aligned}$$

Replacing Equation (2.13) and the inequality above in Equation (2.12), we get:

$$\partial_t \{J_t\} \leq -3\mathcal{E}_t + \underbrace{\frac{1}{n}\int_{\mathbb{R}^d} \sum_{i=1}^{n} \|\nabla_\theta U_{\mathbf{X}_i}(\theta)\|_2^2 \left[\frac{m_t(\mathbf{X}_i|\theta)}{\nu_n(\mathbf{X}_i)} - 1\right]^2 \mathrm{d}n_t(\theta)}_{\Delta_t}. \tag{2.14}$$

We then focus on the second term of the right-hand side. For this purpose, we consider a strictly positive function $g(t)$, which will be fixed later and we split Δ_t into two terms as:

$$\begin{aligned}\Delta_t =& \frac{1}{n}\int_{\mathbb{R}^d} \sum_{i=1}^{n} \|\nabla_\theta U_{\mathbf{X}_i}(\theta)\|_2^2 \mathbb{1}_{\|\nabla_\theta U_{\mathbf{X}_i}(\theta)\|_2 \leq g(t)} \left[\frac{m_t(\mathbf{X}_i|\theta)}{\nu_n(\mathbf{X}_i)} - 1\right]^2 \mathrm{d}n_t(\theta) \\ &+ \frac{1}{n}\int_{\mathbb{R}^d} \sum_{i=1}^{n} \|\nabla_\theta U_{\mathbf{X}_i}(\theta)\|_2^2 \mathbb{1}_{\|\nabla_\theta U_{\mathbf{X}_i}(\theta)\|_2 > g(t)} \left[\frac{m_t(\mathbf{X}_i|\theta)}{\nu_n(\mathbf{X}_i)} - 1\right]^2 \mathrm{d}n_t(\theta) \\ =& \Delta_{1,t} + \Delta_{2,t}.\end{aligned} \tag{2.15}$$

We study each term separately.

- Study of $\Delta_{1,t}$. We introduce a weighted distance $\mathbb{L}^2$ between the conditional distribution of X_t given $\theta_t = \theta$ and the measure ν_n. This distance is denoted

by I_t and is defined as:

$$I_t = \frac{1}{n}\int_{\mathbb{R}^d}\sum_{i=1}^{n}\left(\frac{m_t(\mathbf{X}_i|\theta)}{\nu_n(\mathbf{X}_i)} - 1\right)^2 \mathrm{d}n_t(\theta). \tag{2.16}$$

This quantity measures the average closeness (with respect to θ) of the conditional law of x given θ at time t to ν_n. Using the definition of I_t, the first term of (2.15) is bounded by:

$$\Delta_{1,t} \leq g^2(t) I_t.$$

In Section 2.D, Proposition 2.D.1, we show that $I_t \leq (n-1)e^{-2\alpha_n t}$. Then, we get that:

$$\Delta_{1,t} \leq n g^2(t) e^{-2\alpha_n t}.$$

- Study of $\Delta_{2,t}$. We recall that for any i, $\nu_n(X_i) = 1/n$ and $m_t(\mathbf{X}_i|\theta) \in [0,1]$, then $\left|\frac{m_t(\mathbf{X}_i|\theta)}{\nu_n(\mathbf{X}_i)} - 1\right| \leq n$, which implies that:

$$\Delta_{2,t} \leq n\int_{\mathbb{R}^d}\sum_{i=1}^{n}\|\nabla_\theta U_{\mathbf{X}_i}(\theta)\|_2^2 \mathbb{1}_{\|\nabla_\theta U_{\mathbf{X}_i}(\theta)\|_2 > g(t)}\mathrm{d}n_t(\theta).$$

The Cauchy-Schwarz inequality with respect to the measure $\nu_n(\mathbf{X}_i) \times \mathrm{d}n_t(\theta)$ leads to:

$$\begin{aligned}\Delta_{2,t} &\leq n\left(\int_{\mathbb{R}^d}\sum_{i=1}^{n}\|\nabla_\theta U_{\mathbf{X}_i}(\theta)\|_2^4 \,\mathrm{d}n_t(\theta)\right)^{\frac{1}{2}}\left(\int_{\mathbb{R}^d}\sum_{i=1}^{n}\mathbb{1}_{\|\nabla_\theta U_{\mathbf{X}_i}(\theta)\|_2 > g(t)}\mathrm{d}n_t(\theta)\right)^{\frac{1}{2}} \\ &= n\left(\sum_{i=1}^{n}\mathbb{E}_{n_t}\left(\|\nabla_\theta U_{\mathbf{X}_i}(\theta_t)\|_2^4\right)\right)^{\frac{1}{2}}\left(\sum_{i=1}^{n}\mathbb{P}_{n_t}\left(\|\nabla_\theta U_{\mathbf{X}_i}(\theta_t)\|_2 > g(t)\right)\right)^{\frac{1}{2}}.\end{aligned} \tag{2.17}$$

Thanks to Propositions 2.1.3 and 1.5.2, we deduce that for any $\theta \in \mathbb{R}^d$:

$$\|\nabla_\theta U_{\mathbf{X}_i}(\theta)\|_2^2 \leq 2(nL + \ell_0) U_{\mathbf{X}_i}(\theta),$$

then:

$$\begin{aligned}\sum_{i=1}^{n}\mathbb{E}_{n_t}\left(\|\nabla_\theta U_{\mathbf{X}_i}(\theta_t)\|_2^4\right) &\leq 4(nL+\ell_0)^2\sum_{i=1}^{n}\mathbb{E}_{n_t}\left(U_{\mathbf{X}_i}^2(\theta_t)\right) \\ &\leq 4n^2(nL+\ell_0)^2\mathbb{E}_{n_t}\left(U_{\nu_n}^2(\theta_t)\right),\end{aligned} \tag{2.18}$$

where we used the relation $\|.\|_2 \leq \|.\|_1$ in $\mathbb{R}^n$. Furthermore,

$$\begin{aligned}\sum_{i=1}^{n} \mathbb{P}_{n_t}\left(\|\nabla_\theta U_{\mathbf{X}_i}(\theta_t)\|_2 > g(t)\right) &\leq \sum_{i=1}^{n} \mathbb{P}_{n_t}\left(2(nL+\ell_0)U_{\mathbf{X}_i}(\theta_t) > g^2(t)\right) \\ &\leq \frac{2(nL+\ell_0)}{g^2(t)} \sum_{i=1}^{n} \mathbb{E}_{n_t}\left(U_{\mathbf{X}_i}(\theta_t)\right) \\ &\leq \frac{2n(nL+\ell_0)}{g^2(t)} \mathbb{E}_{n_t}\left(U_{\nu_n}(\theta_t)\right), \qquad (2.19)\end{aligned}$$

where we used Markov inequality in the second line. Replacing (2.18) and (2.19) in (2.17), we get that:

$$\begin{aligned}\Delta_{2,t} &\leq \frac{n^{5/2}[2(nL+\ell_0)]^{3/2}}{g(t)} \left[\mathbb{E}_{n_t}(U^2_{\nu_n}(\theta_t))\mathbb{E}_{n_t}\left(U_{\nu_n}(\theta_t)\right)\right]^{1/2} \\ &\leq Cn^{\frac{11}{2}} \left(d\log^{2\beta}(n)\right)^{\frac{3(1+r)}{2}} g^{-1}(t),\end{aligned}$$

where C is a positive constant obtained when we apply Proposition 2.E.1 with $\alpha = 2$ and $\alpha = 1$ to upper the moments of $U_{\nu_n}(\theta_t)$.

Return to Δ_t in (2.15), we have proven that:

$$\Delta_t \leq ng^2(t)e^{-2\alpha_n t} + Cn^{\frac{11}{2}} \left(d\log^{2\beta}(n)\right)^{\frac{3(1+r)}{2}} g^{-1}(t).$$

Optimizing with respect to $g(t)$, we deduce that:

$$\Delta_t \leq Cn^4 \left(d\log^{2\beta}(n)\right)^{1+r} e^{-\frac{2\alpha_n t}{3}}, \quad \forall t \geq 0.$$

We conclude the proof by replacing the bound of Δ_t in (2.14). □

2.4 Functional inequalities under $\mathcal{H}^{\mathbf{r}}_{\mathbf{KL}}(\mathfrak{c}, L)$ hypothesis

We proved the following link between the derivative of the entropy J_t and the Dirichlet form $\mathcal{E}_t$,

$$\partial_t\{J_t\} \leq -3\mathcal{E}_t + c_{n,d}e^{-\frac{2\alpha_n}{3}t}, \quad t > 0,$$

where $c_{n,d}$ is an explicit function of n and d. This section is devoted to finding a lower bound of $\mathcal{E}_t$ depending on J_t, which will allow us to obtain a functional inequality for J_t. This section is based on the results studied in Section 1.5 taking into account

$$J_t = Ent_{\mu_n}(h_t) \quad \text{and} \quad \mathcal{E}_t = \mathcal{E}_{\mu_n}\left(\sqrt{h_t}\right),$$

where $h_t(\theta) = n_t(\theta)/\mu_n(\theta)$.

Under the $\mathcal{H}^{\mathbf{r}}_{\mathbf{KL}}(\mathfrak{c}, L)$ hypothesis, we are able to describe two situations: if $r = 0$ then the probability measure μ_n is strongly log-concave and satisfies an LSI, in particular we will obtain a link between $\mathcal{E}_t$ and J_t; however, if $0 < r < 1$, then

the probability measure μ_n is weakly log-concave and satisfies a WLSI. Then it is important to prove that the function $\sqrt{h_t}$ is bounded and to determine the bound for $Osc^2\left(\sqrt{h_t}\right)$. We then study each situation separately.

Strongly log-concave case

The following proposition is an immediate consequence of Proposition 2.1.3 and Theorem 1.5.5, therefore we omit its proof.

Proposition 2.4.1. *Assume $\mathcal{H}_{\pi_0}(\ell_0)$ and that each $\theta \mapsto -\log p_\theta(\mathbf{X}_i)$ satisfies $\mathcal{H}^{\mathbf{r}}_{\mathbf{KL}}(\mathfrak{c}, L)$ with $r = 0$. Then μ_n verifies an LSI with constant $C_{\mathrm{LSI}} = \frac{2}{\mathfrak{c}n}$, which is independent of d. In particular:*

$$J_t \leq C_{\mathrm{LSI}} \mathcal{E}_t.$$

It remains to state the functional inequality for J_t resulting from Propositions 2.3.3 and 2.4.1. This is done in Section 2.5.

Weakly log-concave case

We recall from Propositions 2.1.3 and 1.5.6 that μ_n is a weakly log-concave measure and then it satisfies a Poincaré inequality with constant $C_{\mathrm{PI}}(\mu_n)$. From now on $C_{\mathrm{PI}}(\mu_n)$ is denoted as C_{PI}. In addition, using part b of Proposition 1.5.6 we deduce that μ_n also satisfies a WLSI with function

$$\varphi(s) = \mathfrak{b} C_{\mathrm{PI}} \log(1/s), \quad 0 < s \leq e^{-1},$$

where $\mathfrak{b} > 0$ and is independent of n and d. Then, for any function $f \in \mathcal{C}^1_b(\mathbb{R}^d)$ and any $s > 0$,

$$Ent_{\mu_n}(f^2) \leq \varphi(s) \mathcal{E}_{\mu_n}(f) + s\, Osc^2(f),$$

where $Osc(f) = \sup f - \inf f$. In order to use WLSI to link J_t and $\mathcal{E}_t$ we have to find a bound of C_{PI}, show that h_t is bounded and then determine a bound of $Osc^2(\sqrt{h_t})$, for any $t \geq 0$.

The next proposition states two lower bounds on the Poincaré constant within the $\mathcal{H}^{\mathbf{r}}_{\mathbf{KL}}(\mathfrak{c}, L)$ framework when $0 < r < 1$. The first is always valid, regardless of the value of $(X_1, \ldots, X_n)$ that may be randomly sampled. The second has to be considered with high probability, with respect to the sampling process $(X_1, \ldots, X_n)$. The proof of Proposition 2.4.2 is deferred to Section 2.B.

Proposition 2.4.2. *Assume $\mathcal{H}^{\beta}_{\min}, \mathcal{H}_{\mathbf{n_0}}(L, \ell_0)$, $\mathcal{H}_{\pi_0}(\ell_0)$ and that $\theta \mapsto -\log p_\theta(x)$ satisfies $\mathcal{H}^{\mathbf{r}}_{\mathbf{KL}}(\mathfrak{c}, L)$, for any x, where $0 < r < 1$. Then,*

a) For any sample $(X_1, \ldots, X_n)$, it holds:

$$C_{\mathrm{PI}} \lesssim_{uc} \left(d \log^{2\beta}(n)\right)^{(1+r)^2}.$$

b) Assume that $\theta \mapsto \mathbb{P}_\theta$ is injective and θ_0 exists such that $(X_1, \dots, X_n) \sim \mathbb{P}_{\theta_0}$. If locally around θ_0, $\theta \mapsto |\theta - \theta_0|^{-\alpha}\mathcal{W}_1(\mathbb{P}_\theta, \mathbb{P}_{\theta_0})$ does not vanish, then:

$$\mathbb{E}_{(X_1,\dots,X_n)\sim\mathbb{P}_{\theta_0}}[C_{\mathrm{PI}}] \lesssim_{uc} \left(\frac{d\log n}{n}\right)^\alpha.$$

We are finally led to show that h_t is a bounded function and then compute an upper bound of the oscillation $Osc\left(\sqrt{h_t}\right)$ for any time $t > 0$. For this purpose, we recall from the proof of Proposition 1.3.8 that if h_0 is bounded then h_t is also bounded and in fact

$$Osc\left(\sqrt{h_t}\right) \leq Osc\left(\sqrt{h_0}\right), \quad t > 0.$$

The next proposition implies the boundedness of h_t over $\mathbb{R}^d$ when n_0 is chosen as the density function of a centered Gaussian random variable with a carefully tuned covariance matrix.

Proposition 2.4.3. *Assume $\mathcal{H}^\beta_{\min}$,$\mathcal{H}_{\mathbf{n_0}}(L, \ell_0)$, $\mathcal{H}_{\pi_0}(\ell_0)$ and that $\theta \mapsto -\log p_\theta(x)$ satisfies $\mathcal{H}^{\mathfrak{r}}_{\mathbf{KL}}(\mathfrak{c}, L)$, for any x. Then h_t is bounded for any $t > 0$ and there exist two positive constants C_1 and C_2, which are independent from n and d and such that:*

$$Osc^2\left(\sqrt{h_t}\right) \lesssim_{uc} (C_1 d/n)^{\frac{dr}{2}} \exp\left(C_2 n (d\log^{2\beta}(n))^{1+r}\right).$$

The proof of Proposition 2.4.3 may be found in Section 2.B.

As a consequence of Propositions 2.4.2 and 2.4.3, we state the following result, which we will not prove.

Proposition 2.4.4. *Assume $\mathcal{H}^\beta_{\min}$,$\mathcal{H}_{\mathbf{n_0}}(L, \ell_0)$, $\mathcal{H}_{\pi_0}(\ell_0)$ and that $\theta \mapsto -\log p_\theta(x)$ satisfies $\mathcal{H}^{\mathfrak{r}}_{\mathbf{KL}}(\mathfrak{c}, L)$, for any x. Then:*

a) μ_n satisfies a Poincaré inequality with constant C_{PI}.

b) For any $t > 0$ and $s > 0$,

$$J_t \leq \varphi(s)\mathcal{E}_t + sO_{n,d},$$

where $O_{n,d} = (C_1 d/n)^{\frac{dr}{2}} \exp\left(C_2 n (d\log^{2\beta}(n))^{1+r}\right)$ and the function φ could be defined as:

$$\varphi(s) = \mathfrak{b} C_{\mathrm{PI}} \log(1/s), \quad 0 < s \leq e^{-1},$$

where $\mathfrak{b}$ is a positive constant independent of n and d.

2.5 Proof of the main result

In Proposition 2.3.3, we proved that for any $t > 0$:

$$\partial_t\{J_t\} \leq -3\mathcal{E}_t + c_{n,d} e^{-\frac{2\alpha_n}{3}t}, \tag{2.20}$$

where $c_{n,d} \lesssim_{uc} n^4 \left(d \log^{2\beta}(n)\right)^{1+r}$. We study the cases $r = 0$ (strongly log-concave case) and $0 < r < 1$ (weakly log-concave case) separately.

Strongly log-concave case

From Proposition 2.4.1, the probability measure μ_n verifies an LSI with constant $C_{\mathrm{LSI}} = \frac{2}{\mathfrak{c}n}$, which is independent of d. Then, in particular:

$$J_t \leq C_{\mathrm{LSI}} \mathcal{E}_t.$$

We combine the inequality above with (2.20) to get that for any $t > 0$,

$$\partial_t \{J_t\} \leq -\frac{3J_t}{C_{\mathrm{LSI}}} + c_{n,d} e^{-\frac{2\alpha_n}{3}t}.$$

Applying Gronwall's lemma, we deduce that:

$$J_t \leq J_0 e^{-\frac{3t}{C_{\mathrm{LSI}}}} + c_{n,d} e^{-\frac{3t}{C_{\mathrm{LSI}}}} \int_0^t e^{\left(\frac{3}{C_{\mathrm{LSI}}} - \frac{2\alpha_n}{3}\right)u} \mathrm{d}u.$$

Now, we upper bound the second term assuming that $\alpha_n \neq \frac{9}{2C_{\mathrm{LSI}}}$, the same bound is obtained if $\alpha_n = \frac{9}{2C_{\mathrm{LSI}}}$. We apply the mean value theorem to the function $y \mapsto -e^{-yt}$ in the interval $\left(\min\left\{\frac{3}{C_{\mathrm{LSI}}}, \frac{2\alpha_n}{3}\right\}, \max\left\{\frac{3}{C_{\mathrm{LSI}}}, \frac{2\alpha_n}{3}\right\}\right)$ to get

$$e^{-\frac{3t}{C_{\mathrm{LSI}}}} \int_0^t e^{\left(\frac{3}{C_{\mathrm{LSI}}} - \frac{2\alpha_n}{3}\right)u} \mathrm{d}u = \frac{e^{-\frac{2\alpha_n t}{3}} - e^{-\frac{3t}{C_{\mathrm{LSI}}}}}{\frac{3}{C_{\mathrm{LSI}}} - \frac{2\alpha_n}{3}} = te^{-at} \leq te^{-\min\left\{\frac{3}{C_{\mathrm{LSI}}}, \frac{2\alpha_n}{3}\right\}t},$$

where $a \in \left(\min\left\{\frac{3}{C_{\mathrm{LSI}}}, \frac{2\alpha_n}{3}\right\}, \max\left\{\frac{3}{C_{\mathrm{LSI}}}, \frac{2\alpha_n}{3}\right\}\right)$.

Using that for any $x \geq 0$, $xe^{-x} \leq e^{-x/2}$, with $x = \min\left\{\frac{3\mathfrak{c}n}{2}, \frac{2\alpha_n}{3}\right\}t$, we obtain that,

$$e^{-\frac{3t}{C_{\mathrm{LSI}}}} \int_0^t e^{\left(\frac{3}{C_{\mathrm{LSI}}} - \frac{2\alpha_n}{3}\right)u} \mathrm{d}u \leq \max\left\{\frac{C_{\mathrm{LSI}}}{3}, \frac{3}{2\alpha_n}\right\} e^{-\min\left\{\frac{3}{2C_{\mathrm{LSI}}}, \frac{\alpha_n}{3}\right\}t}.$$

We proved that:

$$J_t \leq \left(J_0 + c_{n,d} \max\left\{\frac{C_{\mathrm{LSI}}}{3}, \frac{3}{2\alpha_n}\right\}\right) e^{-\min\left\{\frac{3}{2C_{\mathrm{LSI}}}, \frac{\alpha_n}{3}\right\}t}, \quad t > 0.$$

A sufficient condition to obtain $J_t \leq \varepsilon$ is that

$$\left(J_0 + c_{n,d} \max\left\{\frac{C_{\mathrm{LSI}}}{3}, \frac{3}{2\alpha_n}\right\}\right) e^{-\min\left\{\frac{3}{2C_{\mathrm{LSI}}}, \frac{\alpha_n}{3}\right\}t} \leq \varepsilon,$$

or equivalently:

$$t \geq \max\left\{\frac{2C_{\mathrm{LSI}}}{3}, \frac{3}{\alpha_n}\right\} \left[\log(\varepsilon^{-1}) + \log\left(J_0 + c_{n,d} \max\left\{\frac{C_{\mathrm{LSI}}}{3}, \frac{3}{2\alpha_n}\right\}\right)\right].$$

Using $C_{\mathrm{LSI}} = \frac{2}{\mathfrak{c}n}$, $J_0 \lesssim_{uc} d$, $c_{n,d} \lesssim_{uc} n^4 d \log^{2\beta}(n)$ and taking $\alpha_n = n$, we deduce

$$t \gtrsim_{uc} n^{-1}\left[\log(\varepsilon^{-1}) + \log\left(n^3 d \log^{2\beta}(n)\right)\right] \implies J_t \leq \varepsilon.$$

Weakly log-concave case

From Proposition 2.4.4, we deduce a differential inequality for J_t. For any $t > 0$ and $0 < s < e^{-1}$:

$$\partial_t \{J_t\} \leq -\frac{3J_t}{\varphi(s)} + \frac{3s\, O_{n,d}}{\varphi(s)} + c_{n,d} e^{-\frac{2\alpha_n}{3}t}.$$

Using Gronwall's lemma and the mean value theorem as in the strongly log-concave case, we get that

$$J_t \leq \inf_s \left\{ J_0 e^{-\frac{3t}{\varphi(s)}} + sO_{n,d} + c_{n,d} \max\left\{\frac{\varphi(s)}{3}, \frac{3}{2\alpha_n}\right\} e^{-\min\left\{\frac{3}{2\varphi(s)}, \frac{\alpha_n}{3}\right\}t} \right\}.$$

We then choose $s = s_t = e^{-A\sqrt{t}}$, with $A > 0$. We observe that $0 < s_t < e^{-1}$ if $t > A^{-2}$, and then:

$$\varphi(s_t) = \mathfrak{b} C_{\mathrm{PI}} \log(1/s_t) = \mathfrak{b} A C_{\mathrm{PI}} \sqrt{t},$$

which leads to

$$J_t \leq J_0 e^{-\frac{3\sqrt{t}}{\mathfrak{b} A C_{\mathrm{PI}}}} + O_{n,d} e^{-A\sqrt{t}} + c_{n,d} \max\left\{\frac{\mathfrak{b} A C_{\mathrm{PI}}\sqrt{t}}{3}, \frac{3}{2\alpha_n}\right\} e^{-\min\left\{\frac{3}{2\mathfrak{b} A C_{\mathrm{PI}}\sqrt{t}}, \frac{\alpha_n}{3}\right\}t}.$$

The choice $A = \sqrt{\frac{3}{\mathfrak{b} C_{\mathrm{PI}}}}$ guarantees that the first two terms are of the same order with respect to t. Moreover, the condition $t > A^{-2}$ becomes $t > \frac{\mathfrak{b} C_{\mathrm{PI}}}{3}$.

Then,

$$J_t \leq (J_0 + O_{n,d}) e^{-\sqrt{\frac{3t}{\mathfrak{b} C_{\mathrm{PI}}}}} + c_{n,d} \max\left\{\sqrt{\frac{\mathfrak{b} C_{\mathrm{PI}} t}{3}}, \frac{3}{2\alpha_n}\right\} e^{-\min\left\{\sqrt{\frac{3}{4\mathfrak{b} C_{\mathrm{PI}} t}}, \frac{\alpha_n}{3}\right\}t}.$$

We assume that $\sqrt{\frac{3}{4\mathfrak{b} C_{\mathrm{PI}} t}} \leq \frac{\alpha_n}{3}$, that is, $t \gtrsim_{uc} \frac{1}{\alpha_n^2 C_{\mathrm{PI}}}$, then

$$\begin{aligned} J_t &\leq \left(J_0 + O_{n,d} + c_{n,d}\sqrt{\frac{\mathfrak{b} C_{\mathrm{PI}} t}{3}}\right) e^{-\sqrt{\frac{3t}{4\mathfrak{b} C_{\mathrm{PI}}}}} \\ &\lesssim_{uc} \left(J_0 + O_{n,d} + c_{n,d} C_{\mathrm{PI}}\right) e^{-\sqrt{\frac{3t}{16\mathfrak{b} C_{\mathrm{PI}}}}}, \end{aligned}$$

where we used the inequality: for any $x \geq 0$, $xe^{-x} \leq e^{-x/2}$), with $x = \sqrt{\frac{3t}{4\mathfrak{b} C_{\mathrm{PI}}}}$.

A sufficient condition to get $J_t \leq \varepsilon$ is that the right hand of the previous equation is less than ε, which is verified if:

$$t \gtrsim_{uc} C_{\mathrm{PI}}\left[\log^2(\varepsilon^{-1}) + \log^2\left(J_0 + O_{n,d} + c_{n,d} C_{\mathrm{PI}}\right)\right].$$

In Proposition 2.4.2 we prove $C_{\mathrm{PI}} \lesssim_{uc} \left(d \log^{2\beta}(n)\right)^{(1+r)^2}$. If instead of using the

constant C_{PI}, we use directly $\kappa \left(d \log^{2\beta}(n)\right)^{(1+r)^2}$, then all the previous computations remain the same, only replacing C_{PI} by its upper bound. Therefore, using the values of $O_{n,d}$, $c_{n,d}$ and the upper bound of J_0, we finally observe that if $t \gtrsim_{uc} \alpha_n^{-2} \left(d \log^{2\beta}(n)\right)^{-(1+r)^2}$ and

$$t \gtrsim_{uc} \left(d \log^{2\beta}(n)\right)^{(1+r)^2} \left[\log^2(\varepsilon^{-1}) + n^2 \left(d \log^{2\beta}(n)\right)^{2(1+r)}\right],$$

then $J_t \leq \varepsilon$.

Appendix to Chapter 2

2.A Properties of U_{ν_n} and $\mathcal{Z}_n$

We start by proving Proposition 2.1.3, which shows how a $\mathcal{H}^{\mathbf{r}}_{\mathbf{KL}}(\mathfrak{c}, L)$ condition on $\theta \mapsto -\log p_\theta(x)$ is translated to U_{ν_n}.

Proof of Proposition 2.1.3. First, we observe that if each $\theta \mapsto \nabla \log p_\theta(\mathbf{X}_i)$ is L-Lipschitz and $\theta \mapsto \nabla \log \pi_0$ is ℓ_0-Lipschitz, then the triangle inequality implies that:

$$\begin{aligned}\|\nabla U_{\nu_n}(\theta_1) - \nabla U_{\nu_n}(\theta_2)\|_2 &\leq \|\nabla U_{\mathbf{X}_1}(\theta_1) - \nabla U_{\mathbf{X}_1}(\theta_2)\|_2 \\ &\leq (nL + \ell_0)\|\theta_1 - \theta_2\|_2,\end{aligned}$$

for any $\theta_1, \theta_2 \in \mathbb{R}^d$. Then $\theta \mapsto \nabla U_{\nu_n}(\theta)$ is $(nL + \ell_0)$-Lipschitz.

Second, we consider the lower bound property on the curvature and observe that for any $\theta \in \mathbb{R}^d$,

$$\begin{aligned}\underline{\lambda}_{\nabla^2 U_{\nu_n}}(\theta) &= \inf_{e \in \mathbb{R}^d : \|e\|_2 = 1} e^T (\nabla^2 U_{\nu_n}(\theta)) e \\ &\geq \frac{1}{n} \sum_{i=1}^n \inf_{e \in \mathbb{R}^d : \|e\|_2 = 1} e^T (\nabla^2 U_{\mathbf{X}_i}(\theta)) e.\end{aligned}$$

The log-concavity of the prior yields:

$$\underline{\lambda}_{\nabla^2 U_{\nu_n}}(\theta) \geq \frac{1}{n} \sum_{i=1}^n \underline{\lambda}_{\nabla^2(-n \log p_\theta(\mathbf{X}_i))} = \sum_{i=1}^n \underline{\lambda}_{\nabla^2(-\log p_\theta(\mathbf{X}_i))}.$$

Then, the $\mathcal{H}^{\mathbf{r}}_{\mathbf{KL}}(\mathfrak{c}, L)$ property applied to each term of the sum above and the condition $\min_{\theta \in \mathbb{R}^d} -\log \pi_0(\theta) > 0$ from $\mathcal{H}_{\pi_\mathbf{0}}(\ell_0)$ yield:

$$\begin{aligned}\underline{\lambda}_{\nabla^2 U_{\nu_n}}(\theta) &\geq \mathfrak{c} \sum_{i=1}^n [-\log p_\theta(\mathbf{X}_i)]^{-r} \\ &\geq \mathfrak{c} n^r \sum_{i=1}^n U_{\mathbf{X}_i}^{-r}(\theta) = \mathfrak{c} n^{1+r} \left(\frac{1}{n} \sum_{i=1}^n U_{\mathbf{X}_i}^{-r}(\theta) \right).\end{aligned}$$

From the Jensen inequality, we finally deduce that:

$$\underline{\lambda}_{\nabla^2 U_{\nu_n}}(\theta) \geq \mathfrak{c} n^{1+r} \left(\frac{1}{n} \sum_{i=1}^n U_{\mathbf{X}_i}^{-r}(\theta) \right) \geq \mathfrak{c} n^{1+r} U_{\nu_n}^{-r}(\theta).$$

We conclude that U_{ν_n} satisfies $\mathcal{H}^{\mathbf{r}}_{\mathbf{KL}}(\mathfrak{c}n^{1+r}, nL+\ell_0)$. The proof is similar when we consider $U_{\mathbf{X}_i}$ instead of U_{ν_n}. □

The following proposition shows the growth of the minimum and the minimizer of U_{ν_n} with respect to d and n.

Proposition 2.A.1. *Assume $\mathcal{H}_{\pi_0}(\ell_0)$, $\mathcal{H}^{\beta}_{\min}$ and that for any x: $\theta \longmapsto -\log p_\theta(x)$ satisfies $\mathcal{H}^{\mathbf{r}}_{\mathbf{KL}}(\mathfrak{c}, L)$. Then:*

$$\|\arg\min U_{\nu_n}\|_2 \lesssim_{uc} (\sqrt{d}\log^{\beta}(n))^{1+r} \quad \textit{and} \quad \min_{\theta\in\mathbb{R}^d} U_{\nu_n}(\theta) \lesssim_{uc} nd\log^{2\beta}(n).$$

Proof. The Proposition 2.1.3 shows that U_{ν_n} satisfies $\mathcal{H}^{\mathbf{r}}_{\mathbf{KL}}(\mathfrak{c}n^{1+r}, nL+\ell_0)$. Hence, we can apply Proposition 1.5.3 with $\theta = 0$ and $\arg\min U_{\nu_n} = \theta^*_n$. We deduce that:

$$\|\theta^*_n\|_2^2 \leq \frac{2}{(1+r)\mathfrak{c}n^{1+r}} U_{\nu_n}^{1+r}(0).$$

To obtain an upper bound of $U_{\nu_n}(0)$ we first find an upper bound of $U_{\mathbf{X}_i}(0)$ using $\mathcal{H}^{\beta}_{\min}$ and once again Proposition 1.5.3, for all i, as follows:

$$U_{\mathbf{X}_i}(0) \leq \min\ U_{\mathbf{X}_i} + \frac{nL+\ell_0}{2}\|\arg\min U_{\mathbf{X}_i}\|_2^2 \lesssim_{uc} nd\log^{2\beta}(n),$$

then $U_{\nu_n}(0) \lesssim_{uc} nd\log^{2\beta}(n)$ and:

$$\|\theta^*_n\|_2^2 \lesssim_{uc} \left(d\log^{2\beta}(n)\right)^{1+r}.$$

The second part comes from $U_{\nu_n}(\theta^*_n) \leq U_{\nu_n}(0)$. □

Proposition 2.A.2. *Assume $\mathcal{H}_{\pi_0}(\ell_0)$, $\mathcal{H}_{\mathbf{n_0}}(L, \ell_0)$, $\mathcal{H}^{\beta}_{\min}$ and that for any x: $\theta \longmapsto -\log p_\theta(x)$ satisfies $\mathcal{H}^{\mathbf{r}}_{\mathbf{KL}}(\mathfrak{c}, L)$. Then the normalizing constant of μ_n verifies the following inequality:*

$$\mathcal{Z}_n \leq 2\left(\frac{2\pi}{\mathfrak{c}n^{1+r}}\right)^{d/2} d^{dr/2}.$$

Proof. In Proposition 2.1.3, we proved that U_{ν_n} satisfies a $\mathcal{H}^{\mathbf{r}}_{\mathbf{KL}}(\mathfrak{c}n^{1+r}, nL+\ell_0)$-condition, then we deduce the result using the same argument as in Proposition 1.5.4 but with $\mathfrak{c}n^{1+r}$ instead of $\mathfrak{c}$. □

2.B Smoothness and boundedness of the semigroup

Proof of Proposition 2.3.2. *a)* We start by proving that $h_0 = n_0(\theta)/\mu_n(\theta)$ is a bounded function. For all $\theta \in \mathbb{R}^d$,

$$\begin{aligned} h_0(\theta) &= (2\pi\sigma^2)^{-d/2}\mathcal{Z}_n e^{-\frac{\|\theta\|_2^2}{2\sigma^2}+U_{\nu_n}(\theta)} \\ &\leq 2\left(\sigma^2\mathfrak{c}n^{1+r}\right)^{-d/2} d^{\frac{dr}{2}} e^{-\frac{\|\theta\|_2^2}{2\sigma^2}+U_{\nu_n}(\theta)}, \end{aligned} \tag{2.21}$$

where we used that n_0 is the density function of a $\mathcal{N}(0_d, \sigma^2 I_d)$ random variable and the bound of $\mathcal{Z}_n$ obtained in Proposition 2.A.2. Let us focus on the exponent of Equation (2.21).

By Proposition 2.1.3, U_{ν_n} satisfies a $\mathcal{H}^{\mathbf{r}}_{\mathbf{KL}}(\mathfrak{c}n^{1+r}, nL+\ell_0)$-condition. Then, we denote $\theta_n^\star = \arg\min U_{\nu_n}$ and apply Proposition 1.5.3 to U_{ν_n}:

$$U_{\nu_n}(\theta) \leq U_{\nu_n}(\theta_n^\star) + \frac{(nL+\ell_0)}{2}\|\theta - \theta_n^\star\|_2^2.$$

The exponent of (2.21) satisfies the following inequality:

$$-\frac{\|\theta\|_2^2}{2\sigma^2} + U_{\nu_n}(\theta) \leq -\frac{\|\theta\|_2^2}{2\sigma^2} + U_{\nu_n}(\theta_n^\star) + \frac{(nL+\ell_0)}{2}\|\theta - \theta_n^\star\|_2^2.$$

For all $\sigma^2 \leq \frac{c_2}{nL+\ell_0}$ where $0 < c_2 < 1$, a straightforward optimization on θ yields:

$$-\frac{\|\theta\|_2^2}{2\sigma^2} + \frac{(nL+\ell_0)}{2}\|\theta - \theta_n^\star\|_2^2 \leq \frac{(nL+\ell_0)}{2(1-c_2)}\|\theta_n^\star\|_2^2,$$

then:

$$\begin{aligned} -\frac{\|\theta\|_2^2}{2\sigma^2} + U_{\nu_n}(\theta) &\leq U_{\nu_n}(\theta_n^\star) + \frac{(nL+\ell_0)}{2(1-c_2)}\|\theta_n^\star\|_2^2 \\ &\leq C_2 n \left(d\log^{2\beta}(n)\right)^{1+r}, \end{aligned} \tag{2.22}$$

where we used Proposition 2.A.1 in the last step and we define C_2 as a constant. Replacing (2.22) in (2.21), we get that:

$$\begin{aligned} \|h_0\|_{\mathbb{L}^\infty_m(\mathbb{R}^d)} &\leq 2\left(\sigma^2 \mathfrak{c} n^{1+r}\right)^{-d/2} d^{dr/2} e^{C_2 n\left(d\log^{2\beta}(n)\right)^{1+r}} \\ &\leq \left(\frac{C_1 d}{n}\right)^{\frac{dr}{2}} e^{C_2 n\left(d\log^{2\beta}(n)\right)^{1+r}}, \end{aligned}$$

where we used $\frac{c_1}{nL+\ell_0} \leq \sigma^2$ from hypothesis $\mathcal{H}_{\mathbf{n_0}}(L, \ell_0)$ and one more time we define C_1 as a universal constant.
<u>b)</u> Using that $x \mapsto \log x$ is increasing and the part *a)*, we have that:

$$J_0 \leq \log\left(\|h_0\|_{\mathbb{L}^\infty_m(\mathbb{R}^d)}\right) \lesssim_{uc} n(d\log^{2\beta}(n))^{1+r} + rd\log(d/n).$$

□

Proof of Proposition 2.4.2. <u>*a)*</u> The proof relies on an argument set up with a "fixed" sample $(X_1, \ldots, X_n)$. Our starting point is Proposition 2.1.3 where we proved that U_{ν_n} satisfies a $\mathcal{H}^{\mathbf{r}}_{\mathbf{KL}}(\mathfrak{c}n^{1+r}, nL+\ell_0)$-condition. Then, we apply Proposition 1.5.3 with $\theta_n^* = \arg\min U_{\nu_n}$ and deduce that:

$$\|\theta - \theta_n^*\|_2^2 \leq \frac{2}{(1+r)\mathfrak{c}n^{1+r}} U_{\nu_n}^{1+r}(\theta).$$

We use the fact that for any distribution μ and any $a \in \mathbb{R}^d$, we have $Var_\mu(\theta) \leq \mathbb{E}_\mu(\|\theta - a\|_2^2)$. Taking $\mu = \mu_n$ and $a = \theta_n^*$, then:

$$\begin{aligned} Var_{\mu_n}(\theta) &\leq \mathbb{E}_{\mu_n}\left(\|\theta - \theta_n^*\|_2^2\right) \\ &\leq \frac{2}{(1+r)\mathfrak{c}n^{1+r}} \mathbb{E}_{\mu_n}[U_{\nu_n}^{1+r}(\theta)]. \end{aligned}$$

We then use the ergodic behavior of $(\theta_t)_{t\geq 0}$ and observe that there exists a constant C independent from n and d such that:

$$\begin{aligned} Var_{\mu_n}(\theta) &\leq \frac{2}{(1+r)\mathfrak{c}n^{1+r}} \limsup_{t\geq 0} \mathbb{E}_{n_t}[U_{\nu_n}^{1+r}(\theta_t)] \\ &\leq C\left(d\log^{2\beta}(n)\right)^{(1+r)^2}, \end{aligned}$$

where the last inequality comes from Proposition 2.E.1.

We now use the Bobkov bound on the Poincaré constant for a log-concave distribution, see Remark 1.2.4, and deduce that a universal constant $\kappa > 0$ exists such that:

$$C_{PI} \leq \kappa\left(d\log^{2\beta}(n)\right)^{(1+r)^2}.$$

b) For the second point, we consider a situation on average over the samples, and the result uses the concentration of the posterior distribution around its mean. We know from Theorem 3 of [Gadat et al., 2022] that a constant $c > 0$ exists such that:

$$\mathbb{E}_{(X_1,\ldots,X_n)\sim\mathbb{P}_{\theta_0}}[Var_{\mu_n}(\theta)] \leq c\epsilon_{n,d},$$

with $\epsilon_{n,d} = \left(\frac{Ld\log n}{n}\right)^{\alpha^{-1}}$. The result follows using the Jensen inequality and the Bobkov bound. □

Proof of Proposition 2.4.3. We proved in Proposition 2.3.2 that h_0 is bounded and

$$\|h_0\|_{\mathbb{L}_m^\infty(\mathbb{R}^d)} \leq \left(\frac{C_1 d}{n}\right)^{\frac{dr}{2}} e^{C_2 n\left(d\log^{2\beta}(n)\right)^{1+r}},$$

where C_1 and C_2 are two positive constants independent of n and d. In addition, from the proof of Proposition 1.3.8 we already know that if h_0 is a bounded function, then h_t is also bounded and

$$Osc\left(\sqrt{h_t}\right) \leq Osc\left(\sqrt{h_0}\right), \quad t > 0.$$

We get the statement taking into account $Osc\left(\sqrt{h_0}\right) \leq \|\sqrt{h_0}\|_{\mathbb{L}_m^\infty(\mathbb{R}^d)}$. □

2.C Regularity of the entropy

Let's start by proving the following result, which will be used several times.

Proposition 2.C.1. *Let $\mathcal{G}_t$ be the diffusion operator under the average effect of X_t, defined in Equation (2.9). If $\mathcal{G}_t^\star$ is the adjoint operator of $\mathcal{G}_t$ in $\mathbb{L}^2(\mathbb{R}^d)$, then for any $t > 0$:*

$$\partial_t\{n_t(\theta)\} = \mathcal{G}_t^\star n_t(\theta). \tag{2.23}$$

Proof. Let us prove (2.23) in the weak sense. Consider $g \in \mathcal{C}^2(\mathbb{R}^d)$. We deduce that:

$$\begin{aligned}
\int_{\mathbb{R}^d} g(\theta)\partial_t\{n_t(\theta)\}\mathrm{d}\theta &= \int_{\mathbb{R}^d} g(\theta)\partial_t\left\{\sum_{i=1}^n m_t(\theta,\mathbf{X}_i)\right\}\mathrm{d}\theta \\
&= \int_{\mathbb{R}^d}\sum_{i=1}^n g(\theta)\partial_t\{m_t(\theta,\mathbf{X}_i)\}\mathrm{d}\theta \\
&= \int_{\mathbb{R}^d}\sum_{i=1}^n \mathcal{L}g(\theta,\mathbf{X}_i)\ m_t(\theta,\mathbf{X}_i)\mathrm{d}\theta,
\end{aligned}$$

where we used the definition of n_t in the first step and Kolmogorov forward equation (2.8) in the last one, where $g(\theta,\mathbf{X}_i) = g(\theta)$, for any i, since $g(\theta)$ does not depend on x. We also observe that $\mathcal{L}_2 g(\theta, X_i) = 0$ and then we only need to compute the remaining term $\mathcal{L}_1 g(\theta, X_i)$:

$$\begin{aligned}
\int_{\mathbb{R}^d} g(\theta)\partial_t\{n_t(\theta)\}\mathrm{d}\theta &= \int_{\mathbb{R}^d}\sum_{i=1}^n \mathcal{L}_1 g(\theta,X_i)\ m_t(\theta,\mathbf{X}_i)\mathrm{d}\theta \\
&= \int_{\mathbb{R}^d}\sum_{i=1}^n \left[-\langle\nabla_\theta g(\theta),\nabla_\theta U_{\mathbf{X}_i}(\theta)\rangle + \Delta_\theta g(\theta)\right] m_t(\theta,\mathbf{X}_i)\mathrm{d}\theta \\
&= -\int_{\mathbb{R}^d}\sum_{i=1}^n \langle\nabla_\theta g(\theta),\nabla_\theta U_{\mathbf{X}_i}(\theta)\rangle m_t(\mathbf{X}_i|\theta)\mathrm{d}n_t(\theta) + \int_{\mathbb{R}^d}\Delta_\theta g(\theta)\mathrm{d}n_t(\theta) \\
&= \int_{\mathbb{R}^d}\mathcal{G}_t g(\theta)\mathrm{d}n_t(\theta),
\end{aligned}$$

where we used the fact that $m_t(\theta,\mathbf{X}_i) = m_t(\mathbf{X}_i|\theta)n_t(\theta)$. Using one more time Remark 3.19 of [Kusuoka and Stroock, 1984], we could verify that the first and second derivatives of n_t with respect to θ are locally bounded; therefore, n_t is differentiable in time and Equation (2.23) is satisfied in the strong sense. □

Now let us focus on proving that J_t is finite. Using the ellipticity of the semigroup generated by $\mathcal{G}_t$, we can use the result of [Hörmander, 1967] and have that for any $t > 0$, $n_t \in \mathcal{C}^\infty(\mathbb{R}^d)$ and the irreducibility yields $\forall t \geq 0$, $n_t > 0$. In Proposition 2.3.2 we proved that $h_0 = n_0(\theta)/\mu_n(\theta)$ is bounded and a standard consequence is that:

$$\sup\left(\frac{n_t}{\mu_n}\right) = \sup h_t \leq \sup h_0.$$

Therefore $J_t < \infty$, for any $t > 0$.

In the following proposition, it is shown that J_t is differentiable and that it is possible to exchange derivative and integral sign.

Proposition 2.C.2. *Assume* $\mathcal{H}_{\pi_0}(\ell_0)$, $\mathcal{H}^{\beta}_{\min}$, $\mathcal{H}_{\mathbf{n_0}}(L,\ell_0)$ *and that each* $\theta \mapsto -\log p_\theta(\mathbf{X}_i)$ *satisfies* $\mathcal{H}^{\mathbf{r}}_{\mathbf{KL}}(\mathfrak{c}, L)$. *Then:*

$$\partial_t\{J_t\} = \int_{\mathbb{R}^d}\left(1+\log\left(\frac{n_t(\theta)}{\mu_n(\theta)}\right)\right)\partial_t\{n_t(\theta)\}\mathrm{d}\theta = \int_{\mathbb{R}^d}\mathcal{G}_t\log\left(\frac{n_t(\theta)}{\mu_n(\theta)}\right)\mathrm{d}n_t(\theta). \quad (2.24)$$

Proof. We need to verify that the function of $t \mapsto \log\left(\frac{n_t(\theta)}{\mu_n(\theta)}\right)n_t(\theta)$ is differentiable, which is an immediate consequence of the fact that $t \mapsto m_t(\theta, x)$ is differentiable for any $(\theta, x) \in \mathbb{R}^d \times \mathcal{X}$. Then so are $t \mapsto n_t(\theta)$ and $t \mapsto \log(n_t(\theta)/\mu_n(\theta))n_t(\theta)$.

The second part of the proof consists of finding an integrable function g such that $\left|\partial_t\left\{\log\left(\frac{n_t(\theta)}{\mu_n(\theta)}\right)n_t(\theta)\right\}\right| \leq g(\theta)$ for any $t > 0$.

For any $\theta \in \mathbb{R}^d$ and any $t > 0$, we observe that:

$$\begin{aligned}\left|\partial_t\left\{\log\left(\frac{n_t(\theta)}{\mu_n(\theta)}\right)n_t(\theta)\right\}\right| &= \left|\left(1+\log\left(\frac{n_t(\theta)}{\mu_n(\theta)}\right)\right)\partial_t\{n_t(\theta)\}\right| \\ &\leq |1+\log(\|h_0\|_\infty)||\partial_t\{n_t(\theta)\}|.\end{aligned}$$

We then have to focus on $\partial_t\{n_t(\theta)\}$. Let us denote by $p_{0,t}(\vartheta,\cdot)$ the density function of $\theta_t|\theta_0 = \vartheta$, then the density of θ_t could be rewritten as:

$$n_t(\theta) = \int_{\mathbb{R}^d} n_0(\vartheta)p_{0,t}(\vartheta,\theta)\mathrm{d}\vartheta$$

where n_0 is the density function of θ_0 and in hypothesis $\mathcal{H}_{\mathbf{n_0}}(L,\ell_0)$ we assume that n_0 is a Gaussian distribution. We recall that $p_{0,t}(\vartheta,\cdot)$ satisfies the Fokker Planck equation:

$$\partial_t\{p_{0,t}(\vartheta,\theta)\} = \mathcal{G}^\star_t p_{0,t}(\vartheta,\theta),$$

where $\mathcal{G}^\star_t$ is the adjoint operator of $\mathcal{G}_t$ in $\mathbb{L}^2(\mathbb{R}^d)$. Using Remark 3.19 of [Kusuoka and Stroock, 1984], for any multi-index $\alpha \in \mathbb{N}^d$, the functions $(t,\theta) \mapsto \frac{\partial^{|\alpha|}}{\partial\theta^\alpha}p_{0,t}(\vartheta,\theta)$ are continuous and bounded by a polynomial of $\|\vartheta\|_2$ for any $t \in [T^{-1}, T]$ and $T > 1$. Therefore, the previous equation is satisfied in the strong sense when $t \in [T^{-1}, T]$. The following steps are true for $t \in [T^{-1}, T]$.

Taking into account that we can exchange derivative with the integral sign to find $\partial_t\{n_t(\theta)\}$ since the spatial derivatives of $p_{0,t}$ are bounded by a polynomial function and n_0 is a Gaussian density, we then observe that $\partial_t\{n_t(\theta)\}$ could be written as follows:

$$\begin{aligned}\partial_t\{n_t(\theta)\} &= \int_{\mathbb{R}^d} n_0(\vartheta)\partial_t\left\{p_{0,t}(\vartheta,\theta)\right\}\mathrm{d}\vartheta \\ &= \int_{\mathbb{R}^d}\mathcal{G}_t n_0(\vartheta)p_{0,t}(\vartheta,\theta)\mathrm{d}\vartheta.\end{aligned}$$

Using the definition of $\mathcal{G}_t$ in Equation (2.9), we deduce that for any $\vartheta \in \mathbb{R}^d$:

$$\mathcal{G}_t n_0(\vartheta) = \frac{1}{\sigma^2}\left(\sum_{i=1}^n\langle\vartheta, \nabla_\vartheta U_{\mathbf{X}_i}(\vartheta)\rangle m_t(\mathbf{X}_i|\vartheta) + \frac{\|\vartheta\|_2^2}{\sigma^2} - d\right)n_0(\vartheta),$$

where $\theta_0 \sim \mathcal{N}(0_d, \sigma^2 I_d)$ and $m_t(\cdot|\vartheta)$ is the conditional distribution of X_t when $\theta_0 = \vartheta$. Note that $m_t(\cdot|\vartheta)$ is the probability distribution of a discrete random variable, then $m_t(\mathbf{X}_i|\vartheta) \leq 1$, for any i.

The Lipschitz regularity of $\nabla_\theta U_{\mathbf{X}_i}(\theta)$ guarantees that there exists a positive polynomial q_2 of degree 2 such that:

$$|\mathcal{G}_t n_0(\vartheta)| \leq q_2(\vartheta) n_0(\vartheta).$$

The bound above and the Cauchy-Schwarz inequality lead to:

$$\begin{aligned}
|\partial_t\{n_t(\theta)\}| &\leq \int_{\mathbb{R}^d} q_2(\vartheta) n_0(\vartheta) p_{0,t}(\vartheta,\theta)\mathrm{d}\vartheta \\
&\leq \left(\int_{\mathbb{R}^d} q_2^2(\vartheta) n_0(\vartheta) p_{0,t}(\vartheta,\theta)\mathrm{d}\vartheta\right)^{\frac{1}{2}} \left(\int_{\mathbb{R}^d} n_0(\vartheta) p_{0,t}(\vartheta,\theta)\mathrm{d}\vartheta\right)^{\frac{1}{2}} \\
&\leq \sqrt{A_T n_t(\theta)},
\end{aligned}$$

where we used that the first integral is bounded by a constant A_T since $p_{0,t}(\vartheta,\theta)$ is bounded by a polynomial of $\|\vartheta\|_2$ and n_0 is the density function of a Gaussian distribution.

We have proven that for any $t \in [T^{-1}, T]$:

$$\begin{aligned}
\left|\partial_t\left\{\log\left(\frac{n_t(\theta)}{\mu_n(\theta)}\right) n_t(\theta)\right\}\right| &\leq (1+\log(\|h_0\|_\infty))\sqrt{A_T n_t(\theta)} \\
&\leq (1+\log(\|h_0\|_\infty))\sqrt{A_T\|h_0\|_\infty}\sqrt{\mu_n(\theta)} = g(\theta).
\end{aligned}$$

The integrability of the function $\theta \mapsto \sqrt{\mu_n(\theta)}$ is a consequence of Propositions 2.1.3 and 1.5.3, from which we deduce that U_{ν_n} satisfies a $\mathcal{H}^{\mathbf{r}}_{\mathbf{KL}}(\mathfrak{c}n^{1+r}, nL+\ell_0)$ condition, then for all $\theta \in \mathbb{R}^d$:

$$\sqrt{\mu_n(\theta)} \leq \mathcal{Z}_n^{-1/2} \exp\left\{-a_r\|\theta - \arg\min U_{\nu_n}\|^{2/(1+r)}\right\},$$

where a_r depends of r and n. The inequality above guarantees that $\int_{\mathbb{R}^d} g(\theta)\mathrm{d}\theta < \infty$.

After proving the exchange of derivative and integral sign when $t \in [T^{-1}, T]$, for any $T > 1$, we can conclude it for any $t > 0$. The second part of the statement is immediate from Proposition 2.C.1. □

2.D Evolution of the weighted $\mathbb{L}^2$ distance I_t

The quantity I_t defined in (2.16) measures how close to ν_n the conditional distribution of $X_t|\theta_t$ is. Then, to study I_t, we first note that it may be rewritten in

a simpler way.

$$\begin{aligned}
I_t &= \frac{1}{n}\int_{\mathbb{R}^d}\sum_{i=1}^n\left(\frac{m_t(\mathbf{X}_i|\theta)}{\nu_n(\mathbf{X}_i)}-1\right)^2 \mathrm{d}n_t(\theta)\\
&= \int_{\mathbb{R}^d}\sum_{i=1}^n\left(\frac{m_t^2(\mathbf{X}_i|\theta)}{\nu_n(\mathbf{X}_i)}-2m_t(\mathbf{X}_i|\theta)+\nu_n(\mathbf{X}_i)\right)\mathrm{d}n_t(\theta)\\
&= \int_{\mathbb{R}^d}\left(\sum_{i=1}^n\frac{m_t^2(\mathbf{X}_i|\theta)}{\nu_n(\mathbf{X}_i)}-1\right)\mathrm{d}n_t(\theta)\\
&= \int_{\mathbb{R}^d}\sum_{i=1}^n\frac{m_t^2(\mathbf{X}_i|\theta)}{\nu_n(\mathbf{X}_i)}\mathrm{d}n_t(\theta)-1.
\end{aligned}$$

Using that $m_t(\mathbf{X}_i|\theta)n_t(\theta)=m_t(\theta,\mathbf{X}_i)$ and $\nu_n(\mathbf{X}_i)=\frac{1}{n}$ for $i=1,2,\ldots,n$, we obtain that:

$$I_t = n\int_{\mathbb{R}^d}\sum_{i=1}^n\frac{m_t^2(\theta,\mathbf{X}_i)}{n_t(\theta)}\mathrm{d}\theta-1. \tag{2.25}$$

The next proposition then assesses how fast I_t decreases to 0 as $t\longrightarrow+\infty$.

Proposition 2.D.1. *Assume $\mathcal{H}_{\pi_0}(\ell_0)$ and for each $\mathbf{X}_i$, $\theta\to-\log p_\theta(\mathbf{X}_i)$ satisfies $\mathcal{H}^{\mathbf{r}}_{\mathbf{KL}}(\mathfrak{c},L)$. Then, for any initial distribution n_0 and $t\geq 0$:*

$$I_t\leq I_0e^{-2\alpha_n t}\leq(n-1)e^{-2\alpha_n t}. \tag{2.26}$$

Proof. Our starting point is Equation (2.25). The existence of $\partial_t\{I_t\}$ and the exchange of derivative and integral signs could be justified as in Section 2.C. We then compute its derivative with respect to t as follows:

$$\begin{aligned}
\partial_t\{I_t\} &= 2n\int_{\mathbb{R}^d}\sum_{i=1}^n\frac{m_t(\theta,\mathbf{X}_i)}{n_t(\theta)}\partial_t\{m_t(\theta,\mathbf{X}_i)\}\mathrm{d}\theta-n\int_{\mathbb{R}^d}\sum_{i=1}^n\frac{m_t^2(\theta,\mathbf{X}_i)}{n_t^2(\theta)}\partial_t\{n_t(\theta)\}\mathrm{d}\theta\\
&= 2n\int_{\mathbb{R}^d}\sum_{i=1}^n m_t(\mathbf{X}_i|\theta)\partial_t\{m_t(\theta,\mathbf{X}_i)\}\mathrm{d}\theta-n\int_{\mathbb{R}^d}\sum_{i=1}^n m_t^2(\mathbf{X}_i|\theta)\partial_t\{n_t(\theta)\}\mathrm{d}\theta.
\end{aligned}$$

Using the Kolmogorov forward equation in the first line and $\mathcal{L}=\mathcal{L}_1+\mathcal{L}_2$ in the second one where $\mathcal{L}_1$ and $\mathcal{L}_2$ are defined in Equation (2.7), we have that:

$$\begin{aligned}
\partial_t\{I_t\} &= 2n\int_{\mathbb{R}^d}\sum_{i=1}^n\mathcal{L}m_t(\mathbf{X}_i|\theta)\ m_t(\theta,\mathbf{X}_i)\mathrm{d}\theta-n\int_{\mathbb{R}^d}\sum_{i=1}^n m_t^2(\mathbf{X}_i|\theta)\partial_t\{n_t(\theta)\}\mathrm{d}\theta\\
&= \underbrace{2n\int_{\mathbb{R}^d}\sum_{i=1}^n\mathcal{L}_1m_t(\mathbf{X}_i|\theta)\ m_t(\theta,\mathbf{X}_i)\mathrm{d}\theta}_{I_{3,t}}+\underbrace{2n\int_{\mathbb{R}^d}\sum_{i=1}^n\mathcal{L}_2m_t(\mathbf{X}_i|\theta)\ m_t(\theta,\mathbf{X}_i)\mathrm{d}\theta}_{I_{1,t}}\\
&\quad\underbrace{-n\int_{\mathbb{R}^d}\sum_{i=1}^n m_t^2(\mathbf{X}_i|\theta)\partial_t\{n_t(\theta)\}\mathrm{d}\theta}_{I_{2,t}}.
\end{aligned} \tag{2.27}$$

Then, $\partial_t\{I_t\}$ may be splitted into three terms that are studied separately.

- Study of $I_{1,t}$. We observe that for any i and $\theta \in \mathbb{R}^d$:

$$\mathcal{L}_2 m_t(\mathbf{X}_i|\theta) = \frac{\alpha_n}{n}\sum_{j=1}^{n}[m_t(\mathbf{X}_j|\theta) - m_t(\mathbf{X}_i|\theta)] = \frac{\alpha_n}{n} - \alpha_n m_t(\mathbf{X}_i|\theta). \tag{2.28}$$

We then use this last equation in the definition of $I_{1,t}$ and obtain that:

$$\begin{aligned} I_{1,t} &= 2\alpha_n \int_{\mathbb{R}^d} \sum_{i=1}^{n} m_t(\theta, \mathbf{X}_i)\mathrm{d}\theta - 2\alpha_n n \int_{\mathbb{R}^d} \sum_{i=1}^{n} m_t(\mathbf{X}_i|\theta) m_t(\theta, \mathbf{X}_i)\mathrm{d}\theta \\ &= 2\alpha_n - 2\alpha_n n \int_{\mathbb{R}^d} \sum_{i=1}^{n} \frac{m_t^2(\theta, \mathbf{X}_i)}{n_t(\theta)}\mathrm{d}\theta \\ &= -2\alpha_n I_t. \end{aligned} \tag{2.29}$$

- Study of $I_{2,t}$. Using the definition of n_t, we get that:

$$\begin{aligned} I_{2,t} &= -n \int_{\mathbb{R}^d} \sum_{i=1}^{n} m_t^2(\mathbf{X}_i|\theta)\partial_t\{n_t(\theta)\}\mathrm{d}\theta \\ &= -n \int_{\mathbb{R}^d} \sum_{i=1}^{n} m_t^2(\mathbf{X}_i|\theta)\partial_t \left\{\sum_{j=1}^{n} m_t(\theta, \mathbf{X}_j)\right\}\mathrm{d}\theta \\ &= -n \int_{\mathbb{R}^d} \sum_{j=1}^{n}\sum_{i=1}^{n} m_t^2(\mathbf{X}_i|\theta)\partial_t\{m_t(\theta, \mathbf{X}_j)\}\mathrm{d}\theta \\ &= -n \int_{\mathbb{R}^d} \sum_{j=1}^{n} \left(\sum_{i=1}^{n} \mathcal{L} m_t^2(\mathbf{X}_i|\theta)\right) m_t(\theta, \mathbf{X}_j)\mathrm{d}\theta \\ &= -n \int_{\mathbb{R}^d} \sum_{i=1}^{n} \mathcal{L} m_t^2(\mathbf{X}_i|\theta)\ \mathrm{d}n_t(\theta). \end{aligned}$$

where we used the Kolmogorov forward equation in the fourth line and again the definition of n_t in the last line. Again, the decomposition $\mathcal{L} = \mathcal{L}_1 + \mathcal{L}_2$ yields:

$$I_{2,t} = -n \int_{\mathbb{R}^d} \sum_{i=1}^{n} \mathcal{L}_1 m_t^2(\mathbf{X}_i|\theta)\ \mathrm{d}n_t(\theta) - n \int_{\mathbb{R}^d} \sum_{i=1}^{n} \mathcal{L}_2 m_t^2(\mathbf{X}_i|\theta)\ \mathrm{d}n_t(\theta).$$

We repeat some similar computations as those developed in Equation (2.28) to study the action of the jump component induced by $\mathcal{L}_2$ on $m_t^2(\mathbf{X}_i|\theta)$. Then, we deduce that:

$$\mathcal{L}_2 m_t^2(\mathbf{X}_i|\theta) = \frac{\alpha_n}{n}\sum_{k=1}^{n} m_t^2(\mathbf{X}_k|\theta) - \alpha_n m_t^2(\mathbf{X}_i|\theta).$$

We use this last equation to show that:

$$\begin{aligned}
I_{2,t} &= -n\int_{\mathbb{R}^d}\sum_{i=1}^n \mathcal{L}_1 m_t^2(\mathbf{X}_i|\theta)\ \mathrm{d}n_t(\theta) - \alpha_n\int_{\mathbb{R}^d}\sum_{i=1}^n\sum_{k=1}^n m_t^2(\mathbf{X}_k|\theta)\ \mathrm{d}n_t(\theta)\\
&\quad +\alpha_n n\int_{\mathbb{R}^d}\sum_{i=1}^n m_t^2(\mathbf{X}_i|\theta)\ \mathrm{d}n_t(\theta)\\
&= -n\int_{\mathbb{R}^d}\sum_{i=1}^n \mathcal{L}_1 m_t^2(\mathbf{X}_i|\theta)\ \mathrm{d}n_t(\theta) - \alpha_n n\int_{\mathbb{R}^d}\sum_{k=1}^n m_t^2(\mathbf{X}_k|\theta)\ \mathrm{d}n_t(\theta)\\
&\quad +\alpha_n n\int_{\mathbb{R}^d}\sum_{i=1}^n m_t^2(\mathbf{X}_i|\theta)\ \mathrm{d}n_t(\theta)\\
&= -n\int_{\mathbb{R}^d}\sum_{i=1}^n \mathcal{L}_1 m_t^2(\mathbf{X}_i|\theta)\ \mathrm{d}n_t(\theta).
\end{aligned}$$

- Study of $I_{2,t}+I_{3,t}$. We observe that this sum involves only $\mathcal{L}_1$, which was defined in Equation (2.7). We first compute:

$$\mathcal{L}_1 m_t(\mathbf{X}_i|\theta) = -\langle\nabla_\theta U_{\mathbf{X}_i}(\theta), \nabla_\theta m_t(\mathbf{X}_i|\theta)\rangle + \Delta_\theta m_t(\mathbf{X}_i|\theta),$$

Then:

$$\begin{aligned}
I_{3,t} &= 2n\int_{\mathbb{R}^d}\sum_{i=1}^n \mathcal{L}_1 m_t(\mathbf{X}_i|\theta)\ m_t(\theta,\mathbf{X}_i)\mathrm{d}\theta\\
&= 2n\int_{\mathbb{R}^d}\sum_{i=1}^n \left[-\langle\nabla_\theta U_{\mathbf{X}_i}(\theta), \nabla_\theta m_t(\mathbf{X}_i|\theta)\rangle + \Delta_\theta m_t(\mathbf{X}_i|\theta)\right] m_t(\theta,\mathbf{X}_i)\mathrm{d}\theta.
\end{aligned}$$

Similarly, we deduce that:

$$\begin{aligned}
\mathcal{L}_1 m_t^2(\mathbf{X}_i|\theta) =& -\langle\nabla_\theta U_{\mathbf{X}_i}(\theta), \nabla_\theta m_t^2(\mathbf{X}_i|\theta),\rangle + \Delta_\theta m_t^2(\mathbf{X}_i|\theta)\\
=& 2m_t(\mathbf{X}_i|\theta)\left(-\langle\nabla_\theta U_{\mathbf{X}_i}(\theta), \nabla_\theta m_t(\mathbf{X}_i|\theta)\rangle + \Delta_\theta m_t(\mathbf{X}_i|\theta)\right)\\
&+ 2\|\nabla_\theta m_t(\mathbf{X}_i|\theta)\|_2^2.
\end{aligned}$$

Using that $m_t(\mathbf{X}_i|\theta)n_t(\theta) = m_t(\theta,\mathbf{X}_i)$, we get that:

$$\begin{aligned}
I_{2,t} &= -n\int_{\mathbb{R}^d}\sum_{i=1}^n \mathcal{L}_1 m_t^2(\mathbf{X}_i|\theta)\ \mathrm{d}n_t(\theta)\\
&= -2n\int_{\mathbb{R}^d}\sum_{i=1}^n \left[-\langle\nabla_\theta U_{\mathbf{X}_i}(\theta), \nabla_\theta m_t(\mathbf{X}_i|\theta)\rangle + \Delta_\theta m_t(\mathbf{X}_i|\theta)\right] m_t(\theta,\mathbf{X}_i)\mathrm{d}\theta\\
&- 2n\int_{\mathbb{R}^d}\sum_{i=1}^n \|\nabla_\theta m_t(\mathbf{X}_i|\theta)\|_2^2\ \mathrm{d}n_t(\theta).
\end{aligned}$$

It is immediate that:

$$I_{2,t}+I_{3,t} = -2n\int_{\mathbb{R}^d}\sum_{i=1}^n \|\nabla_\theta m_t(\mathbf{X}_i|\theta)\|_2^2\ \mathrm{d}n_t(\theta) \le 0.$$

Gathering this last inequality with (2.29) into Equation (2.27) yields:

$$\partial_t\{I_t\} \leq -2\alpha_n I_t.$$

We conclude with a direct application of the Gronwall lemma while observing that $I_0 \leq n-1$. □

2.E Moments upper bounds

Proposition 2.E.1. *Assume $\mathcal{H}_{\mathbf{n_0}}(L,\ell_0)$, $\mathcal{H}_{\pi_0}(\ell_0)$, $\mathcal{H}^{\beta}_{\min}$ and that for each $\mathbf{X}_i$, $\theta \mapsto -\log p_\theta(\mathbf{X}_i)$ satisfies $\mathcal{H}^{\mathbf{r}}_{\mathbf{KL}}(\mathfrak{c},L)$. Then:*

a) Three positive constants C_1, C_2 and C_3, independent from n and d, exist such that for any $t>0$:

$$\mathbb{E}_{n_t}\left[e^{\frac{(1+r)n\mathfrak{c}}{16}\frac{1}{1+r}(\|\theta_t\|_2^2+1)^{\frac{1}{1+r}}}\right] \leq C_1\left(d\log^{2\beta}(n)\right)^r e^{C_2 nd\log^{2\beta}(n)} + C_3^d e^{\frac{(1+r)n\mathfrak{c}}{16}\frac{1}{1+r}}.$$

b) For any $t>0$ and for any $\alpha \geq 1$:

$$\mathbb{E}_{n_t}[U^{\alpha}_{\nu_n}(\theta_t)] \lesssim_{uc} n^{\alpha}\left(d\log^{2\beta}(n)\right)^{\alpha(1+r)}.$$

Proof of a). The proof is based on a Lyapunov argument. Consider the twice differentiable function:

$$f(\theta) = \exp\left(\frac{a}{2}(\|\theta\|_2^2+1)^{\rho}\right), \quad \theta \in \mathbb{R}^d,$$

where $0<\rho<1$ and $a>0$ are two constants to fix later on. For any $\theta \in \mathbb{R}^d$, the gradient of f is:

$$\nabla f(\theta) = a\rho(\|\theta\|_2^2+1)^{\rho-1}f(\theta)\theta,$$

and the Laplacian of f satisfies the following inequality:

$$\begin{aligned}\Delta f(\theta) &= a\rho(\|\theta\|_2^2+1)^{\rho-2}f(\theta)\left[a\rho(\|\theta\|_2^2+1)^{\rho}\|\theta\|_2^2 + d(\|\theta\|_2^2+1) + 2(\rho-1)\|\theta\|_2^2\right]\\ &\leq a\rho(\|\theta\|_2^2+1)^{\rho-1}f(\theta)\left[a\rho(\|\theta\|_2^2+1)^{\rho}+d\right],\end{aligned}$$

where we used that $0<\rho<1$ and $\|\theta\|_2^2 \leq \|\theta\|_2^2+1$.

We then deduce that for any $t > 0$ and $\theta \in \mathbb{R}^d$:

$$\begin{aligned}\frac{\mathcal{G}_t f(\theta)}{f(\theta)} &= \frac{1}{f(\theta)}\left[-\sum_{i=1}^n \langle \nabla U_{\mathbf{X}_i}, \nabla f(\theta)\rangle m_t(\mathbf{X}_i|\theta) + \Delta f(\theta)\right] \\ &\leq a\rho(\|\theta\|_2^2+1)^{\rho-1}\Big[-\sum_{i=1}^n \langle \theta, \nabla_\theta U_{\mathbf{X}_i}(\theta)\rangle m_t(\mathbf{X}_i|\theta) + a\rho(\|\theta\|_2^2+1)^\rho + d\Big] \\ &\leq a\rho(\|\theta\|_2^2+1)^{\rho-1}\left[-\sum_{i=1}^n \left(U_{\mathbf{X}_i}(\theta) - U_{\mathbf{X}_i}(0)\right) m_t(\mathbf{X}_i|\theta) + a\rho(\|\theta\|_2^2+1)^\rho + d\right],\end{aligned} \tag{2.30}$$

where we used the convexity of U_x for any position x.

First considerations: For any i, we denote by $\theta_i = \arg\min U_{\mathbf{X}_i}$ and from hypothesis $\mathcal{H}^\beta_{\min}$, there exist two positive constants $\mathcal{K}_1$ and $\mathcal{K}_2$ independent of n and d such that:

$$\max_i \|\theta_i\|_2^2 \leq \mathcal{K}_1 d \log^{2\beta}(n) \quad \text{and} \quad \max_i U_{\mathbf{X}_i}(\theta_i) \leq \mathcal{K}_2 d \log^{2\beta}(n).$$

So, in order to lower bound the term $\sum_{i=1}^n \left(U_{\mathbf{X}_i}(\theta) - U_{\mathbf{X}_i}(0)\right) m_t(\mathbf{X}_i|\theta)$, let us establish the bounds of $U_{\mathbf{X}_i}(\theta)$ and $U_{\mathbf{X}_i}(0)$ separately.

• In Proposition 2.1.3 we proved that each non-negative function $U_{\mathbf{X}_i}$ satisfies a $\mathcal{H}^{\mathbf{r}}_{\mathbf{KL}}(\mathfrak{c}n^{1+r}, nL+\ell_0)$-condition, then we are able to apply Proposition 1.5.3 and obtain that for any $U_{\mathbf{X}_i}$:

$$U_{\mathbf{X}_i}(\theta) \geq n\left[\frac{(1+r)\mathfrak{c}}{2}\right]^{\frac{1}{1+r}} \|\theta-\theta_i\|_2^{\frac{2}{1+r}}.$$

Since $\frac{2}{1+r} > 1$, the Jensen inequality yields $(u+v)^{\frac{2}{1+r}} \leq 2^{\frac{1-r}{1+r}}\left[u^{\frac{2}{1+r}} + v^{\frac{2}{1+r}}\right]$, for all $(u,v) \in \mathbb{R}_+^2$ and we deduce that for any $\theta \in \mathbb{R}^d$:

$$\begin{aligned}\|\theta-\theta_i\|_2^{\frac{2}{1+r}} &\geq 2^{\frac{r-1}{1+r}}\|\theta\|_2^{\frac{2}{1+r}} - \|\theta_i\|_2^{\frac{2}{1+r}} \\ &\geq 2^{\frac{r-1}{1+r}}\|\theta\|_2^{\frac{2}{1+r}} - \left(\mathcal{K}_1 d\log^{2\beta}(n)\right)^{\frac{1}{1+r}}.\end{aligned}$$

Then we use this inequality to obtain a lower bound of $U_{\mathbf{X}_i}$:

$$\begin{aligned}U_{\mathbf{X}_i}(\theta) &\geq 2n\left[\frac{(1+r)\mathfrak{c}}{8}\right]^{\frac{1}{1+r}}\|\theta\|_2^{\frac{2}{1+r}} - n\left[\frac{(1+r)\mathfrak{c}}{2}\right]^{\frac{1}{1+r}}(\mathcal{K}_1 d\log^{2\beta}(n))^{\frac{1}{1+r}} \\ &\geq \frac{n\mathfrak{c}^{\frac{1}{1+r}}}{4}\|\theta\|_2^{\frac{2}{1+r}} - n\mathfrak{c}^{\frac{1}{1+r}}(\mathcal{K}_1 d\log^{2\beta}(n))^{\frac{1}{1+r}},\end{aligned}$$

where we used some uniform upper bounds when $r \in [0,1)$.

• An upper bound of $\max_i U_{\mathbf{X}_i}(0)$ comes from Proposition 2.1.3 and 1.5.3 as follows:

$$\begin{aligned} U_{\mathbf{X}_i}(0) &\leq U_{\mathbf{X}_i}(\theta_i) + \frac{nL+\ell_0}{2}\|\theta_i\|_2^2 \\ &\leq \left(\mathcal{K}_2 + \frac{\mathcal{K}_1(nL+\ell_0)}{2}\right) d\log^{2\beta}(n) \\ &\leq \mathcal{K} nd\log^{2\beta}(n), \end{aligned}$$

where $\mathcal{K}$ is a constant independent of n and d and could proportionally change from line to line.

Using the previous bounds and the fact that $\sum_{i=1}^n m_t(\mathbf{X}_i|\theta) = 1$, it yields:

$$\sum_{i=1}^n \left(U_{\mathbf{X}_i}(\theta) - U_{\mathbf{X}_i}(0)\right) m_t(\mathbf{X}_i|\theta) \geq \frac{n\mathfrak{c}^{\frac{1}{1+r}}}{4}\|\theta\|_2^{\frac{2}{1+r}} - \mathcal{K}nd\log^{2\beta}(n).$$

We omit the term $n\mathfrak{c}^{\frac{1}{1+r}}(\mathcal{K}_1 d\log^{2\beta}(n))^{\frac{1}{1+r}}$ since it could be bounded by $\mathcal{K}nd\log^{2\beta}(n)$ if we proportionally change the value of $\mathcal{K}$. Returning to the inequality (2.30), we now get

$$\frac{\mathcal{G}_t f(\theta)}{f(\theta)} \leq a\rho(\|\theta\|_2^2+1)^{\rho-1}\left[-\frac{n\mathfrak{c}^{\frac{1}{1+r}}}{4}\|\theta\|_2^{\frac{2}{1+r}} + \mathcal{K}nd\log^{2\beta}(n) + a\rho(\|\theta\|_2^2+1)^{\rho}\right].$$

Values of a and ρ: We choose $\rho = \frac{1}{1+r}$ and deduce that:

$$\frac{\mathcal{G}_t f(\theta)}{f(\theta)} \leq \frac{a}{1+r}(\|\theta\|_2^2+1)^{-\frac{r}{1+r}}\left[-\frac{n\mathfrak{c}^{\frac{1}{1+r}}}{4}\|\theta\|_2^{\frac{2}{1+r}} + \mathcal{K}nd\log^{2\beta}(n) + \frac{a}{1+r}(\|\theta\|_2^2+1)^{\frac{1}{1+r}}\right],$$

using the inequality $(\|\theta\|_2^2+1)^{\frac{1}{1+r}} \leq \|\theta\|_2^{\frac{2}{1+r}} + 1$, then:

$$\frac{\mathcal{G}_t f(\theta)}{f(\theta)} \leq \frac{a}{1+r}(\|\theta\|_2^2+1)^{-\frac{r}{1+r}}\left[-\left(\frac{n\mathfrak{c}^{\frac{1}{1+r}}}{4} - \frac{a}{1+r}\right)\|\theta\|_2^{\frac{2}{1+r}} + \mathcal{K}nd\log^{2\beta}(n)\right].$$

Let us fix $a = \frac{n(1+r)\mathfrak{c}^{\frac{1}{1+r}}}{8}$, then for any $t > 0$ and $\theta \in \mathbb{R}^d$,

$$\frac{\mathcal{G}_t f(\theta)}{f(\theta)} \leq -\frac{n^2\mathfrak{c}^{\frac{2}{1+r}}}{64}(\|\theta\|_2^2+1)^{-\frac{r}{1+r}}\left[\|\theta\|_2^{\frac{2}{1+r}} - \mathcal{K}d\log^{2\beta}(n)\right]. \tag{2.31}$$

Lyapunov contraction: We study two complementary situations, and below we denote by $K_{n,d}$ the radius of the key compact set involved by the previous Lyapunov contraction:

$$K_{n,d}^{\frac{2}{1+r}} = Cd\log^{2\beta}(n).$$

• When $\|\theta\|_2$ is large enough ($\|\theta\|_2 \geq K_{n,d}$), we use the fact that for any two fixed constants $k \geq 0$ and $0 \leq r < 1$, the function $x \mapsto -(x+1)^{-\frac{r}{1+r}}(x^{\frac{1}{1+r}} - k)$, for $x \geq k^{1+r}$, is decreasing and we observe that a large enough $C > 0$ independent of n

and d exists such that:

$$\begin{aligned}\|\theta\|_2^{\frac{2}{1+r}} \geq Cd\log^{2\beta}(n) &\Longrightarrow \frac{\mathcal{G}_t f(\theta)}{f(\theta)} \leq -\frac{n^2\mathfrak{c}^{\frac{2}{1+r}}}{64}\frac{(C-\mathcal{K})}{2C^r}(d\log^{2\beta}(n))^{1-r} \\ &\Longrightarrow \frac{\mathcal{G}_t f(\theta)}{f(\theta)} \leq -\frac{n^2\mathfrak{c}^{\frac{2}{1+r}}}{128}(d\log^{2\beta}(n))^{1-r} = -a_{n,d}. \qquad (2.32)\end{aligned}$$

• When $\|\theta\|_2$ is upper bounded ($\|\theta\|_2 \leq K_{n,d}$), we use the upper bound stated in Equation (2.31) and obtain that a universal C_1 (whose value may change from line to line) exists such that :

$$\begin{aligned}\|\theta\|_2^{\frac{2}{1+r}} \leq Cd\log^{2\beta}(n) &\Longrightarrow \mathcal{G}_t f(\theta) \leq C_1 n^2 d\log^{2\beta}(n) f(\theta) \\ &\Longrightarrow \mathcal{G}_t f(\theta) \leq C_1 n^2 d\log^{2\beta}(n)\exp\left(C\mathfrak{c}^{\frac{1}{1+r}} nd\log^{2\beta}(n)\right) \\ &\Longrightarrow \mathcal{G}_t f(\theta) \leq b_{n,d}e^{\delta_{n,d}}, \qquad (2.33)\end{aligned}$$

where we denoted $b_{n,d} = C_1 n^2 d\log^{2\beta}(n)$ and $\delta_{n,d} = C\mathfrak{c}^{\frac{1}{1+r}} nd\log^{2\beta}(n)$.

We then use Equations (2.32) and (2.33) as follows. We define the function $\psi_{n,d}$ as $\psi_{n,d}(t) = \mathbb{E}_{n_t}[f(\theta_t)]$. The existence of $\partial_t\{\psi_{n,d}(t)\}$ and the following equality are justified as in Section 2.C:

$$\begin{aligned}\partial_t\{\psi_{n,d}(t)\} &= \mathbb{E}_{n_t}[\mathcal{G}_t f(\theta_t)] = \mathbb{E}_{n_t}\left[\mathcal{G}_t f(\theta_t)\left(\mathbb{1}_{\|\theta_t\|_2\geq K_{n,d}} + \mathbb{1}_{\|\theta_t\|_2\leq K_{n,d}}\right)\right] \\ &\leq \mathbb{E}_{n_t}\left[-a_{n,d}f(\theta_t)\mathbb{1}_{\|\theta_t\|_2\geq K_{n,d}} + b_{n,d}e^{\delta_{n,d}}\mathbb{1}_{\|\theta_t\|_2\leq K_{n,d}}\right] \\ &\leq -a_{n,d}\psi_{n,d}(t) + a_{n,d}\sup_{\|\theta\|_2\leq K_{n,d}} f(\theta) + b_{n,d}e^{\delta_{n,d}} \\ &\leq -a_{n,d}\psi_{n,d}(t) + (a_{n,d}+b_{n,d})e^{\delta_{n,d}}.\end{aligned}$$

We apply the Gronwall Lemma and obtain that:

$$\forall t > 0, \qquad \psi_{n,d}(t) \leq \left(1 + \frac{b_{n,d}}{a_{n,d}}\right)e^{\delta_{n,d}} + \psi_{n,d}(0)e^{-a_{n,d}t}. \qquad (2.34)$$

From the $\mathcal{H}_{\mathbf{n_0}}(L,\ell_0)$ hypothesis, n_0 is a Gaussian distribution, the we find an upper bound of $\psi_{n,d}(0) = \mathbb{E}_{n_0}[f(\theta_0)] = \int_{\mathbb{R}^d} f(\theta)\mathrm{d}n_0(\theta)$ as follows:

$$\begin{aligned}\psi_{n,d}(0) &= \left(2\pi\sigma^2\right)^{-\frac{d}{2}}\int_{\mathbb{R}^d} e^{\frac{a}{2}\left(\|\theta\|_2^2+1\right)^{\frac{1}{1+r}} - \frac{\|\theta\|_2^2}{2\sigma^2}}\mathrm{d}\theta \\ &\leq \left(2\pi\sigma^2\right)^{-\frac{d}{2}} e^{\frac{a}{2}}\int_{\mathbb{R}^d} e^{-\frac{\|\theta\|_2^2}{2}\left(\frac{1}{\sigma^2}-a\right)}\mathrm{d}\theta,\end{aligned}$$

if $\sigma^2 < \frac{1}{a} = \frac{8}{n(1+r)\mathfrak{c}^{\frac{1}{1+r}}}$ then the integral above is finite. From Remark 2.1.1, we verify that $c_2 < 1 \leq \frac{8L}{(1+r)\mathfrak{c}^{\frac{1}{1+r}}}$, which guarantees that $\sigma^2 < \frac{1}{a}$ and:

$$\psi_{n,d}(0) \leq \left(1 - a\sigma^2\right)^{-\frac{d}{2}} e^{\frac{a}{2}} \leq C_3^d e^{\frac{(1+r)n\mathfrak{c}^{\frac{1}{1+r}}}{16}},$$

where C_3 is a constant independent of n and d.

Finally, using the value of $a_{n,d}$ and $b_{n,d}$ in (2.34), we deduce that for any $t > 0$,

$$\mathbb{E}_{n_t}\left[e^{\frac{(1+r)n\mathfrak{c}^{\frac{1}{1+r}}}{16}(\|\theta_t\|_2^2+1)^{\frac{1}{1+r}}}\right] \leq C_1\left(d\log^{2\beta}(n)\right)^r e^{C_2 nd\log^{2\beta}(n)} + C_3^d e^{\frac{(1+r)n\mathfrak{c}^{\frac{1}{1+r}}}{16}}.$$

where C_2 is another universal constant, which concludes the proof. □

Proof of b). We consider $\alpha > 1$ and below, $C > 0$ refers to a constant independent from n and d, whose value may change from line to line. Our starting point is the upper bound of the exponential moments obtained in $i)$. Proposition 2.1.3 shows that U_{ν_n} satisfies $\mathcal{H}^{\mathbf{r}}_{\mathbf{KL}}(\mathfrak{c}n^{1+r}, nL + \ell_0)$, then thanks to Proposition 1.5.3:

$$\begin{aligned}\mathbb{E}_{n_t}[U^\alpha_{\nu_n}(\theta_t)] &\leq \mathbb{E}_{n_t}\left[\left(U_{\nu_n}(\theta_n^*) + Cn\|\theta_t - \theta_n^*\|_2^2\right)^\alpha\right]\\ &\leq \mathbb{E}_{n_t}\left[\left(U_{\nu_n}(\theta_n^*) + Cn\|\theta_n^*\|_2^2 + Cn\|\theta_t\|_2^2\right)^\alpha\right],\end{aligned}$$

where $\theta_n^* = \arg\min U_{\nu_n}$.

By using Proposition 2.A.1 and the inequality derived from the Jensen inequality $(a+b)^\alpha \leq c_\alpha(a^\alpha + b^\alpha)$ for $(a,b) \in \mathbb{R}_+^2$ and $\alpha \geq 1$, we obtain that:

$$\begin{aligned}\mathbb{E}_{n_t}[U^\alpha_{\nu_n}(\theta_t)] &\leq C\mathbb{E}_{n_t}\left[\left(nd\log^{2\beta}(n) + n\left(d\log^{2\beta}(n)\right)^{1+r} + n\|\theta_t\|_2^2\right)^\alpha\right]\\ &\leq Cn^\alpha\left[\left(d\log^{2\beta}(n)\right)^{\alpha(1+r)} + \mathbb{E}_{n_t}\left(\|\theta_t\|_2^{2\alpha}\right)\right]. \qquad (2.35)\end{aligned}$$

Let us focus on the moment of order 2α. It could be rewritten as:

$$\begin{aligned}\mathbb{E}_{n_t}\left(\|\theta_t\|_2^{2\alpha}\right) &= k^{-\alpha(1+r)}\mathbb{E}_{n_t}\left(\log^{\alpha(1+r)}\left(e^{k\|\theta_t\|_2^{\frac{2}{1+r}}}\right)\right)\\ &\leq k^{-\alpha(1+r)}\mathbb{E}_{n_t}\left(\log^{\alpha(1+r)}\left(e^{\alpha(1+r)-1+k\|\theta_t\|_2^{\frac{2}{1+r}}}\right)\right).\end{aligned}$$

The Jensen inequality and the concavity of $x \mapsto \log^p(x)$ on $[e^{p-1}, +\infty[$ when $p \geq 1$ yield:

$$\begin{aligned}\mathbb{E}_{n_t}\left(\|\theta_t\|_2^{2\alpha}\right) &\leq k^{-\alpha(1+r)}\log^{\alpha(1+r)}\left(\mathbb{E}_{n_t}\left(e^{\alpha(1+r)-1+k\|\theta_t\|_2^{\frac{2}{1+r}}}\right)\right)\\ &\leq k^{-\alpha(1+r)}\left[\alpha(1+r) - 1 + \log\left(\mathbb{E}_{n_t}\left(e^{k\|\theta_t\|_2^{\frac{2}{1+r}}}\right)\right)\right]^{\alpha(1+r)}\\ &\leq k^{-\alpha(1+r)}\left[\alpha(1+r) - 1 + \log\left(\mathbb{E}_{n_t}\left(e^{k(\|\theta_t\|_2^2+1)^{\frac{1}{1+r}}}\right)\right)\right]^{\alpha(1+r)},\end{aligned}$$

where we used in the last inequality that $\|\theta\|_2^2 \leq \|\theta\|_2^2 + 1$.

We then apply $i)$ in Proposition 2.E.1, we choose $k = \frac{(1+r)n\mathfrak{c}^{\frac{1}{1+r}}}{16}$ and obtain that:

$$\begin{aligned}
\mathbb{E}_{n_t}\left(\|\theta_t\|_2^{2\alpha}\right) &\leq \frac{C}{n^{\alpha(1+r)}}\left[1+\log\left(\mathbb{E}_{n_t}\left(e^{\frac{(1+r)n\mathfrak{c}^{\frac{1}{1+r}}}{16}(\|\theta_t\|_2^2+1)^{\frac{1}{1+r}}}\right)\right)\right]^{\alpha(1+r)} \\
&\leq \frac{C}{n^{\alpha(1+r)}}\left[1+\log\left[C_1\left(d\log^{2\beta}(n)\right)^r e^{C_2 nd\log^{2\beta}(n)} + C_3^d e^{\frac{(1+r)n\mathfrak{c}^{\frac{1}{1+r}}}{16}}\right]\right]^{\alpha(1+r)} \\
&\leq C\left(d\log^{2\beta}(n)\right)^{\alpha(1+r)}, \qquad (2.36)
\end{aligned}$$

where we used in the previous lines simple algebra and $\log(a+b) \leq \log(2)+\log(a)+\log(b)$ when $a \geq 1$ and $b \geq 1$. Replacing (2.36) in (2.35), we conclude that:

$$\mathbb{E}_{n_t}[U^{\alpha}_{\nu_n}(\theta_t)] \leq Cn^{\alpha}\left(d\log^{2\beta}(n)\right)^{\alpha(1+r)}.$$

□

Chapter 3

Discretization of Langevin Diffusion

3.1 Introduction

This chapter is devoted to the study of the Unadjusted Langevin Algorithm to sample a probability distribution over $\mathbb{R}^d$. This probability measure is absolutely continuous with respect to the Lebesgue measure and its density function can be written as

$$\mu(\theta) = \frac{e^{-U(\theta)}}{\mathcal{Z}}, \theta \in \mathbb{R}^d, \quad \text{with} \quad \mathcal{Z} = \int_{\mathbb{R}^d} e^{-U(\theta)} \mathrm{d}\theta, \tag{3.1}$$

where $U : \mathbb{R}^d \to \mathbb{R}$ is a $\mathcal{C}^2(\mathbb{R}^d)$ convex function and ∇U is $L-$Lipschitz. To solve the sampling problem raised by (3.1), a popular alternative is the over-damped Langevin diffusion associated with U

$$\mathrm{d}\vartheta_t = -\nabla U(\vartheta_t)\mathrm{d}t + \sqrt{2}\mathrm{d}B_t, \quad t \geq 0, \tag{3.2}$$

where $(B_t)_{t\geq 0}$ is a $d-$dimensional standard Brownian Motion. Standard results on Markov processes guarantee the strong existence and ergodicity of the continuous-time process $(\vartheta_t)_{t\geq 0}$ that converges in many senses (for example: Wasserstein, $\mathbb{L}^2_\mu(\mathbb{R}^d)$, entropy, total variation) towards its unique invariant distribution μ (see [Pavliotis, 2016]). However, in most cases, it is not possible to simulate an exact solution of (3.2), and then a discretization of (3.2) is used instead. In addition, numerical solutions associated with these discretizations define Markov kernels for which μ is no longer invariant. Therefore, quantifying the error introduced by these approximations is crucial to justify their use to sample from μ.

The most popular approach is certainly the first-order Euler-Maruyama scheme, known as the Unadjusted Langevin Algorithm (ULA, for short). With a fixed step size $h > 0$, ULA is defined as

$$\theta_{(k+1)h} = \theta_{kh} - h\nabla U(\theta_{kh}) + \sqrt{2h}\xi_{k+1}, \quad k \geq 0, \tag{3.3}$$

where $(\xi_k)_{k\geq 1}$ are standard Gaussian random variables in $\mathbb{R}^d$, mutually independent and independent of θ_0.

In this chapter, we assume that U is a weakly convex function that satisfies a

Kurdyka-Łojasiewicz hypothesis. We will assess our convergence results with the help of the relative entropy (or Kullback-Leibler divergence) of the distribution of θ_{kh} with respect to the target μ. Our framework is therefore similar to some extent to the recent contributions [Erdogdu and Hosseinzadeh, 2021, Chewi et al., 2022a].

We consider the probability distribution of θ_{kh}, thanks to the explicit Euler-Maruyama scheme (3.3) that involves an elliptic Gaussian convolution, for any initial distribution m_0, the law of θ_{kh} is absolutely continuous with respect to the Lebesgue measure, and we denote by m_k the associated density function, which is indeed infinitely differentiable as soon as $k \geq 1$. We then define the relative entropy of m_k with respect to the target measure μ as follows

$$J_k = \int_{\mathbb{R}^d} \log\left(\frac{m_k(\theta)}{\mu(\theta)}\right) \mathrm{d}m_k(\theta). \tag{3.4}$$

At each iteration k, J_k measures the divergence between the instantaneous law of the process at time kh and the (presumably) invariant distribution μ: a small value of J_k induces the closeness of m_k to μ. Of course, since we are studying the *Unadjusted* Langevin Algorithm, for a fixed value of h, m_k corresponds to the distribution of a Markov chain with an invariant measure μ_h and $\mu_h \neq \mu$. Consequently, and contrary to the MALA modification (see, for example, [Roberts and Rosenthal, 1998]), we cannot expect that $J_k \longrightarrow 0$ when $k \longrightarrow \infty$ without any fine-tuning of the parameter h, which must be chosen carefully to balance the bias and the ergodicity of the chain.

In what follows, for a given small ε, we will carefully address the choice of the step size h_ε, and of the simulation horizon k_ε to guarantee an ε smallness of J_k within our framework of weak KL convexity. In particular, we will focus on the impact of d on the computational cost, as well as the one of r involved in $\mathcal{H}^{\mathbf{r}}_{\mathbf{KL}}(\mathfrak{c}, L)$.

Structure of the chapter: In Section 3.2, the hypotheses and main convergence results are presented and compared with those previously used in the literature. Section 3.3 is devoted to proving Theorem 3.2.6, the main result of the chapter. Section 3.4 has a more probabilistic approach. In this section, we show that in the weakly log-concave situation, μ satisfies a p-WLSI, where $p > 2$ is a fixed constant. This new functional inequality was introduced and studied in Chapter 1. In the present chapter, we derive quantitative estimates in terms of the dimension d and the convexity parameter r.

3.2 Assumptions and main result

Main assumptions on U

We are interested in studying the convergence of the ULA in a weakly log-concave framework, that is, when U is a convex function but not necessarily strongly convex. [Vempala and Wibisono, 2019] was the first work to obtain the order of convergence of entropy using log-Sobolev and Talagrand inequalities. However, LSI generally requires convexity (see, for example, [Bobkov, 1999, Bakry et al., 2014]) to achieve reasonable dimension dependence, and even strong convexity to be dimension-free,

as discussed in Chapter 1. So, once again, we have chosen to parameterize this lack of strong convexity with the help of the Kurdyka-Łojasiewicz inequality. This hypothesis allows us to understand the influence of the lack of strong convexity and not simply assume an LSI.

For the sake of completeness, we recall the definition of hypothesis $\mathcal{H}^{\mathbf{r}}_{\mathbf{KL}}(\mathfrak{c}, L)$. For further details on this hypothesis and its consequences, we refer to Section 1.5. For any $\mathcal{C}^2(\mathbb{R}^d)$ function V, the spectrum of the Hessian matrix of V is denoted as $Sp(\nabla^2 V(\theta))$, and, if V is convex, then

$$\underline{\lambda}_{\nabla^2 V}(\theta) = \inf Sp(\nabla^2 V(\theta)).$$

Hypothesis ($\mathcal{H}^{\mathbf{r}}_{\mathbf{KL}}(\mathfrak{c}, L)$)**.** *We say that a function $V : \mathbb{R}^d \to \mathbb{R}$ satisfies a $\mathcal{H}^{\mathbf{r}}_{\mathbf{KL}}(\mathfrak{c}, L)$-condition if:*

a) V is a $\mathcal{C}^2(\mathbb{R}^d)$ function,

b) V is convex and $\min_{\theta \in \mathbb{R}^d} V(\theta) = V(\theta^*) > 0$,

c) ∇V is L-Lipschitz and

d) there exist some constants $0 \leq r < 1$ and $\mathfrak{c} > 0$ such that

$$\mathfrak{c} V^{-r}(\theta) \leq \underline{\lambda}_{\nabla^2 V(\theta)}, \quad \forall \theta \in \mathbb{R}^d.$$

Throughout this chapter, we assume the existence of three fixed constants $\mathfrak{c} > 0$, $L > 0$, and $0 \leq r < 1$ such that U satisfies $\mathcal{H}^{\mathbf{r}}_{\mathbf{KL}}(\mathfrak{c}, L)$. One of the advantages of assuming $\mathcal{H}^{\mathbf{r}}_{\mathbf{KL}}(\mathfrak{c}, L)$ is that it is possible to obtain upper and lower bounds of $U(\theta)$ through $\min U$ and powers of $\|\theta - \arg\min U\|_2$. Therefore, we need to specify the dependence of $\min U$ and $\arg\min U$ with respect to the dimension d, in order to later understand the dimension dependence of the convergence result.

Similarly to Chapter 2, we will assume that the minimizer of the function U is contained in a ball of radius that only depends on d. Furthermore, we consider that $\min U$ is at most of the order d. For this purpose, we recall the notation $a \lesssim_{uc} b$ ($a \gtrsim_{uc} b$), which means $a \leq cb$ (respectively $a \geq cb$) where c is a positive constant independent of d.

Hypothesis 3.2.1 ($\mathcal{H}_{\min}$)**.** *We have*

$$\|\arg\min U\|_2 \lesssim_{uc} \sqrt{d} \quad \textit{and} \quad \min_{\theta \in \mathbb{R}^d} U(\theta) \lesssim_{uc} d.$$

The assumption $\mathcal{H}_{\min}$ is not restrictive and includes translations as simple as $U(\theta) = (1 + \|\theta - 1_d\|_2^2)^{\frac{1}{1+r}}$ where $1_d = (1, \ldots, 1) \in \mathbb{R}^d$.

Main assumptions on the probability densities

We specify the initial distribution m_0, which is chosen by the user. We decided to use a Gaussian distribution which is a standard option and has the advantage of providing simpler computations. We denote $0_d = (0, \ldots, 0) \in \mathbb{R}^d$ and I_d the identity matrix of size d.

Hypothesis 3.2.2 ($\mathcal{H}^{\mathbf{r}}_{\mathbf{m_0}}(\mathfrak{c}, L)$)**.** *A positive constant σ^2 exists such that m_0 is the density function of a random variable $\mathcal{N}(0_d, \sigma^2 I_d)$ and*

$$\sigma^2 \leq \min\left\{1/(2L), 4/\mathfrak{c}^{\frac{1}{1+r}}\right\}.$$

Remark 3.2.3. *Conditions $\sigma^2 \leq 1/(2L)$ and $\sigma^2 \leq 4/\mathfrak{c}^{\frac{1}{1+r}}$ will be used in the proof of Propositions 3.2.4 and 3.4.1, respectively, to upper bound initial terms determined by m_0.*

Although the term J_0 is commonly ignored to describe the order of convergence of J_k, even with a simple choice such as an initial Gaussian distribution, we will see below that J_0 depends on the dimension d in the strongly log-concave case, and also depends on d and r in our weakly log-concave case. For the sake of completeness, the proof of Proposition 3.2.4 may be found in Appendix 3.A.

Proposition 3.2.4. *Assume $\mathcal{H}^{\mathbf{r}}_{\mathbf{m_0}}(\mathfrak{c}, L)$ and that U satisfies $\mathcal{H}^{\mathbf{r}}_{\mathbf{KL}}(\mathfrak{c}, L)$ and $\mathcal{H}_{\min}$. Then*

$$J_0 = \int_{\mathbb{R}^d} \log\left(\frac{m_0(\theta)}{\mu(\theta)}\right) \mathrm{d}m_0(\theta) \lesssim_{uc} d(1 + r\log d).$$

In the following, we present the most significant hypothesis regarding the probability density of θ_{kh}.

Hypothesis 3.2.5 ($\mathcal{H}_{\mathbf{m_k}}(A_d, b)$)**.** *Two positive constants A_d (which could depend on d) and $0 < b < 1$ exist such that the densities $(m_k)_{k\geq 1}$ are uniformly bounded as*

$$m_k(\theta) \leq A_d e^{-bU(\theta)}, \quad \theta \in \mathbb{R}^d, \quad k \geq 1.$$

Let us briefly comment on this assumption. This bound is immediately verified for quadratic functions. For example, when $U(\theta) = \frac{1}{2}\|\theta\|_2^2$ and $h \leq \frac{1}{2}\left(1 - \frac{1}{\sigma^2}\right)$, this bound is immediately obtained where $A_d = (2\pi)^{-\frac{d}{2}}$ and $b = [2(\sigma^2 + \frac{4}{3})]^{-1}$.

However, this hypothesis is more difficult to verify in a weakly log-concave case. One could expect a polynomial dependency with d of A_d, even in the limiting case $r = 1$, that is, $\log A_d \lesssim_{uc} d$. We have chosen to leave this important axis of research open, as it is not the main topic of this chapter.

Entropic convergence result

Let us present the principal result of this chapter, which proves a decrease in the relative entropy of m_k towards the correct measure μ, as the number of iterations increases and the step size h of the scheme decreases.

Theorem 3.2.6. *Assume $\mathcal{H}^{\mathbf{r}}_{\mathbf{m_0}}(\mathfrak{c}, L)$ and that U satisfies $\mathcal{H}^{\mathbf{r}}_{\mathbf{KL}}(\mathfrak{c}, L)$ and $\mathcal{H}_{\min}$.*

a) If $r = 0$, then μ satisfies an LSI with constant $C_{\mathrm{LSI}} = 2/\mathfrak{c}$, which is independent of d. Moreover, if $h \lesssim_{uc} \varepsilon d^{-1}$ and

$$k \gtrsim_{uc} \varepsilon^{-1} d \log(d/\varepsilon),$$

then $J_k \leq \varepsilon$.

b) If $0 < r < 1$, then μ satisfies a Poincaré inequality with constant C_{PI} and for any $p > 2$, there exist three constants $a_p > 0$, $b_p > 0$ and $s_p > 0$ such that μ also satisfies p-WLSI with function

$$\varphi_p(s) = b_p C_{\mathrm{PI}} \log(a_p/s), \quad 0 < s < s_p.$$

Moreover, assume $\mathcal{H}_{\mathbf{m_k}}(A_d, b)$. If $h \lesssim_{uc} \varepsilon(d^{1+(1+r)^2} \log(A_d/\varepsilon))^{-1}$ and

$$k \gtrsim_{uc} \varepsilon^{-1} d^{1+2(1+r)^2} \log^2(A_d/\varepsilon) \log(d/\varepsilon),$$

then $J_k \leq \varepsilon$.

When we compare to the result stated in Theorem 2.2.2 in the same setting ($\mathcal{H}^{\mathbf{r}}_{\mathbf{KL}}(\mathfrak{c}, L)$ and entropic convergence) but only studies the *continuous-time* stochastic Langevin diffusion, we observe that the computational time becomes Poly(d) ε^{-1}, while it was Poly(d) $\log^2(\varepsilon^{-1})$ in Theorem 2.2.2. Again, this exemplifies the price to pay to turn the continuous-time diffusion into a tractable numerical discrete algorithm. Thanks to Theorem 3.2.6 it follows that $J_k \leq \varepsilon$ after approximately

$$k \gtrsim_{uc} \varepsilon^{-1} d^{3+2(1+r)^2}$$

iterations, where $\log(A_d)$ is considered proportional to d.

State of the art comparison

In Table 3.2.1, we present the iteration complexity obtained previously by [Durmus et al., 2019] and [Vempala and Wibisono, 2019] when U is strongly convex and its gradient is Lipschitz. We observe that we have recovered the state-of-the-art result. All of these works have used different strategies to prove the order of convergence.

Work	Iteration cost for Entropy $J_k \leq \varepsilon$
[Durmus et al., 2019]	
[Vempala and Wibisono, 2019]	$\varepsilon^{-1} d$
This chapter	

Table 3.2.1: Comparison of iteration complexity when U is a strongly convex function with Lipschitz gradient.

In the literature there are different works that study the ULA when the potential U is weakly log-concave, we need to adapt their hypotheses to ours to be able to compare the results. We focus on [Cheng and Bartlett, 2018], [Durmus et al., 2019],[Erdogdu and Hosseinzadeh, 2021] and [Chewi et al., 2022a]. The hypothesis of said works were adapted to our $\mathcal{H}^{\mathbf{r}}_{\mathbf{KL}}(\mathfrak{c}, L)$ context where $r \in (0, 1)$ and assuming $\mathcal{H}_{\min}$ and $\mathcal{H}^{\mathbf{r}}_{\mathbf{m_0}}(\mathfrak{c}, L)$. In particular, we pay special attention to the dependency with ε, the dimension d and the parameter r. Table 3.2.2 shows a summary of these results, omitting the logarithmic terms $\log(d)$ and $\log(\varepsilon^{-1})$.

Work	Iteration cost for Entropy $J_k \leq \varepsilon$
[Cheng and Bartlett, 2018]	$\varepsilon^{-3} d^{1+2(1+r)}$
[Durmus et al., 2019]	$\varepsilon^{-2} d^{1+2(1+r)}$
[Erdogdu and Hosseinzadeh, 2021] and [Chewi et al., 2022a]	$\varepsilon^{-1} d^{3.5+\frac{2r(2+r)}{1+r}}$
This chapter	$\varepsilon^{-1} d^{3+2(1+r)^2}$

Table 3.2.2: Comparison of the iteration complexity when U satisfies $\mathcal{H}^{\mathbf{r}}_{\mathbf{KL}}(\mathfrak{c}, L)$, $0 < r < 1$ and $\mathcal{H}_{\min}$ and the initial distribution is defined as in $\mathcal{H}^{\mathbf{r}}_{\mathbf{m_0}}(\mathfrak{c}, L)$, admitting that $\log(A_d) \lesssim_{uc} d$.

• When compared with the bounds obtained in [Durmus et al., 2019] (and in [Cheng and Bartlett, 2018]), we obtain a much better dependency with ε and a degradation with the dimension d. When paying a specific attention to the approach of Corollary 7 of [Durmus et al., 2019], the improvement from ε^{-2} to ε^{-1} comes from a direct study of the entropy in our chapter, and not from a general catch-all inequality between entropy and the Wasserstein distance. We refer to [Villani et al., 2009] for a complete presentation of the Wasserstein distance. That being said, the coupling strategy of [Durmus et al., 2019] generates a better dependency with respect to d than ours and an obvious computation shows that the bound stated in [Durmus et al., 2019] is better than the one of this chapter for some very high-dimensional settings: $\log(d) \geq (2(1+r+r^2))^{-1} \log(\varepsilon^{-1})$. In contrast, when ε is chosen smaller than $d^{-2(1+r+r^2)}$, our bound is tighter.

Furthermore, the result of [Durmus et al., 2019] does not involve the distribution of ULA in iteration k, but a Césaro average in all iterations of ULA, which is an additional numerical difficulty compared to simply construct one trajectory and take the value of the procedure attained at the final iteration.

• In order to compare our result with the one that would be obtained in [Erdogdu and Hosseinzadeh, 2021], let us consider, for example, $U(\theta) = (1 + \|\theta - 1_d\|_2^2)^{\frac{1}{1+r}}$, where $r \in (0, 1)$ and adjust their α-dissipativity and ζ-growth of the gradient assumption. They establish that there exist two constants $a > 0$ and $b > 0$ such that $\langle \nabla U(\theta), \theta \rangle \geq a\|\theta\|_2^\alpha - b$, for all $\theta \in \mathbb{R}^d$. If we take $\theta = \arg\min U$, then $\|\arg\min U\|_2 \leq (b/a)^{\frac{1}{\alpha}}$, so the dependence on d is not explicit or does not exist. Then, for comparison, we take a as a universal constant and $b = d^{\frac{\alpha}{2}} = d^{\frac{1}{1+r}}$ in the particular case that was studied. Furthermore, they assume that there exist two positive constants ζ and M such that $\|\nabla U(\theta)\|_2 \leq M(1 + \|\theta\|_2^\zeta)$, for all $\theta \in \mathbb{R}^d$. Using the specific function U, we get $M \propto \sqrt{d}$.

After specifying the dependence on d of the constants b and M, Corollary 4 of [Erdogdu and Hosseinzadeh, 2021] shows that an ε-error is reached after

$$k \gtrsim_{uc} \varepsilon^{-1} d^{3.5+\frac{2r(2+r)}{1+r}} \mathrm{Poly}(\log(d), \log(\varepsilon^{-1}))$$

iterations. The same result is obtained for [Chewi et al., 2022a], only they obtained their result in Rényi divergence.

When comparing the results, we observe that the *modified* log-Sobolev inequality in [Erdogdu and Hosseinzadeh, 2021](and in [Chewi et al., 2022a]) leads to a better dependence on d, as the terms in that inequality are multiplied, while in the p-WLSI they are added. Another potential factor is our use of the Bobkov bound for the Poincaré constant C_{PI} associated with μ, where we use $C_{\mathrm{PI}} \lesssim_{uc} d^{(1+r)^2}$. Obtaining a tighter bound on this constant would enhance the result. Then, even though our result is not as sharp as those in [Erdogdu and Hosseinzadeh, 2021] and [Chewi et al., 2022a], it provides an alternative perspective that warrants further investigation to potentially obtain sharper bounds.

3.3 Proof of the main result

Similarly to [Vempala and Wibisono, 2019] and with the objective of studying the order of convergence of J_k, for any $k \geq 0$, we define a continuous-time process $(\theta_{kh+t})_{0\leq t\leq h}$ in the time interval $[kh, (k+1)h]$, such that the distributions at the ends of the interval coincide with m_k and m_{k+1}, respectively. The relative entropy of the density of the said continuous process with respect to the law μ will allow us to obtain the order of convergence of J_k.

We consider one fixed step k of (3.3), then the output $\theta_{(k+1)h}$ is in fact the output at time h of the stochastic differential equation

$$\mathrm{d}\theta_{kh+t} = -\nabla U(\theta_{kh})\mathrm{d}t + \sqrt{2}\mathrm{d}W_t, \quad t \in [0, h], \tag{3.5}$$

where $(W_t)_{0\leq t\leq h}$ is a standard Brownian Motion in $\mathbb{R}^d$. Indeed, at time $t \in [0, h]$, the solution of (3.5) is

$$\theta_{kh+t} = \theta_{kh} - t\nabla U(\theta_{kh}) + \sqrt{2}W_t, \tag{3.6}$$

so if $t = h$, then $\mathrm{Law}(\theta_{kh} - h\nabla U(\theta_{kh}) + \sqrt{2}W_h) = \mathrm{Law}(\theta_{(k+1)h})$.

The process $(\theta_{kh+t}|\theta_{kh})_{0\leq t\leq h}$ that evolves following equation (3.5) is a continuous-time Markov process associated with the generator $\mathcal{L}^k$ of the Markov semigroup, where

$$\mathcal{L}^k f(\theta) = -\langle \nabla U(\theta_{kh}), \nabla f(\theta)\rangle + \Delta f(\theta),$$

for all $f \in \mathcal{C}^2(\mathbb{R}^d)$ and $\theta \in \mathbb{R}^d$.

Let us denote

$$\mathrm{Law}(\theta_{kh+t}) = n_{k,t} \quad \text{and} \quad \mathrm{Law}(\theta_{kh+t}|\theta_{kh}) = n_{t|k}.$$

Using (3.6), we have that for any $\eta \in \mathbb{R}^d$, $n_{t|k}(\cdot|\eta)$ is the density function of a $\mathcal{N}(\eta - t\nabla U(\eta), 2tI_d)$ random variable, then $n_{t|k}$ is $\mathcal{C}^\infty(\mathbb{R}^d)$, $n_{k,t}$ is also $\mathcal{C}^\infty(\mathbb{R}^d)$ and $\partial_t\{n_{t|k}\}$ exists.

We recall that if $\mathcal{L}^{k*}$ denotes the adjoint operator of $\mathcal{L}^k$, the Kolmogorov forward

equation yields

$$\partial_t\{n_{t|k}(\theta|\eta)\} = \mathcal{L}^{k*}n_{t|k}(\theta|\eta), \quad (\theta,\eta)\in\mathbb{R}^d\times\mathbb{R}^d.$$

This equality is used in the proof of Lemma 3.3.1.

Similarly to the definition of J_k in (3.4), we denote the relative entropy of $n_{k,t}$ with respect to the target measure μ as

$$J_{k,t} = \int_{\mathbb{R}^d}\log\left(\frac{n_{k,t}(\theta)}{\mu(\theta)}\right)\mathrm{d}n_{k,t}(\theta),$$

for any $k\geq 0$ and $t\in[0,h]$. Observe that, in particular $n_{k,0}=m_k$ and $n_{k,h}=m_{k+1}$, then $J_{k,0}=J_k$ and $J_{k,h}=J_{k+1}$. We also define the Dirichlet form of $\sqrt{n_{k,t}/\mu}$ (proportional to the Fisher information of $n_{k,t}$ with respect to μ) as

$$\mathcal{E}_{k,t} = \int_{\mathbb{R}^d}\left\|\nabla\left(\sqrt{\frac{n_{k,t}(\theta)}{\mu(\theta)}}\right)\right\|_2^2\mathrm{d}\mu(\theta) = \frac{1}{4}\int_{\mathbb{R}^d}\left\|\nabla\log\left(\frac{n_{k,t}(\theta)}{\mu(\theta)}\right)\right\|_2^2\mathrm{d}n_{k,t}(\theta),$$

for any $k\geq 0$ and $t\in[0,h]$. We recall that in the proof of Lemma 3 of [Vempala and Wibisono, 2019] it was obtained the following link between $\partial_t\{J_{k,t}\}$ and $\mathcal{E}_{k,t}$

$$\partial_t\{J_{k,t}\}\leq -3\mathcal{E}_{k,t}+\mathbb{E}_{m_kn_{t|k}}[\|\nabla U(\theta_{kh+t})-\nabla U(\theta_{kh})\|_2^2],$$

where the second term is the discretisation error which in the strongly log-concave context was controlled using Talagrand's (or equivalently log-Sobolev) inequality. In general, we are not in the strongly log-concave context, then we control differently the error term in Lemma 3.3.1 , the proof may be found in Appendix 3.B . Moreover, we want to emphasize that the strategy used to prove Lemma 3.3.1 is equivalent to that of [Vempala and Wibisono, 2019] but is based on the infinitesimal generator $\mathcal{L}^k$ and the Kolmogorov forward equation.

Lemma 3.3.1. *Assume that ∇U is L-Lipschitz and $h\leq 1/(4L)$. Then, for any $k\geq 0$ and $t\in[0,h]$,*

$$\partial_t\{J_{k,t}\}\leq -2\mathcal{E}_{k,t}+4dL^2t.$$

To construct a differential inequality using the above result, we have to use a functional inequality that links $\mathcal{E}_{k,t}$ and $J_{k,t}$. If $r=0$, μ is strongly log-concave, then it is possible to apply an LSI. However, if $0<r<1$, μ is weakly log-concave and the first alternative is trying to apply the weak log-Sobolev inequality (WLSI). However, one of its terms depends on $n_{k,t}/\mu$ being bounded, and, in general, we do not have such a result. So, let us use p-WLSI.

From now on, we study the cases $r=0$ and $0<r<1$ separately.

- Case $r=0$.

  Since μ is a strongly log-concave measure, we follow the strategy used in [Vempala and Wibisono, 2019], which is based on applying LSI to $f_{k,t}^2 = n_{k,t}/\mu$, see the details about functional inequalities in an $\mathcal{H}^{\mathbf{r}}_{\mathbf{KL}}(\mathfrak{c},L)$-context in

Section 1.5. Using Theorem 1.5.5, μ satisfies an LSI with constant $C_{\mathrm{LSI}} = 2/\mathfrak{c}$, which is independent of d. In particular,

$$J_{k,t} \leq C_{\mathrm{LSI}} \mathcal{E}_{k,t}, \tag{3.7}$$

We combine Lemma 3.3.1 and inequality (3.7) to get that for any $k \geq 0$,

$$\partial_t \{J_{k,t}\} \leq -\frac{2}{C_{\mathrm{LSI}}} J_{k,t} + 4dL^2 h, \quad t \in (0,h),$$

where we used $t \leq h$ in the second term of the right hand side. We recall that $J_{k,0} = J_k$ and $J_{k,h} = J_{k+1}$, for any k. Then if we apply Gronwall's lemma, it results that for any $k \geq 0$,

$$J_{k+1} \leq J_k e^{-\frac{2h}{C_{\mathrm{LSI}}}} + 2dL^2 h C_{LSI} \left(1 - e^{-\frac{2h}{C_{\mathrm{LSI}}}}\right).$$

Using a recursive argument, Proposition 3.2.4 to upper bound J_0 and $C_{\mathrm{LSI}} = 2/\mathfrak{c}$, we get that

$$J_k \leq J_0 e^{-\frac{2kh}{C_{\mathrm{LSI}}}} + 2dL^2 h C_{\mathrm{LSI}} \left(1 - e^{-\frac{2kh}{C_{\mathrm{LSI}}}}\right) \leq Cd\left[e^{-k\mathfrak{c}h} + h\right],$$

where C is a constant independent of d. Given $\varepsilon > 0$ small, a sufficient condition for $J_k \leq \varepsilon$ is

$$h \lesssim_{uc} \varepsilon d^{-1} \quad \text{and} \quad k \gtrsim_{uc} \varepsilon^{-1} d \log(d/\varepsilon).$$

- Case $0 < r < 1$.

In the weakly log-concave case, we apply Theorem 1.5.6. Then μ satisfies a Poincaré inequality with constant C_{PI} and p-WLSI, for any $p > 2$. The choice of the value of p is studied in Section 3.4, so we will rely on Proposition 3.4.4 and refer to said Section for further details.

From now on, $p = \begin{cases} \frac{1}{1-b} & \text{if } 0 < b < \frac{1}{2} \\ 3 & \text{if } \frac{1}{2} < b < 1 \end{cases}$. In Proposition 3.4.4, we proved that for any $k \geq 0$ and $t \in [0,h]$,

$$J_{k,t} \leq \varphi_p(s) \mathcal{E}_{k,t} + sO_d, \quad s > 0, \tag{3.8}$$

where

$$\varphi_p(s) = b_p C_{\mathrm{PI}} \log(a_p/s), \quad 0 < s < s_p,$$

a_p, b_p and s_p are positive constants depending on p, $C_{\mathrm{PI}} \lesssim_{uc} d^{(1+r)^2}$, and $\log(O_d) \lesssim_{uc} \log\left(\frac{A_d}{(1-h(2+L^2))^{\frac{d}{2}}}\right) + d(1 + r\log(d))$.

We are now ready to derive a good differential inequality for $J_{k,t}$ by combining

Lemma 3.3.1 and Equation (3.8), then for s small,

$$\partial_t \{J_{k,t}\} \leq -\frac{2J_{k,t}}{\varphi_p(s)} + \frac{sO_d}{\varphi_p(s)} + 4dL^2h.$$

where we used $t \leq h$. Applying the Gronwall lemma, for any $k \geq 0$ we have that

$$J_{k+1} \leq J_k e^{-\frac{2h}{\varphi_p(s)}} + \left(sO_d + 2dL^2h\varphi_p(s)\right)\left(1 - e^{-\frac{2h}{\varphi_p(s)}}\right).$$

We recall one more time that $J_{k,0} = J_k$ and $J_{k,h} = J_{k+1}$, for any k. Using a recursive argument and the result $J_0 \lesssim_{uc} d(1 + r\log d)$ proved in Proposition 3.2.4, we get that

$$J_k \leq J_0 e^{-\frac{2kh}{\varphi_p(s)}} + sO_d + 2dL^2h\varphi_p(s).$$

If $h \leq \frac{1}{2(2+L^2)}$, then $\log(O_d) \leq \log(A_d) + d(1 + r\log(d))$, but, using that $\log(A_d) \gtrsim_{uc} d$, then, we just consider the first term.

Given $\varepsilon > 0$ small, a sufficient condition for $J_k \leq \varepsilon$ is

$$s = \frac{\varepsilon}{3A_d}, \qquad h \lesssim_{uc} \frac{\varepsilon}{dC_{\mathrm{PI}}\log(A_d/\varepsilon)},$$

and

$$k \gtrsim_{uc} \varepsilon^{-1} dC_{\mathrm{PI}}^2 \log^2(A_d/\varepsilon)\log((1 + r\log(d))d/\varepsilon),$$

which concludes the proof.

3.4 Technical results on p -WLSI under $\mathcal{H}^{\mathbf{r}}_{\mathbf{KL}}(\mathfrak{c}, L)$ hypothesis

This section is devoted to providing justification for inequality (3.8) in Section 3.3. This inequality is obtained by applying a functional inequality to the function $\sqrt{n_{k,t}/\mu}$. We refer to the results of Section 1.5, in particular, to Theorem 1.5.6, where we proved that if U satisfies $\mathcal{H}^{\mathbf{r}}_{\mathbf{KL}}(\mathfrak{c}, L)$ with $0 < r < 1$, then μ satisfies a p-WLSI for any $p > 2$, with function

$$\varphi(s) = b_p C_{\mathrm{PI}} \log(a_p/s), \quad 0 < s < s_p,$$

where a_p, b_p, and s_p are three positive constants depending on p. This functional inequality was introduced and studied in Section 1.4.

Throughout this section, we choose the value of p and find a bound on the Poincaré constant C_{PI}. We pay special attention to the dependence of all the constants involved in the inequalities in terms of the parameter r and the dimension d.

Poincaré constant

From Chapter 1, we recall that μ is a log-concave measure, then it satisfies a Poincaré inequality with constant C_{PI}. Using the Bobkov bound of the Poincaré constant, we get

$$C_{\mathrm{PI}} \leq 4K^2 Var_{\mu}(\theta),$$

which is stated in Remark 1.2.4. Following the strategy of Proposition 2.4.2, we derive a bound on the Poincaré constant depending on d and r. The proof of Proposition 3.4.1 is deferred to Appendix 3.C.

Proposition 3.4.1. *Assume $\mathcal{H}^{\mathbf{r}}_{\mathbf{m_0}}(\mathfrak{c}, L)$ and that U satisfies hypotheses $\mathcal{H}^{\mathbf{r}}_{\mathbf{KL}}(\mathfrak{c}, L)$ and $\mathcal{H}_{\min}$, with $0 < r < 1$. Then*

$$C_{\mathrm{PI}} \lesssim_{uc} d^{(1+r)^2}.$$

This bound may introduce an error in terms of d. Therefore, during the discussion of the main result, we also present the order of convergence of the ULA depending on the Poincaré constant.

Bound of the density $n_{k,t}$

The discrete Euler-Maruyama scheme (3.3) involves a Gaussian convolution kernel at each iteration, of variance $2h$. Then, as was said above, the distribution of θ_{kh} has an infinitely differentiable density m_k with respect to the Lebesgue measure for any $k \geq 1$.

In [Konakov and Mammen, 2000], the authors investigate local bounds for the transition probabilities of diffusion processes, and as a result, we deduce that $m_k(\theta) \leq D_1 e^{-D_2\|\theta\|_2^2}$ whenever U is bounded and D_1 and D_2 are positive constants which could depend on d, k and h. This approach does not apply in our context and since this is not the primary focus of our study, we decided to assume $\mathcal{H}_{\mathbf{m_k}}(A_d, b)$ and leave a more detailed exploration of this direction for future work.

The bound assumed in hypothesis $\mathcal{H}_{\mathbf{m_k}}(A_d, b)$ is extended to the density $n_{k,t}$ as follows.

Proposition 3.4.2. *Assume $\mathcal{H}_{\mathbf{m_k}}(A_d, b)$ and $h \leq \frac{1-4(\frac{1}{2}-b)^2}{3+L^2}$. Then, for any $k \geq 0$ and $t \in (0, h)$,*

$$\forall \theta \in \mathbb{R}^d, \qquad n_{k,t}(\theta) \leq A_{d,h} e^{-bU(\theta)},$$

where $A_{d,h} = A_d/(1 - h(2 + L^2))^{\frac{d}{2}}$.

For the sake of readability, the proof of Proposition 3.4.2 is deferred to Appendix 3.C.

In general $n_{k,t}/\mu$ is not a bounded function; however, the result of Proposition 3.4.2 allows us to find a value of p for which $\sqrt{n_{k,t}/\mu} \in \mathbb{L}^p_{\mu}(\mathbb{R}^d)$, which is a necessary condition to apply p-WLSI to $\sqrt{n_{k,t}/\mu}$.

p-WLSI

The purpose of the following proposition is to determine the value of the parameter p, for which $\sqrt{n_{k,t}/\mu}$ belongs to $\mathbb{L}^p_\mu(\mathbb{R}^d)$, for any $k \geq 0$ and $t \in [0, h]$.

Proposition 3.4.3. *Assume $\mathcal{H}^{\mathbf{r}}_{\mathbf{m_0}}(\mathfrak{c}, L)$, $\mathcal{H}_{\mathbf{m_k}}(A_d, b)$ and that U satisfies $\mathcal{H}^{\mathbf{r}}_{\mathbf{KL}}(\mathfrak{c}, L)$ and $\mathcal{H}_{\min}$ with $0 < r < 1$. Then, for any $k \geq 0$,*

$$\log\left(\left\|\sqrt{\frac{n_{k,t}}{\mu}}\right\|_{\mathbb{L}^p_\mu(\mathbb{R}^d)}\right) \lesssim_{uc} \log(A_{d,h}) + d(1 + r\log(d)),$$

where $p = \begin{cases} \frac{1}{1-b} & \text{if } 0 < b < \frac{1}{2} \\ 3 & \text{if } \frac{1}{2} < b < 1 \end{cases}$.

The proof of Proposition 3.4.3 is deferred to Appendix 3.C. This result then allows us to apply p-WLSI with $p = \frac{1}{1-b}$ if b is small or $p = 3$ if b is large. The constants a_p, b_p, and s_p can be computed depending on the value of p.

In conclusion, we can state the following proposition, which we will not prove and which collects the results of this section and Theorem 1.5.6.

Proposition 3.4.4. *Assume $\mathcal{H}^{\mathbf{r}}_{\mathbf{m_0}}(\mathfrak{c}, L)$, $\mathcal{H}_{\mathbf{m_k}}(A_d, b)$, U satisfies $\mathcal{H}^{\mathbf{r}}_{\mathbf{KL}}(\mathfrak{c}, L)$ and $\mathcal{H}_{\min}$, and $h \lesssim_{uc} \min\{1/L, 1/d\}$. Then,*

a) μ satisfies a Poincaré inequality with constant $C_{\mathrm{PI}} \lesssim_{uc} d^{(1+r)^2}$.

b) μ satisfies a p-WLSI with $p = \begin{cases} \frac{1}{1-b} & \text{if } 0 < b < \frac{1}{2} \\ 3 & \text{if } \frac{1}{2} < b < 1 \end{cases}$ *and*

$$\varphi_p(s) = b_p C_{\mathrm{PI}} \log(a_p/s), \quad 0 < s < s_p,$$

where a_p, b_p and s_p are positive constants.

c) As a consequence, for any $k \geq 0$ and $t \in [0, h]$,

$$J_{k,t} \leq \varphi_p(s)\mathcal{E}_{k,t} + sO_d, \quad s > 0,$$

where $\log(O_d) \lesssim_{uc} \log(A_d) + d(1 + r\log(d))$ *if* $h \leq \frac{1-4(\frac{1}{2}-b)^2}{3+L^2}$.

Appendix to Chapter 3

3.A Proofs of the results in Section 3.2

Proof of Proposition 3.2.4

The initial relative entropy is defined as

$$J_0 = \int_{\mathbb{R}^d} \log\left(\frac{m_0(\theta)}{\mu(\theta)}\right) \mathrm{d}m_0(\theta).$$

Replacing m_0 by the density of a random variable $\mathcal{N}(0_d, \sigma^2 I_d)$ and $\mu(\theta) = \frac{1}{\mathcal{Z}} e^{-U(\theta)}$, we observe that

$$J_0 = \log\left(\frac{\mathcal{Z}}{(2\pi\sigma^2)^{\frac{d}{2}}}\right) + \int_{\mathbb{R}^d} \left(U(\theta) - \frac{\|\theta\|_2^2}{2\sigma^2}\right) \mathrm{d}m_0(\theta).$$

Let us study each term separately.

- From Proposition 1.5.4, we deduce that the first term is upper bounded as follows:
$$\log\left(\frac{\mathcal{Z}}{(2\pi\sigma^2)^{\frac{d}{2}}}\right) \leq \log\left(\frac{2d^{\frac{dr}{2}}}{(\mathfrak{c}\sigma^2)^{\frac{d}{2}}}\right) \lesssim_{uc} d(1 + r\log d).$$

- Let be $\theta^* = \arg\min U$, applying Proposition 1.5.3, we have that
$$U(\theta) - \frac{\|\theta\|_2^2}{2\sigma^2} \leq \frac{L}{2}\|\theta - \theta^*\|_2^2 - \frac{\|\theta\|_2^2}{2\sigma^2} + \min U.$$
If $\sigma^2 < \frac{1}{L}$, then the function $\theta \mapsto \frac{L}{2}\|\theta - \theta^*\|_2^2 - \frac{\|\theta\|_2^2}{2\sigma^2} + \min U$ attains its global maximum at $\frac{\sigma^2 L}{\sigma^2 L - 1}\theta^*$ and
$$U(\theta) - \frac{\|\theta\|_2^2}{2\sigma^2} \leq \frac{L}{2(1-\sigma^2 L)}\|\theta^*\|_2^2 + \min U \lesssim_{uc} d,$$
where we used hypothesis $\mathcal{H}_{\min}$ in the last step. Therefore,
$$\int_{\mathbb{R}^d} \left(U(\theta) - \frac{\|\theta\|_2^2}{2\sigma^2}\right) \mathrm{d}m_0(\theta) \lesssim_{uc} d.$$

Putting both parts together, the result is proven.

3.B Proofs of the results in Section 3.3

Proof of Lemma 3.3.1

We begin by formulating the following lemma to control the expectation of $\|\nabla U\|_2^2$ under a change of measure. This result will be useful for the proof of the Lemma 3.3.1. Its proof is taken from [Chewi et al., 2022a] and we include it for the sake of completeness.

Lemma 3.B.1. *Assume that ∇U is L-Lipschitz. For any probability measure m, it holds that*

$$\mathbb{E}_m\left(\|\nabla U(\theta)\|_2^2\right) \leq 4\mathbb{E}_\mu\left(\left\|\nabla\left(\sqrt{m(\theta)/\mu(\theta)}\right)\right\|_2^2\right) + 2dL.$$

Proof of Lemma 3.B.1. We introduce the infinitesimal generator $\mathcal{L}$ defined for any real-valued function $f \in \mathcal{C}^2(\mathbb{R}^d)$ by

$$\mathcal{L}f(\theta) = -\langle \nabla U(\theta), \nabla f(\theta)\rangle + \Delta f(\theta), \tag{3.9}$$

which is associated to the diffusion (3.2). Applying (3.9) to U, we observe that $\mathcal{L}U = \|\nabla U\|_2^2 - \Delta U$. We use integration by parts then,

$$\begin{aligned}
\mathbb{E}_m\left[\|\nabla U(\theta)\|_2^2\right] &= \mathbb{E}_m[\mathcal{L}U(\theta)] + \mathbb{E}_m[\Delta U(\theta)] \\
&\leq \int_{\mathbb{R}^d} \mathcal{L}U(\theta)\frac{m(\theta)}{\mu(\theta)}\mathrm{d}\mu(\theta) + dL \\
&= \int \langle \nabla U(\theta), \nabla\left(\frac{m(\theta)}{\mu(\theta)}\right)\rangle \mathrm{d}\mu(\theta) + dL \\
&\leq 2\int \langle \sqrt{\frac{m(\theta)}{\mu(\theta)}}\nabla U(\theta), \nabla\left(\sqrt{\frac{m(\theta)}{\mu(\theta)}}\right)\rangle \mathrm{d}\mu(\theta) + dL \\
&\leq \frac{1}{2}\mathbb{E}_m\left[\|\nabla U(\theta)\|_2^2\right] + 2\mathbb{E}_\mu\left[\left\|\nabla\left(\sqrt{\frac{m(\theta)}{\mu(\theta)}}\right)\right\|_2^2\right] + dL.
\end{aligned}$$

Rearrange this inequality to obtain the desired result. □

The Lemma 3.3.1 recalls the link obtained in [Vempala and Wibisono, 2019] between the derivative of the entropy of $\frac{n_{k,t}}{\mu}$ and the Dirichlet form of $\sqrt{\frac{n_{k,t}}{\mu}}$. However, we use the Lemma 3.B.1 to control the discretization error as it appears in [Balasubramanian et al., 2022].

Proof of Lemma 3.3.1. Let k be fixed. Using the logic of Proposition 2.C.2, we could show that $\partial_t\{J_{k,t}\}$ exists and

$$\partial_t\{J_{k,t}\} = \int_{\mathbb{R}^d}\left(1 + \log\left(\frac{n_{k,t}(\theta)}{\mu(\theta)}\right)\right)\partial_t\{n_{k,t}(\theta)\}\mathrm{d}\theta.$$

Now, we will follow an argument equivalent to the one used in Lemma 3 of [Vempala and Wibisono, 2019]. This argument is based on the infinitesimal

generator $\mathcal{L}^k$ and the Kolmogorov forward equation. We recall that $n_{t|k}$ and m_k are the density functions of θ_{kh+t} conditional to θ_{kh} and θ_{kh}, respectively, then we replace

$$n_{k,t}(\theta) = \int_{\mathbb{R}^d} n_{t|k}(\theta|\eta)\mathrm{d}m_k(\eta)$$

in the equation for $\partial_t\{J_{k,t}\}$ to get that

$$\begin{aligned}
\partial_t\{J_{k,t}\} &= \int_{\mathbb{R}^d}\left(1+\log\left(\frac{n_{k,t}(\theta)}{\mu(\theta)}\right)\right)\partial_t\left\{\int_{\mathbb{R}^d} n_{t|k}(\theta|\eta)\mathrm{d}m_k(\eta)\right\}\mathrm{d}\theta \\
&= \int_{\mathbb{R}^d\times\mathbb{R}^d}\left(1+\log\left(\frac{n_{k,t}(\theta)}{\mu(\theta)}\right)\right)\partial_t\{n_{t|k}(\theta|\eta)\}\mathrm{d}m_k(\eta)\mathrm{d}\theta \\
&= \int_{\mathbb{R}^d\times\mathbb{R}^d}\left(1+\log\left(\frac{n_{k,t}(\theta)}{\mu(\theta)}\right)\right)\partial_t\{n_{t|k}(\theta|\eta)\}\mathrm{d}\theta\mathrm{d}m_k(\eta) \\
&= \int_{\mathbb{R}^d\times\mathbb{R}^d}\mathcal{L}^k\log\left(\frac{n_{k,t}(\theta)}{\mu(\theta)}\right)\mathrm{d}n_{t|k}(\theta|\eta)\mathrm{d}m_k(\eta),
\end{aligned}$$

where we used the Fubini theorem and the Kolmogorov forward equation in the last two steps, respectively.

We introduce the infinitesimal generator $\mathcal{L}$ given in (3.9) and De Bruijn's identity (see Proposition 1.2.17),

$$\begin{aligned}
\partial_t\{J_{k,t}\} &= \int_{\mathbb{R}^d}\mathcal{L}\log\left(\frac{n_{k,t}(\theta)}{\mu(\theta)}\right)\mathrm{d}n_{k,t}(\theta) \\
&\quad + \int_{\mathbb{R}^d\times\mathbb{R}^d}\left(\mathcal{L}^k-\mathcal{L}\right)\log\left(\frac{n_{k,t}(\theta)}{\mu(\theta)}\right)\mathrm{d}n_{t|k}(\theta|\eta)\mathrm{d}m_k(\eta) \\
&= \int_{\mathbb{R}^d}\mathcal{L}\log\left(\frac{n_{k,t}(\theta)}{\mu(\theta)}\right)\mathrm{d}n_{k,t}(\theta) \\
&\quad + \int_{\mathbb{R}^d\times\mathbb{R}^d}\left\langle\nabla U(\theta)-\nabla U(\eta),\nabla\log\left(\frac{n_{k,t}(\theta)}{\mu(\theta)}\right)\right\rangle\mathrm{d}n_{t|k}(\theta|\eta)\mathrm{d}m_k(\eta) \\
&= -4\mathcal{E}_{k,t} + \int_{\mathbb{R}^d\times\mathbb{R}^d}\left\langle\nabla U(\theta)-\nabla U(\eta),\nabla\log\left(\frac{n_{k,t}(\theta)}{\mu(\theta)}\right)\right\rangle\mathrm{d}n_{t|k}(\theta|\eta)\mathrm{d}m_k(\eta).
\end{aligned} \tag{3.10}$$

Let us study the second term of (3.10). Using Young's inequality: for any $a,b\in\mathbb{R}^d$ and $\epsilon>0$, $\langle a,b\rangle \le \frac{\epsilon}{2}\|a\|_2^2+\frac{1}{2\epsilon}\|b\|_2^2$, where ϵ will be fixed later on, we observe that

$$\begin{aligned}
&\int_{\mathbb{R}^d\times\mathbb{R}^d}\left\langle\nabla U(\theta)-\nabla U(\eta),\nabla\log\left(\frac{n_{k,t}(\theta)}{\mu(\theta)}\right)\right\rangle\mathrm{d}n_{t|k}(\theta|\eta)\mathrm{d}m_k(\eta) \\
&\le \frac{\epsilon}{2}\int_{\mathbb{R}^d\times\mathbb{R}^d}\|\nabla U(\theta)-\nabla U(\eta)\|_2^2\,\mathrm{d}n_{t|k}(\theta|\eta)\mathrm{d}m_k(\eta) + \frac{1}{2\epsilon}\int_{\mathbb{R}^d}\left\|\nabla\log\left(\frac{n_{k,t}(\theta)}{\mu(\theta)}\right)\right\|_2^2\mathrm{d}n_{k,t}(\theta) \\
&\le \frac{L^2\epsilon}{2}\underbrace{\int_{\mathbb{R}^d\times\mathbb{R}^d}\|\theta-\eta\|_2^2\,\mathrm{d}n_{t|k}(\theta|\eta)\mathrm{d}m_k(\eta)}_{\Delta} + \frac{2}{\epsilon}\mathcal{E}_{k,t},
\end{aligned} \tag{3.11}$$

where we used that ∇U is $L-$Lipschitz.

Note that the error term Δ can be written as follows

$$\Delta = \int_{\mathbb{R}^d \times \mathbb{R}^d} \|\theta - \eta\|_2^2 \, \mathrm{d}n_{t|k}(\theta|\eta) \mathrm{d}m_k(\eta) = \mathbb{E}_{m_k n_{t|k}} \left[\|\theta_{kh+t} - \theta_{kh}\|_2^2 \right].$$

Moreover, the solution of (3.5) at time t is such that

$$\mathrm{Law}(\theta_{kh+t} - \theta_{kh}) = \mathrm{Law}(-t\nabla U(\theta_{kh}) + \sqrt{2t}\xi),$$

where ξ is a standard Gaussian random variable in $\mathbb{R}^d$ independent of θ_{kh}. If γ_1 denotes the density function of ξ, then the error term is

$$\Delta = \mathbb{E}_{m_k,\gamma_1} \left[\left\| -t\nabla U(\theta_{kh}) + \sqrt{2t}\xi \right\|_2^2 \right] = t^2 \mathbb{E}_{m_k} \left[\|\nabla U(\theta_{kh})\|_2^2 \right] + 2dt. \tag{3.12}$$

Next, since ∇U is L-Lipschitz,

$$\begin{aligned} \|\nabla U(\theta_{kh})\|_2 &\leq \|\nabla U(\theta_{kh+t})\|_2 + L\|\theta_{kh+t} - \theta_{kh}\|_2 \\ &\leq \|\nabla U(\theta_{kh+t})\|_2 + tL\|\nabla U(\theta_{kh})\|_2 + \sqrt{2t}L\|\xi\|_2, \end{aligned}$$

for $h < 1/L$, we can rearrange this inequality to obtain an upper bound for $\|\nabla U(\theta_{kh})\|_2$ as follows

$$\|\nabla U(\theta_{kh})\|_2 \leq \frac{1}{1-tL}\|\nabla U(\theta_{kh+t})\|_2 + \frac{\sqrt{2t}L}{1-tL}\|\xi\|_2.$$

Taking squares and expectations above, then using that $(a+b)^2 \leq 2(a^2+b^2)$, it results

$$\begin{aligned} \mathbb{E}_{m_k} \left[\|\nabla U(\theta_{kh})\|_2^2 \right] &\leq \frac{2}{(1-tL)^2} \mathbb{E}_{n_{k,t}} \left[\|\nabla U(\theta_{kh+t})\|_2^2 \right] + \frac{4dL^2 t}{(1-tL)^2} \\ &\leq \frac{8}{(1-tL)^2} \mathcal{E}_{k,t} + \frac{4dL(1+tL)}{(1-tL)^2}, \end{aligned} \tag{3.13}$$

where we used Lemma 3.B.1 with probability measure $m = n_{k,t}$.

Plugging (3.11), (3.12) and (3.13) into (3.10), we observe that

$$\partial_t \{J_{k,t}\} \leq -4 \left(1 - \frac{1}{2\epsilon} - \frac{\epsilon t^2 L^2}{(1-tL)^2} \right) \mathcal{E}_{k,t} + \epsilon d L^2 t \frac{(1+3t^2L^2)}{(1-tL)^2}.$$

If we assume $h \leq \frac{1}{4L}$, then $1 - \frac{1}{2\epsilon} - \frac{\epsilon t^2 L^2}{(1-tL)^2} \geq 1 - \frac{1}{2\epsilon} - \frac{\epsilon}{9}$. We optimize ϵ so that $1 - \frac{1}{2\epsilon} - \frac{\epsilon}{9} \geq \frac{1}{2}$, then we get $\epsilon = \frac{3}{2}$ and finally conclude that

$$\partial_t \{J_{k,t}\} \leq -2\mathcal{E}_{k,t} + 4dL^2 t.$$

□

3.C Proofs of the results in Section 3.4

Proof of Proposition 3.4.1

We recall the Bobkov bound on the Poincaré constant C_{PI} for the log-concave probability μ:

$$C_{\mathrm{PI}} \leq 4K^2 Var_\mu(\theta),$$

where K is a universal constant; see Remark 1.2.4. We prove Proposition 3.4.1 using the Bobkov bound applied to μ and following the same strategy as in Proposition 2.4.2.

Proof. Since U satisfies hypothesis $\mathcal{H}^{\mathbf{r}}_{\mathbf{KL}}(\mathfrak{c}, L)$, Proposition 1.5.3 implies

$$\|\theta - \theta^*\|_2^2 \leq \frac{2}{(1+r)\mathfrak{c}} U^{1+r}(\theta), \quad \forall \theta \in \mathbb{R}^d,$$

where $\theta^* = \arg\min U$. We use the fact that for any distribution m, $Var_m(\theta) \leq \mathbb{E}_m[\|\theta - a\|_2^2]$, for any $a \in \mathbb{R}^d$, then

$$Var_\mu(\theta) \leq \int_{\mathbb{R}^d} \|\theta - \theta^*\|_2^2 \mathrm{d}\mu \leq \frac{2}{(1+r)\mathfrak{c}} \mathbb{E}_\mu[U^{1+r}(\theta)].$$

We formulate the following lemma to control the moments of U along the Langevin dynamics (3.2). This lemma is a particular case of Proposition 2.E.1, so its proof will be omitted.

Lemma 3.C.1. *Assume that U satisfies $\mathcal{H}^{\mathbf{r}}_{\mathbf{KL}}(\mathfrak{c}, L)$ and $\mathcal{H}_{\min}$. Then for any $\alpha \geq 1$ and any $t > 0$*

$$\mathbb{E}_{p_t}[U^\alpha(\vartheta_t)] \lesssim_{uc} d^{\alpha(1+r)},$$

where $p_t = Law(\vartheta_t)$ and ϑ_t is the Langevin dynamics in (3.2) *with initial distribution $p_0 = \mathcal{N}(0_d, \sigma^2 I_d)$.*

The ergodic behavior of $(\vartheta_t)_{t\geq 0}$ yields

$$Var_\mu(\theta) \leq \frac{2}{(1+r)\mathfrak{c}} \limsup_{t\geq 0} \mathbb{E}_{p_t}[U^{1+r}(\vartheta_t)] \lesssim_{uc} d^{(1+r)^2}.$$

We conclude using the Bobkov bound. □

Proof of Proposition 3.4.2

We first establish several preliminary considerations that will be used later.

The convexity of U guarantees that

$$U(z+y) \geq U(y) + \langle z, \nabla U(y)\rangle, \quad \forall y, z \in \mathbb{R}^d. \tag{3.14}$$

The L-smoothness of U implies that for any $z, y \in \mathbb{R}^d$,

$$\begin{aligned}\|\nabla U(y)\|_2 &\leq \|\nabla U(y) - \nabla U(z+y)\|_2 + \|\nabla U(z+y)\|_2 \\ &\leq L\|z\|_2 + \|\nabla U(z+y)\|_2.\end{aligned}$$

Taking square and using Young's inequality, for any $\epsilon > 0$,

$$\begin{aligned}\|\nabla U(y)\|_2^2 &\leq (L\|z\|_2 + \|\nabla U(z+y)\|_2)^2 \\ &\leq L^2\|z\|_2 + 2L\langle z, \nabla U(y)\rangle + \|\nabla U(z+y)\|_2^2 \\ &\leq \left(1 + \frac{1}{\epsilon}\right) L^2\|z\|_2^2 + (1+\epsilon)\|\nabla U(z+y)\|_2^2,\end{aligned}$$

to finally deduce

$$\|\nabla U(z+y)\|_2^2 \geq \frac{1}{1+\epsilon}\|\nabla U(y)\|_2^2 - \frac{L^2}{\epsilon}\|z\|_2^2. \tag{3.15}$$

On the other hand,

$$\langle z, \nabla U(z+y) - \nabla U(y)\rangle \leq L\|z\|_2^2, \quad z, y \in \mathbb{R}^d. \tag{3.16}$$

The second part of the proof is dedicated to proving the main result. For any $t \in [0, h]$ and $k \geq 0$

$$n_{k,t}(y) = \frac{1}{(4\pi t)^{\frac{d}{2}}} \int_{\mathbb{R}^d} m_k(x) e^{-\frac{1}{4t}\|y-x+t\nabla U(x)\|_2^2} \mathrm{d}x, \quad y \in \mathbb{R}^d.$$

From hypothesis $\mathcal{H}_{\mathbf{m_k}}(A_d, b)$, we have that

$$\begin{aligned}n_{k,t}(y) &\leq \frac{A}{(4\pi t)^{\frac{d}{2}}} \int_{\mathbb{R}^d} e^{-bU(x) - \frac{1}{4t}\|y-x+t\nabla U(x)\|_2^2} \mathrm{d}x \\ &\leq \frac{A}{(4\pi t)^{\frac{d}{2}}} \int_{\mathbb{R}^d} e^{-bU(z+y) - \frac{1}{4t}\|z-t\nabla U(z+y)\|_2^2} \mathrm{d}z,\end{aligned}$$

where we used the change of variable $z = x - y$ in the last step. For any $z, y \in \mathbb{R}^d$, the exponent satisfies the following inequality for any $\epsilon > 0$,

$$\begin{aligned}&- bU(z+y) - \frac{1}{4t}\|z - t\nabla U(z+y)\|_2^2 \\ &= -bU(z+y) - \frac{1}{4t}\left(\|z\|_2^2 - 2t\langle z, \nabla U(z+y)\rangle + t^2\|\nabla U(z+y)\|_2^2\right) \\ &\leq -bU(y) - \frac{t}{4(1+\epsilon)}\|\nabla U(y)\|_2^2 - b\langle z, \nabla U(y)\rangle - \frac{(1 - \frac{t^2L^2}{\epsilon})}{4t}\|z\|_2^2 + \frac{1}{2}\langle z, \nabla U(z+y)\rangle,\end{aligned}$$

where we replaced (3.14) and (3.15). Moreover, (3.16) implies that

$$
\begin{aligned}
&- bU(z+y) - \frac{1}{4t}\|z - t\nabla U(z+y)\|_2^2 \\
&\leq -bU(y) - \frac{t}{4(1+\epsilon)}\|\nabla U(y)\|_2^2 + \left(\frac{1}{2} - b\right)\langle z, \nabla U(y)\rangle - \frac{(1 - \frac{t^2L^2}{\epsilon})}{4t}\|z\|_2^2 \\
&+ \frac{1}{2}\langle z, \nabla U(z+y) - \nabla U(y)\rangle \\
&\leq -bU(y) - \frac{t}{4(1+\epsilon)}\|\nabla U(y)\|_2^2 + \left(\frac{1}{2} - b\right)\langle z, \nabla U(y)\rangle - \frac{(1 - 2t - \frac{t^2L^2}{\epsilon})}{4t}\|z\|_2^2.
\end{aligned}
$$

We suppose that $2t + \frac{t^2L^2}{\epsilon} < 1$, then,

$$
\begin{aligned}
&n_{k,t}(y) \\
&\leq \frac{Ae^{-bU(y) - \frac{t}{4(1+\epsilon)}\|\nabla U(y)\|_2^2}}{(1 - 2t - \frac{t^2L^2}{\epsilon})^{\frac{d}{2}}} \int_{\mathbb{R}^d} \left(\frac{1 - 2t - \frac{L^2t^2}{\epsilon}}{4\pi t}\right)^{\frac{d}{2}} e^{\left(\frac{1}{2} - b\right)\langle z, \nabla U(y)\rangle - \frac{(1 - 2t - \frac{t^2L^2}{\epsilon})}{4t}\|z\|_2^2} \mathrm{d}z \\
&\leq \frac{A}{(1 - 2t - \frac{t^2L^2}{\epsilon})^{\frac{d}{2}}} \exp\left\{-bU(y) - \frac{t}{4(1+\epsilon)}\|\nabla U(y)\|_2^2 + \frac{t\left(\frac{1}{2} - b\right)^2}{1 - 2t - \frac{t^2L^2}{\epsilon}}\|\nabla U(y)\|_2^2\right\},
\end{aligned}
$$

where the integral is the moment-generating function of a random variable $\mathcal{N}\left(0_d, \frac{2t}{1 - 2t - \frac{L^2t^2}{\epsilon}} I_d\right)$.

We chose $\epsilon = t$, then

$$
-\frac{t}{4(1+\epsilon)} + \frac{t\left(\frac{1}{2} - b\right)^2}{1 - 2t - \frac{t^2L^2}{\epsilon}} = -\frac{t}{4(1+t)} + \frac{t\left(\frac{1}{2} - b\right)^2}{1 - t(2 + L^2)} \leq 0,
$$

if $t \leq \frac{1 - 4(\frac{1}{2} - b)^2}{2 + L^2 + 4(\frac{1}{2} - b)^2}$ where $0 < b < 1$.

Using $\epsilon = t$ and $t \in [0, h]$, we conclude

$$
n_{k,t}(y) \leq \frac{A}{(1 - h(2 + L^2))^{\frac{d}{2}}} e^{-bU(y)}.
$$

For simplicity, we just consider $h \leq \frac{1 - 4(\frac{1}{2} - b)^2}{3 + L^2}$.

Proof of Proposition 3.4.3

The proof will be written for the particular case in which $t = 0$, that is, $n_{k,t} = m_k$. Although the proof for $n_{k,t}$ is equivalent.

We have

$$
\forall \theta \in \mathbb{R}^d, \quad \forall k \geq 0, \qquad m_k(\theta) \leq A_d e^{-bU(\theta)}.
$$

Using this inequality and the definition of U, we have that for any $p > 2$,

$$\left\| \sqrt{\frac{m_k}{\mu}} \right\|^p_{\mathbb{L}^p_\mu(\mathbb{R}^d)} = \int_{\mathbb{R}^d} \left| \frac{m_k(\theta)}{\mu(\theta)} \right|^{\frac{p}{2}} \mathrm{d}\mu(\theta) \leq A_d^{\frac{p}{2}} \mathcal{Z}^{\frac{p}{2}-1} \int_{\mathbb{R}^d} e^{-\left(1-\frac{p(1-b)}{2}\right)U(\theta)} \mathrm{d}\theta.$$

We have to choose a value of p below $\frac{2}{1-b}$ to guarantee the convergence of the above integral. For simplicity, we only prove the case $b < \frac{1}{2}$. We chose $p = \frac{1}{1-b}$.

In this case, the integral is

$$\int_{\mathbb{R}^d} e^{-\left(1-\frac{p(1-b)}{2}\right)U(\theta)} \mathrm{d}\theta = \int_{\mathbb{R}^d} e^{-\frac{1}{2}U(\theta)} \mathrm{d}\theta \leq 2d^{\frac{dr}{2}} \left(2^{2+r}\pi\mathfrak{c}^{-1}\right)^{\frac{d}{2}},$$

where we used the same type of computations as those used in the proof of Proposition 1.5.4. Similarly, we prove $\log(\mathcal{Z}) \geq \frac{d}{2}\log(2\pi/L) - \min U \gtrsim_{uc} d$, where we also used the hypothesis $\mathcal{H}_{\min}$.

Let us study the term $\log\left(\left\| \sqrt{\frac{m_k}{\mu}} \right\|^p_{\mathbb{L}^p_\mu(\mathbb{R}^d)}\right)$. Then,

$$\begin{aligned}\log\left(\left\| \sqrt{\frac{m_k}{\mu}} \right\|^p_{\mathbb{L}^p_\mu(\mathbb{R}^d)}\right) &\leq \frac{\log(A_d)}{2(1-b)} - \frac{1-2b}{2(1-b)}\log(\mathcal{Z}) + \log\left(2d^{\frac{dr}{2}}\left(2^{2+r}\pi\mathfrak{c}^{-1}\right)^{\frac{d}{2}}\right) \\ &\lesssim_{uc} \log(A_d) + d(1 + r\log(d)),\end{aligned}$$

which concludes the proof.

Chapter 4

Illustrative examples and numerical experiments

4.1 Introduction

This chapter is divided into two parts: Section 4.2 is devoted to the Stochastic Gradient Langevin Dynamics (SGLD) and Section 4.3 focuses on the Unadjusted Langevin algorithm (ULA). Both processes were introduced in Chapters 2 and 3, respectively. We recall that SGLD is a continuous-time process, while ULA is a discrete-time algorithm. The main objective of this chapter is to present an algorithm, some illustrative examples, and a simulation study for each process.

Section 4.2 summarizes some important definitions and properties of the continuous-time SGLD and then presents two examples. The first one has a pedagogical character: the potential function is quadratic, allowing for some terms to be computed explicitly. We also discuss the choice of the parameter α_n associated with the process. The second example is based on a Bayesian logistic regression problem, for which we establish a posterior distribution whose potential satisfies the hypothesis $\mathcal{H}^{\mathbf{r}}_{\mathbf{KL}}(\mathfrak{c}, L)$. In this context, it is not possible to find the explicit solution of the SGLD, therefore we rely on the Euler-Maruyama approximation with a small step size h, to graphically show the decay of the error of the Césaro average under the influence of dimension d and parameter r, which is associated with concavity.

Similarly, Section 4.3 discusses some basic ideas about ULA and provides two numerical experiments. The first example studies Bayesian logistic regression, but now using ULA and two data sets taken from the UCI repository. Once again, it is possible to observe the influence of the parameter r in hypothesis $\mathcal{H}^{\mathbf{r}}_{\mathbf{KL}}(\mathfrak{c}, L)$ through the mean squared error. The second example focuses on graphically showing the decay of the entropy when the hypotheses of Theorem 3.2.6 are verified. Since it is not possible to compute the entropy explicitly, we use an estimator to verify how it decreases when $t \to \infty$.

4.2 Stochastic Gradient Langevin Dynamics

We star this section by recalling some important facts and an equivalent description of the definition of continuous-time SGLD. For more details, we refer the reader to Section 2.1 and the discussion in Section 2.2. Using a Bayesian approach, we are interested in estimating the value θ^* of the parameter $\theta \in \mathbb{R}^d$ associated with the probability density p_θ.

Let $n \geq 1$. We observe n i.i.d. observations $\mathbf{X}_1, \mathbf{X}_2, \ldots, \mathbf{X}_n$ distributed according to p_{θ^*}. Given a prior distribution π_0 on $\mathbb{R}^d$, the posterior distribution μ_n is then defined as:

$$\mu_n(\theta) \propto \pi_0(\theta) \times \prod_{i=1}^{n} p_\theta(\mathbf{X}_i) \propto \exp(-U_{\nu_n}(\theta)), \quad \theta \in \mathbb{R}^d.$$

A popular alternative to estimate θ, is to sample μ_n using the Langevin Dynamics (LD)

$$\mathrm{d}\vartheta_t = -\nabla U_{\nu_n}(\vartheta_t)\mathrm{d}t + \sqrt{2}\mathrm{d}B_t, \quad t \geq 0,$$

where $(B_t)_{t\geq 0}$ is a standard Brownian motion in $\mathbb{R}^d$. Under some mild assumptions, μ_n is the unique invariant distribution of the LD.

In our work, we have decided to use SGLD instead of LD, since SGLD replaces the role of the function U_{ν_n} (which depends on all observations $\mathbf{X}_1, \mathbf{X}_2, \ldots, \mathbf{X}_n$) by U_{X_t} where X_t is a jump process that takes the value of a single unique observation chosen uniformly over $\mathbf{X}_1, \mathbf{X}_2, \ldots, \mathbf{X}_n$ during exponential times. Let us recall the formal definition of SGLD that appears in Chapter 2.

Consider two sequences of i.i.d. random variables $\left(T_j^{(n)}\right)_{j\geq 1}$ and $\left(V_j^{(n)}\right)_{j\geq 1}$, such that for any $j \geq 1$, the distribution of $T_j^{(n)}$ is exponential with mean α_n^{-1} and $V_j^{(n)}$ is uniformly distributed in $[n] = \{1, 2, \ldots, n\}$. We define $\tau_0^{(n)} = 0$ and denote

$$\tau_j^{(n)} = \sum_{k=1}^{j} T_k^{(n)}, \quad j \geq 1.$$

We define the process $(X_t)_{t\geq 0}$ as a jump process taking values in $\{\mathbf{X}_1, \ldots, \mathbf{X}_n\}$ such that:

$$X_t = \mathbf{X}_{V_j^{(n)}}, \qquad \text{if } \tau_{j-1}^{(n)} \leq t < \tau_j^{(n)}, \quad j \geq 1. \tag{4.1}$$

In simple terms, at times $\tau_0^{(n)}, \tau_1^{(n)}, \tau_2^{(n)}, \ldots$, the process $(X_t)_{t\geq 0}$ takes the value of an observation from $\mathbf{X}_1, \mathbf{X}_2, \ldots, \mathbf{X}_n$. We then define the continuous-time SGLD as the solution $(\theta_t, X_t)_{t\geq 0}$ of the SDE

$$\begin{aligned} &\mathrm{d}\theta_t = -\nabla_\theta U_{X_t}(\theta_t)\mathrm{d}t + \sqrt{2}\mathrm{d}B_t, \quad t > 0, \\ &\theta_0 \sim \mathcal{N}(0_d, \sigma^2 I_d), \end{aligned} \tag{4.2}$$

where $(B_t)_{t\geq 0}$ is a multivariate standard Brownian Motion and σ^2 is a positive constant that satisfies hypothesis $\mathcal{H}_{\mathbf{n_0}}(L, \ell_0)$.

We present the continuous-time SGLD algorithm in Algorithm 4.2.1, which is built using the Langevin over-damped diffusion (4.2).

Data: $(\mathbf{X}_1, \dots, \mathbf{X}_n)$ i.i.d. observations distributed according to p_{θ^*} , n_0 initial distribution, π_0 prior distribution

1 $\tau_0^{(n)} = 0$
2 Generate θ_0 according to n_0
3 **for** $j = 1, 2, \dots$ **do**
4 Pick $V_j^{(n)}$ uniformly in $[n] = \{1, \dots, n\}$
5 Generate $T_j^{(n)}$ according to an Exponential distribution with mean α_n^{-1}
6 $\tau_j^{(n)} = \tau_{j-1}^{(n)} + T_j^{(n)}$
7 $\theta_{\tau_j^{(n)}} = \theta_{\tau_{j-1}^{(n)}} - \int_{\tau_{j-1}^{(n)}}^{\tau_j^{(n)}} \nabla_\theta U_{\mathbf{X}_{V_j^{(n)}}}(\theta_s)\mathrm{d}s + \sqrt{2}B_{T_j^{(n)}}$
8 **end**
9 **return** $\lim_{j\to\infty} \theta_{\tau_j^{(n)}}$

Algorithm 4.2.1: Continuous-time SGLD.

The computational cost of Algorithm 4.2.1 increases with the value of n and it may be possible that all observations are not available at the initial time. So, an equivalent idea is to consider a new jump process $(Y_t)_{t\geq 0}$ such that, for all $t \in [0, \tau_n^{(n)})$, $X_t = Y_t$ in law. As we discussed in Section 2.2, we define $(Y_t)_{t\in[0,\tau_n^{(n)})}$ as an online algorithm. That is, at the initial instant of time an observation $\mathbf{X}_1$ becomes available and Y_t takes that value until time $\tau_1^{(n)}$, when $\mathbf{X}_2$ becomes available, and then Y_t takes that new value until time $\tau_2^{(n)}$, and so on, until $\tau_n^{(n)}$.

Formally, we define $(Y_t)_{t\in[0,\tau_n^{(n)})}$ as

$$Y_t = \mathbf{X}_j, \qquad \text{if } \tau_{j-1}^{(n)} \leq t < \tau_j^{(n)}, \quad 1 \leq j \leq n. \tag{4.3}$$

Using that the observations $\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_n$ are i.i.d. , we immediately deduce the property

$$\forall t \in [0, \tau_n^{(n)}), \quad X_t = Y_t \text{ in law.}$$

Then, we are able to replace Y_t by X_t in (4.2) to get that

$$\begin{aligned} &\mathrm{d}\theta_t = -\nabla_\theta U_{\mathbf{X}_j}(\theta_t)\mathrm{d}t + \sqrt{2}\mathrm{d}B_t, \quad \tau_{j-1}^{(n)} \leq t < \tau_j^{(n)}, 1 \leq j \leq n, \\ &\theta_0 \sim \mathcal{N}(0_d, \sigma^2 I_d), \end{aligned} \tag{4.4}$$

and introduce Algorithm 4.2.2, which is an alternative when n is large. Note that the observations are chosen sequentially in (4.4), which allows in Algorithm 4.2.2 to discharge step 4 of Algorithm 4.2.1.

Although it does not appear explicitly, we observe that a unique single observation $\mathbf{X}_j$ is required at time $\tau_{j-1}^{(n)}$, for $1 \leq j \leq n$. Therefore, we can describe the SGLD as an online algorithm on $[0, \tau_n^{(n)})$.

As we proved in Theorem 2.2.2, the continuous-time SGLD could be used to sample the measure μ_n, since the distribution of θ_t converges (in entropy) to μ_n, when $t \to \infty$. However, in practice, we do not have the explicit distribution of θ_t, but a sequence $(\theta_{\tau_j^{(n)}})_{j\geq 0}$.

Data: $(\mathbf{X}_1, \ldots, \mathbf{X}_n)$ i.i.d. observations distributed according to p_{θ^*} , n_0 initial distribution, π_0 prior distribution

1 $\tau_0^{(n)} = 0$
2 Generate θ_0 according to n_0
3 **for** $j = 1, 2, \ldots$ **do**
4 Generate $T_j^{(n)}$ according to an Exponential distribution with mean α_n^{-1}
5 $\tau_j^{(n)} = \tau_{j-1}^{(n)} + T_j^{(n)}$
6 $\theta_{\tau_j^{(n)}} = \theta_{\tau_{j-1}^{(n)}} - \int_{\tau_{j-1}^{(n)}}^{\tau_j^{(n)}} \nabla_\theta U_{\mathbf{X}_j}(\theta_s)\mathrm{d}s + \sqrt{2}B_{T_j^{(n)}}$
7 **end**
8 **return** $\lim_{j\to n} \theta_{\tau_j^{(n)}}$

Algorithm 4.2.2: Continuous-time SGLD (sequential alternative where step 4 of Algorithm 4.2.1 is ommited).

Then, a popular choice of estimator which requires only one path of $\left(\theta_{\tau_j^{(n)}}\right)_{j\geq 0}$ is the Césaro average

$$\widehat{\theta}_{n,\tau_j^{(n)}} = \frac{1}{j}\sum_{i=0}^{j} \theta_{\tau_i^{(n)}}, \quad j \geq 0.$$

For Algorithm 4.2.2, the Césaro average is well defined if $j < n$. To verify the convergence of the estimator towards the real value θ^* of the parameter θ, we rely on the squared error

$$e_{n,\tau_j^{(n)}} = \left\|\widehat{\theta}_{n,\tau_j^{(n)}} - \theta^*\right\|_2^2, \quad j \geq 0, \tag{4.5}$$

when we only generate a single path. While, when we generate N trajectories, $N > 1$, we then consider the mean squared error (MSE for short)

$$\mathrm{MSE}\left(\tau_j^{(n)}\right) = \frac{1}{N}\sum_{l=1}^{N} \left\|\widehat{\theta}^l_{n,\tau_j^{(n)}} - \theta^*\right\|_2^2, \quad j \geq 0, \tag{4.6}$$

where $\left(\widehat{\theta}^l_{n,\tau_j^{(n)}}\right)_{j\geq 0}$ is the Césaro average of the l-th path, for $1 \leq l \leq N$.

After clarifying these concepts, let us study two examples.

A pedagogical example

We illustrate our algorithms with a basic example in which the solution of continuous-time SGLD is explicit. We consider a fix value $\theta^* \in \mathbb{R}^d$, p_θ the probability density of a Gaussian random variable with mean θ and covariance matrix I_d, and constant $n \geq 1$. We observe n i.i.d. random variables $\mathbf{X}_1, \mathbf{X}_2, \ldots, \mathbf{X}_n$ with distribution p_{θ^*}.

We suppose that the prior distribution π_0 is a standard normal density function,

then the posterior distribution is

$$\mu_n(\theta) \propto \exp\left\{-\frac{1}{n}\sum_{i=1}^{n}\frac{1+n}{2}\left\|\theta - \frac{n}{1+n}\mathbf{X}_i\right\|_2^2\right\}, \quad \theta \in \mathbb{R}^d.$$

Using our notation above, we deduce that

$$U_x(\theta) = \frac{1+n}{2}\left\|\theta - \frac{nx}{1+n}\right\|_2^2, \quad \theta \in \mathbb{R}^d, \quad x \in \mathbb{R}^d,$$

and then the continuous-time SGLD is the solution of the stochastic differential equation

$$\begin{aligned}
&\mathrm{d}\theta_t = -\nabla_\theta U_{\mathbf{X}_{V_j^{(n)}}}(\theta_t)\mathrm{d}t + \sqrt{2}\mathrm{d}B_t, \quad \tau_{j-1}^{(n)} \le t < \tau_j^{(n)}, \quad j \ge 1,\\
&\theta_0 \sim \mathcal{N}(0_d, \sigma^2 I_d),
\end{aligned}$$

where we defined $\left(V_j^{(n)}\right)_{j\ge1}$, $\left(T_j^{(n)}\right)_{j\ge1}$ and $(\tau_j^{(n)})_{1\le j\le n}$ as above and we have chosen $\sigma^2 = \frac{1}{2(n+1)}$, to satisfy the hypothesis $\mathcal{H}_{\mathbf{n_0}}(L, \ell_0)$. Using the explicit function U, the previous equation in the time interval $[\tau_{j-1}^{(n)}, \tau_j^{(n)})$, for any $j \ge 1$, takes the form

$$\mathrm{d}\theta_t = (1+n)\left(\frac{n}{1+n}\mathbf{X}_{V_j^{(n)}} - \theta_t\right)\mathrm{d}t + \sqrt{2}\mathrm{d}B_t,$$

whose exact solution at time $t \in (\tau_{j-1}^{(n)}, \tau_j^{(n)})$ is

$$\theta_t = \theta_{\tau_{j-1}^{(n)}} e^{-(1+n)(t-\tau_{j-1}^{(n)})} + \frac{n}{1+n}\mathbf{X}_{V_j^{(n)}}(1 - e^{-(1+n)(t-\tau_{j-1}^{(n)})}) + \sqrt{2}e^{-(1+n)t}\int_{\tau_{j-1}^{(n)}}^{t} e^{(1+n)s}\mathrm{d}B(s).$$

We take $t = \tau_j^{(n)}$ and deduce that

$$\left(\theta_{\tau_j^{(n)}} | \theta_{\tau_{j-1}^{(n)}}, \mathbf{X}_{V_j^{(n)}}, \tau_j^{(n)}, \tau_{j-1}^{(n)}\right) \sim \mathcal{N}(\mu_j, \sigma_j^2)$$

where, for any $j \ge 1$,

$$\begin{aligned}
\mu_j &= \theta_{\tau_{j-1}^{(n)}} e^{-(1+n)T_j^{(n)}} + \frac{n}{1+n}\mathbf{X}_{V_j^{(n)}}(1 - e^{-(1+n)T_j^{(n)}}),\\
\sigma_j^2 &= \frac{1}{1+n}\left(1 - e^{-2(1+n)T_j^{(n)}}\right).
\end{aligned}$$

Similarly, we could deduce the distribution of $\left(\theta_t | \theta_{\tau_{j-1}^{(n)}}, \mathbf{X}_{V_j^{(n)}}, \tau_j^{(n)}, \tau_{j-1}^{(n)}\right)$ for any $t \in (\tau_{j-1}^{(n)}, \tau_j^{(n)})$, $j \ge 1$. Although the conditional distribution of θ_t is known, it seems almost impossible to find a general expression of n_t the distribution of θ_t. Therefore, we observe that even in a simple case it is very difficult to find J_t explicitly.

Returning to μ_n, we verify that we can rewrite

$$\mu_n(\theta) \propto \exp\left\{-\frac{1+n}{2}\left\|\theta - \frac{n}{1+n}\overline{\mathbf{X}}_n\right\|_2^2\right\}, \quad \theta \in \mathbb{R}^d,$$

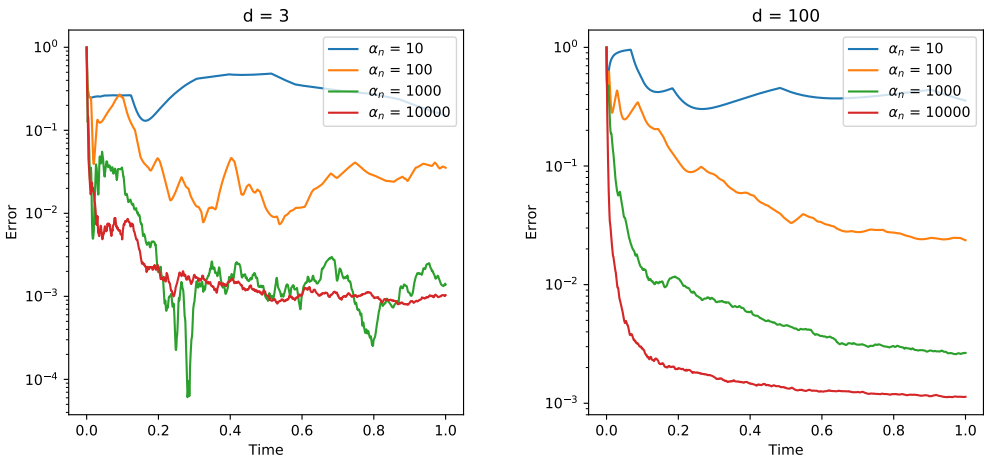
where $\overline{\mathbf{X}}_n = \frac{1}{n}\sum_{i=1}^n \mathbf{X}_i$. Then U_{ν_n} satisfies a $\mathcal{H}^{\mathbf{r}}_{\mathbf{KL}}(\mathfrak{c}, L)$-condition with $r = 0$ and μ_n satisfies an LSI with constant $\frac{2}{n+1}$. Theorem 2.2.2 shows that for any $\varepsilon > 0$, if $\alpha_n = n$ and

$$t \gtrsim_{uc} n^{-1}\left[\log(\varepsilon^{-1}) + \log(d) + \log(n)\right],$$

then $J_t \leq \varepsilon$.

Let us now illustrate Algorithm 4.2.1. To do this, we first generate a single path $(\theta_{\tau_j^{(n)}})_{j\geq 0}$ and verify that the squared error, defined in (4.5), decreases as time increases. We consider $n = 1000$ i.i.d. observations, the real value $\theta^* = (1, 1, \ldots, 1) = 1_d$ of the parameter θ and two values of the dimension d, $d = 3$ and $d = 100$. Theorem 2.2.2 states that the optimal choice of α_n to obtain the convergence in entropy of the SGLD is $\alpha_n = n$. Although here we only observe the decay of the squared error of a path and not the entropy itself, let us compare different choices of α_n.

We simulate a path of the continuous-time SGLD for different values of α_n. In Figure 4.2.1, we observe the decay of the squared error of the Césaro average for each path. A large value of α_n means that the process (X_t) has a large number of jumps and therefore the error decreases rapidly, although as shown in Table 4.2.1, the computational cost is higher. On the other hand, a small value of α_n means that the process (X_t) has few jumps, the run-time is small, but the error is greater. As we expected, it seems that the value of $\alpha_n = n$ is the one that manages to balance the computational cost and a relatively small error, which is consistent with the result of Theorem 2.2.2.



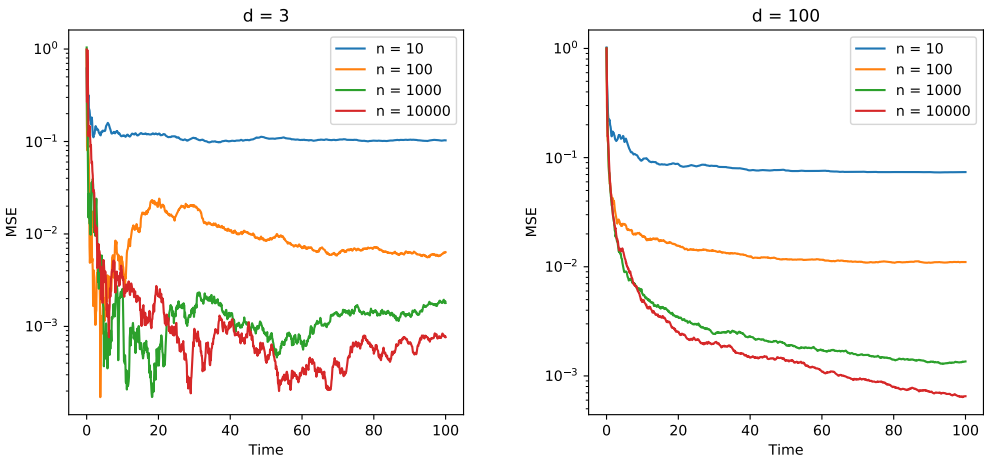
(a) (b)

Figure 4.2.1: Squared error of the Césaro average of a path of the continuous-time SGLD where $n = 1000$, $\theta^* = 1_d$ and d is 3 and 100, respectively.

On the other hand, we generate $N = 100$ SGLD paths for different values of n and d, to corroborate the influence of the number of observations on the order of convergence (of the mean squared error). Each of the paths was built with the corresponding $\alpha_n = n$ and we can observe in Figure 4.2.2 that the MSE is lower as n increases.

α_n	$d = 3$		$d = 100$	
	error	Runtime (in s)	error	Runtime (in s)
10	10^{-1}	0.25	10^{-1}	1.13
10^2	10^{-2}	0.28	10^{-2}	1.25
10^3	10^{-3}	0.37	10^{-3}	2.04
10^4	10^{-3}	1.31	10^{-3}	9.92

Table 4.2.1: Order of the squared error divided by d of the Césaro average at time $t = 1$ of a path of the continuous-time SGLD where $n = 1000$ and $\theta^* = 1_d$.



(a) (b)

Figure 4.2.2: Mean squared error of $N = 100$ path of the continuous-time SGLD where $\alpha_n = n$, $\theta^* = 1_d$ and d is 3 and 100, respectively.

Bayesian logistic regression

The second example is an application of the continuous-time SGLD to the Bayesian logistic regression. To do this, we rely on the example studied in [Durmus et al., 2019]. We consider $n \geq 1$ i.i.d. observations (X_1, Y_1), (X_2, Y_2), ..., (X_n, Y_n) such that $X_1, X_2, \ldots, X_n$ are d-dimensional input variables and $Y_1, \ldots, Y_n$ are binary output responses. Logistic regression framework amounts to assume the responses to be distributed as Bernoulli random variables such that

$$Y_i \sim \text{Ber}\left(\phi(\theta^\top X_i)\right), \quad i \in \{1, \ldots, n\},$$

where ϕ is the logit function defined by $\phi(x) = (1 + e^{-x})^{-1}$, $x \in \mathbb{R}$ and $\theta \in \mathbb{R}^d$ is the parameter of interest. We consider a prior distribution given by the following

density with respect to Lebesgue measure on $\mathbb{R}^d$,

$$p_r(\theta) \propto \exp\left\{-\left(1+\|\theta\|_2^2\right)^{\frac{1}{1+r}}\right\}, \quad \theta \in \mathbb{R}^d, \quad 0 \leq r < 1.$$

The variations of p_r with r enable to address situations from strongly log-concave posterior distributions ($r = 0$) to weakly log-concave ones ($0 < r < 1$). This choice of prior leads to the log-concave posterior distribution of θ which is given by the density

$$\mu_n(\theta) = p_r\left(\theta|(X_1, Y_1), \ldots, (X_n, Y_n)\right) \propto \exp\left\{-\sum_{i=1}^n \ell_i(\theta) - \left(1+\|\theta\|_2^2\right)^{\frac{1}{1+r}}\right\},$$

for $\theta \in \mathbb{R}^d$ and $0 \leq r < 1$, where the log-likelihoods are given by

$$\ell_i(\theta) = \log\left(1+\exp\left((1-2Y_i)\theta^\top X_i\right)\right).$$

We now introduce the potential

$$U_{\nu_n}(\theta) = \sum_{i=1}^n \ell_i(\theta) + \left(1+\|\theta\|_2^2\right)^{\frac{1}{1+r}}, \quad \theta \in \mathbb{R}^d,$$

which verifies a $\mathcal{H}^{\mathbf{r}}_{\mathbf{KL}}(\mathfrak{c}, L)$-condition. Since $U_{\nu_n}(\theta) = \frac{1}{n}\sum_{i=1}^n U_{(X_i,Y_i)}(\theta)$, then

$$U_{(X_i,Y_i)}(\theta) = n\ell_i(\theta) + \left(1+\|\theta\|_2^2\right)^{\frac{1}{1+r}}, \quad \theta \in \mathbb{R}^d.$$

The associated gradient defined by $\nabla U_{(X_i,Y_i)}(\theta) = (\partial_1 U_{(X_i,Y_i)}(\theta), \ldots, \partial_d U_{(X_i,Y_i)}(\theta))^\top$ is given by

$$\partial_j U_{(X_i,Y_i)}(\theta) = n(1-2Y_i)\phi\left((1-2Y_i)\theta^\top X_i\right) X_{ij} + \frac{2}{1+r}\left(1+\|\theta\|_2^2\right)^{\frac{-r}{1+r}}\theta_j,$$

where X_{ij} and θ_j are the j-th component of X_i and θ respectively, for any $1 \leq i, j \leq n$ and $\theta \in \mathbb{R}^d$. Using the definition of $(\tau_i^{(n)})_{1\leq i\leq n}$ and Algorithm 4.2.2, the continuous-time SGLD evolves according to the equation

$$\mathrm{d}\theta_t = -\left(n(1-2Y_i)\phi\left((1-2Y_i)\theta_t^\top X_i\right) X_i + \frac{2}{1+r}\left(1+\|\theta_t\|_2^2\right)^{\frac{-r}{1+r}}\theta_t\right)\mathrm{d}t + \sqrt{2}\mathrm{d}B_t,$$

for any $t \in (\tau_{i-1}^{(n)}, \tau_i^{(n)})$, $1 \leq i \leq n$ and $\theta_0 \sim \mathcal{N}(0_d, \sigma^2 I_d)$ where $\sigma^2 = \frac{1}{n\max_{i,j} X_{ij}^2}$. As we commented before, the choice of σ^2 is according to hypothesis $\mathcal{H}_{\mathbf{n_0}}(L, \ell_0)$. We conclude that $J_t \leq \varepsilon$ if one of the following two conditions is satisfied

1. $r = 0$, $\alpha_n = n$ and $t \gtrsim_{uc} n^{-1}\left[\log(\varepsilon^{-1}) + \log(d) + \log(n)\right]$,

2. $0 < r < 1$, $\alpha_n > 0$ and

$$t \gtrsim_{uc} \max\left\{\alpha_n^{-2} C_{PI}^{-1}(\mu_n), C_{PI}(\mu_n)\left[\log^2(\varepsilon^{-1}) + n^2\left(d\log^{2\beta}(n)\right)^{2(1+r)}\right]\right\},$$

where $C_{PI}(\mu_n)$ is the Poincaré constant of μ_n.

The solution of the continuous-time process is not explicit; we then approximate it using the Euler-Maruyama discretization with a sufficiently small step size $h > 0$. That is,

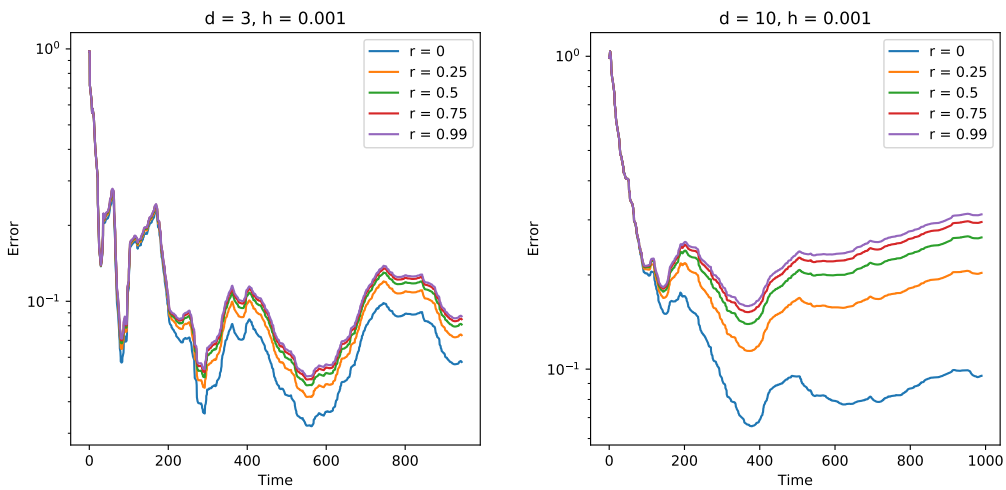
$$\theta_{\tau_{i-1}^{(n)}+(k+1)h} = \theta_{\tau_{i-1}^{(n)}+kh} - h\nabla U_{(X_i,Y_i)}\left(\theta_{\tau_{i-1}^{(n)}+kh}\right) + \sqrt{2h}\xi_{k+1,i}, \tag{4.7}$$

for any $k \geq 0$ and $1 \leq i \leq n$ such that $\tau_{i-1}^{(n)} + kh \leq \tau_i^{(n)}$ and $(\xi_{k,i})_{k\geq 1, 1\leq i\leq n}$ is a sequence of i.i.d. standard Gaussian random variables in $\mathbb{R}^d$. For the numerical experiments, we chose $\theta^* = 1_d$, $n = 1000$, and $\alpha_n = 1$. So, we generate the i.i.d. observations (X_1, Y_1), (X_2, Y_2), $\ldots$, (X_n, Y_n) such that $X_i \sim \mathcal{N}(0_d, I_d)$ and $Y_i \sim \text{Ber}\left(\phi((\theta^*)^\top X_i)\right)$, for $1 \leq i \leq n$. The Césaro average associated with the sequence $\left(\theta_{\tau_{i-1}^{(n)}+kh}\right)_{k,i}$ is

$$\widehat{\theta}^h_{\tau_{i-1}^{(n)}+kh} = \frac{1}{N_{k,i}} \sum_{k',i'} \theta_{\tau_{i'-1}^{(n)}+k'h}, \quad 1 \leq i \leq n, \quad k \geq 0,$$

where the sum is considered for k' and i' such that $\tau_{i'-1}^{(n)} + k'h \leq \tau_{i-1}^{(n)} + kh$ and $N_{k,i}$ is the number of times $\tau_{i'-1}^{(n)} + k'h$ below $\tau_{i-1}^{(n)} + kh$.

We focus on the convergence of the process with respect to r and d. Then, for any value of $r \in \{0, 0.25, 0.5, 0.75, 0.99\}$ and $d \in \{3, 10\}$, we first generate a path using the recursive equation (4.7), the sequential alternative of 4.2.2, and then we compute the squared error of the Césaro average.



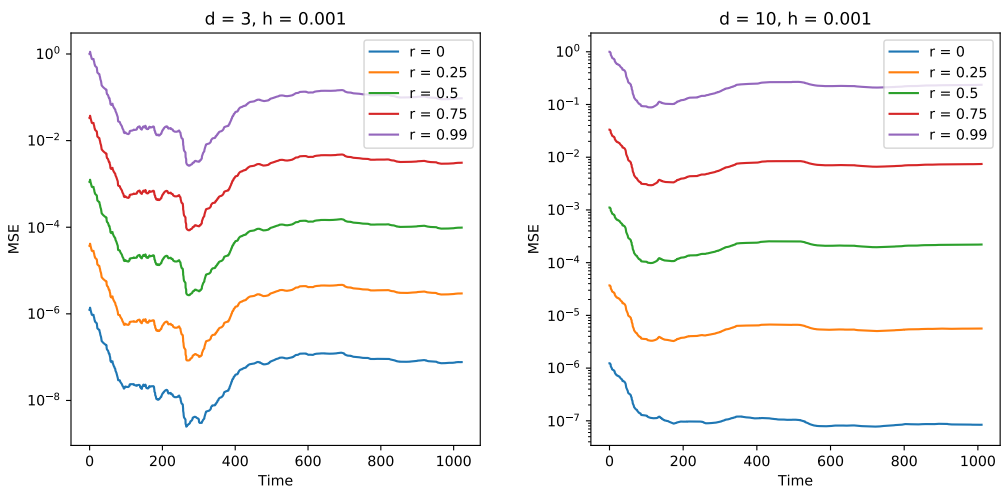
(a) (b)

Figure 4.2.3: Squared error of the Césaro average of a path of the continuous-time SGLD where $n = 1000$, $\alpha_n = 1$ and $\theta^* = 1_d$.

In Figure 4.2.3, we observe that the error decreases. When t is large, the curves are ordered according to the value of r, the curve for $r = 0$ is the one with the least

error since it corresponds to the strongly log-concave situation. The constant h is 10^{-3}, while α_n remained constant and equal to 1 in order to be able to compare all the trajectories for each value of r. We emphasize that we are generating the paths using the same seed, which guarantees that the difference between the curves is a consequence of the parameter r.

Similarly, we can now generate $N = 30$ SGLD paths and confirm in Figure 4.2.4, the decay of the MSE for the different values of r. Once again, the curves appear to be ordered with respect to the value of r, where the curve corresponding to $r = 0$ is the one with the lowest MSE.



(a) (b)

Figure 4.2.4: Mean squared error of $N = 30$ paths of the continuous-time SGLD where $n = 1000$, $\alpha_n = 1$ and $\theta^* = 1_d$.

These graphs agree with the results established in our Theorem 2.2.2 and show that the SGLD process is an alternative to using Langevin Dynamics, which requires using all n observations.

4.3 Unadjusted Langevin Algorithm

In this section, we study two numerical experiments based in the unadjusted Langevin algorithm. Let us recall some essential definitions and refer to Chapter 3 for further details. The Langevin Dynamics (LD) is defined by the stochastic differential equation

$$\mathrm{d}\vartheta_t = -\nabla U(\vartheta_t)\mathrm{d}t + \sqrt{2}\mathrm{d}B_t, \quad t \geq 0, \tag{4.8}$$

where $(B_t)_{t\geq 0}$ is a standard Brownian motion in $\mathbb{R}^d$, $U : \mathbb{R}^d \to \mathbb{R}$ is a $\mathcal{C}^2(\mathbb{R}^d)$ convex function and ∇U is L-Lipschitz. Under these conditions, the continuous-time process $(\vartheta_t)_{t\geq 0}$ converges in many senses (Wasserstein, $\mathbb{L}^2_\mu(\mathbb{R}^d)$, entropy, total variation, ...) towards its invariant measure

$$\mu(\theta) \propto e^{-U(\theta)}, \quad \theta \in \mathbb{R}^d.$$

In general, the solution of the LD is not explicit, which leads to using a numerical approximation of (4.8). In particular, we focus on its Euler-Maruyama approximation, also known as ULA and defined as

$$\begin{aligned} \theta_{(k+1)h} &= \theta_{kh} - h\nabla U(\theta_{kh}) + \sqrt{2h}\xi_{k+1}, \quad k \geq 0, \\ \theta_0 &= \sigma_0 \xi_0, \end{aligned} \tag{4.9}$$

where $h > 0$ is a sufficiently small step size, $(\xi_k)_{k\geq 0}$ is a sequence of i.i.d. standard Gaussian random variables in $\mathbb{R}^d$ and σ_0 is chosen according to hypothesis $\mathcal{H}^{\mathbf{r}}_{\mathbf{m_0}}(\mathfrak{c}, L)$. We now present Algorithm 4.3.1 which is based on the recursive equation 4.9.

Data: h step size, σ_0 parameter of the initial distribution, U potential
$\theta_0 = \sigma_0 \xi_0$
for $k = 0, 1, \ldots$ **do**
  $\theta_{(k+1)h} = \theta_{kh} - h\nabla U(\theta_{kh}) + \sqrt{2h}\xi_k$
end
return $\lim_{k\to\infty} \theta_{kh}$

Algorithm 4.3.1: ULA

We denote as J_k the relative entropy between the distribution of θ_{kh} and μ. In Theorem 3.2.6, we proved that for any $\varepsilon > 0$ sufficiently small, if k is large and h is small, then $J_k \leq \varepsilon$.

Next, we study the application of the ULA to the Bayesian logistic regression problem presented before, but using two real datasets and a simulation study in a synthetic situation where we can estimate numerically and verify the decay of the entropy.

In our examples, we again relay on the Césaro average associated with a path $(\theta_{kh})_{k\geq 0}$, defined as

$$\widehat{\theta}_{kh} = \frac{1}{k+1} \sum_{j=0}^{k} \theta_{jh}, \quad k \geq 0,$$

and the mean squared error (MSE) of N paths, that is, for any $k \geq 0$ at time kh,

$$\mathrm{MSE}(kh) = \frac{1}{N} \sum_{l=1}^{N} \|\widehat{\theta}^l_{kh} - \theta^*\|_2^2,$$

where $\left(\widehat{\theta}^l_{kh}\right)_{k\geq 0}$ is the Césaro average of the l-th path, $1 \leq l \leq N$.

Bayesian logistic regression

We are interested in applying ULA to the Bayesian logistic regression problem studied in [Durmus et al., 2019] (we refer also to [Held and Holmes, 2006, Gramacy and Polson, 2012, Park and Hastie, 2007]).

In the context of the Bayesian logistic regression presented in the previous section, we recall that $(X_1, Y_1), \ldots, (X_n, Y_n)$ are observation i.i.d. where $X_1, \ldots, X_n$

are d-dimensional input variables and $Y_1, \dots, Y_n$ are binary output responses such that $Y_i \sim \text{Ber}\left(\phi(\theta^\top X_i)\right)$, for all $1 \leq i \leq n$, ϕ is the logit function and θ is the parameter of interest.

We consider the following prior distribution on $\mathbb{R}^d$,

$$p_r(\theta) \propto \exp\left\{-(1+\|\theta\|_2^2)^{\frac{1}{1+r}}\right\}, \quad \theta \in \mathbb{R}^d,$$

where $r \in [0,1)$. This choice of prior results in a log-concave posterior distribution for θ, described by the density

$$\mu(\theta) = p_r\left(\theta|(X_1,Y_1),\dots,(X_n,Y_n)\right) \propto \exp\left\{-\sum_{i=1}^n \ell_i(\theta) - (1+\|\theta\|_2^2)^{\frac{1}{1+r}}\right\},$$

where $\ell_i(\theta) = \log\left(1+\exp\left((1-2Y_i)\theta^\top X_i\right)\right)$, $\theta \in \mathbb{R}^d$. The posterior distribution is strongly log-concave if $r=0$ and weakly log-concave if $0<r<1$. The potential is defined by

$$U_r(\theta) = \sum_{i=1}^n \ell_i(\theta) + (1+\|\theta\|_2^2)^{\frac{1}{1+r}},$$

which verifies a $\mathcal{H}^{\mathbf{r}}_{\mathbf{KL}}(\mathfrak{c}, L)$-condition. The associated gradient defined by $\nabla U_r(\theta) = (\partial_1 U_r(\theta), \dots, \partial_d U_r(\theta))^\top$ is given by

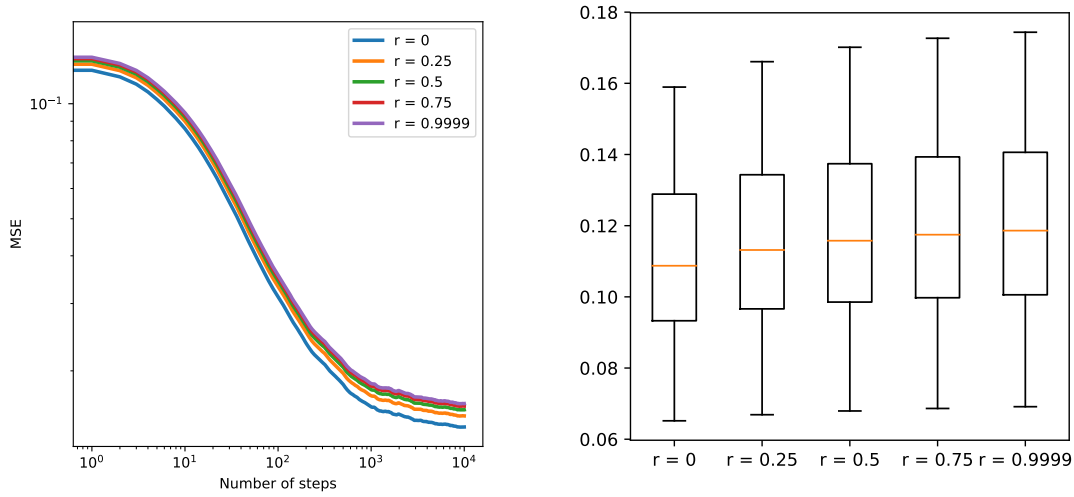
$$\partial_j U_r(\theta) = \sum_{i=1}^n (1-2Y_i)\phi\left((1-2Y_i)\theta^\top X_i\right) X_{ij} + \frac{2}{1+r}\left(1+\|\theta\|_2^2\right)^{\frac{-r}{1+r}} \theta_j,$$

where X_{ij} and θ_j are the j-th component of X_i and θ respectively, for any $1 \leq j \leq d$. Given a step size $h>0$ and an initial variance factor $\sigma_0^2 = \frac{1}{n \max_{i,j} X_{ij}^2}$, we introduce the sequence $(\theta_{kh})_{k\geq 0}$ defined by the following recursion,

$$\begin{aligned}
&\theta_{(k+1)h} = \theta_{kh} - h\nabla U_r(\theta_{kh}) + \sqrt{2h}\xi_{k+1}, \ k \geq 0, \\
&\theta_0 = \sigma_0 \xi_0,
\end{aligned}$$

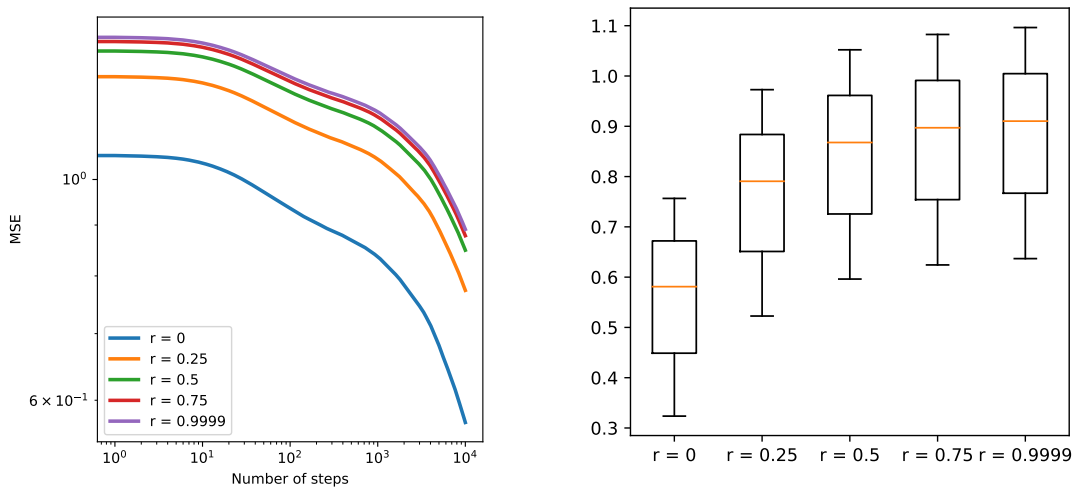
where $(\xi_k)_{k\geq 0}$ is a sequence of i.i.d. standard Gaussian random variables in $\mathbb{R}^d$.

We consider two data sets from the UCI repository: the Heart Disease Data Set with Dimensions $n = 214$ and $d = 13$ (see [Janosi et al., 1988]) and the Musk Data Set with Dimensions $n = 6598$ and $d = 166$ (see [Chapman and Jain, 1994]). In both cases, we standardize the data and run 100 independent ULA trajectories for different values of r. For each data set and for each r, we compared the results with a *true value* obtained by performing $K = 10^6$ steps with $h = 10^{-4}$ for the heart disease data set ($K = 10^5$ and $h = 10^{-5}$ for the Musk data set) and taking the Césaro average over the last 10^5 (respectively 10^4) values. The results are presented in Figures 4.3.1 and 4.3.2.



(a) (b)

Figure 4.3.1: (a) Mean squared error of the first component estimator at time k, $\widehat{\theta_k^{(1)}}$, for Heart disease dataset where we run 100 trajectories with step size $h = 10^{-3}$ and initial standard deviation factor $\sigma = 0.2$. (b) Boxplot of the absolute error of the estimator at time $K = 10^4$.



(a) (b)

Figure 4.3.2: (a) Mean squared error of the first component estimator at time k, $\widehat{\theta_k^{(1)}}$, for Musk dataset where we run 100 trajectories with step size $h = 10^{-5}$ and initial standard deviation factor $\sigma = 0.2$. (b) Boxplot of the absolute error of the estimator at time $K = 10^4$.

We observe that in both cases the mean squared error has a tendency to decrease as the number of ULA steps increases. The convergence of the error for the Musk

dataset is slower also due to the large size of the dimension d. In the strongly convex situation the convergence is faster, and the curve is apparently different from the rest of the curves. Likewise, the absolute error is much lower in the case $r = 0$. These numerical results show the complexity of the study when dealing with a weakly log-concave distribution compared to a strongly log-concave one.

Synthetic example and entropy convergence

We are interested in verifying the decay of the entropy obtained in Theorem 3.2.6 by simulation. It is important to mention that so far we have not found a numerical study in the related literature in which entropy is estimated.

We consider $0 \leq r < 1$ and define the potential function $U_r : \mathbb{R}^d \to \mathbb{R}$ as

$$U_r(\theta) = (1 + \|\theta\|^2)^{\frac{1}{1+r}},$$

which verifies a $\mathcal{H}^{\mathbf{r}}_{\mathbf{KL}}(\mathfrak{c}, L)$-condition. Then its associated gradient is given by $\nabla U_r(\theta) = \frac{2}{1+r} U_r^{-r}(\theta)\theta$. This choice of potential leads to the log-concave density function

$$\mu_r(\theta) = \frac{e^{-U_r(\theta)}}{\mathcal{Z}},$$

where the normalizing constant could be computed using a change to d-dimensional spherical coordinates and, in the case $0 < r < 1$, we approximated the integral for the radial coordinate x,

$$\mathcal{Z}_r = \int_{\mathbb{R}^d} e^{-U_r(\theta)} \mathrm{d}\theta = \begin{cases} \pi^{d/2} e^{-1}, & \text{if } r = 0, \\ \frac{d\pi^{d/2}}{\Gamma(d/2+1)} \int_0^\infty x^{d-1} e^{-(1+x^2)^{\frac{1}{1+r}}} \mathrm{d}x, & \text{if } 0 < r < 1. \end{cases}$$

Once again, the value of r allows us to describe from strongly log-concave distributions ($r = 0$) to weakly log-concave ones ($0 < r < 1$).

Given a step size $h > 0$ and an initial standard deviation factor $\sigma > 0$, we introduce the sequence $(\theta_{kh})_{k\geq 0}$ defined by the following recursion,

$$\theta_0 = \sigma\xi_0 \quad \text{and} \quad \theta_{(k+1)h} = \theta_{kh} - \frac{2h}{1+r} U_r^{-r}(\theta_{kh})\theta_{kh} + \sqrt{2h}\xi_{k+1}, \ k \geq 0,$$

where $(\xi_k)_{k\geq 0}$ is a sequence of i.i.d. standard Gaussian random variables in $\mathbb{R}^d$.

We recall that the entropy at any step $k \geq 0$ is defined in (3.4) as

$$J_k = \int_{\mathbb{R}^d} \log\left(\frac{m_k(\theta)}{\mu_r(\theta)}\right) \mathrm{d}m_k(\theta),$$

where m_k is the density function of θ_{kh}. Then, if we observe $N \geq 1$ independent trajectories $(\theta^1_{kh})_{k\geq 0}, (\theta^2_{kh})_{k\geq 0}, \ldots, (\theta^N_{kh})_{k\geq 0}$, we can estimate at any step k, the density function m_k as follows

$$\widehat{m}^N_k(\theta) = \frac{1}{N\delta} \sum_{i=1}^N \gamma_1\left(\frac{\theta - \theta^i_{kh}}{\delta}\right),$$

where γ_1 is the standard normal density function and the bandwidth δ is chosen so that it minimizes the mean integrated squared error (for a more in-depth study see [Scott, 2015, Silverman, 2018]).

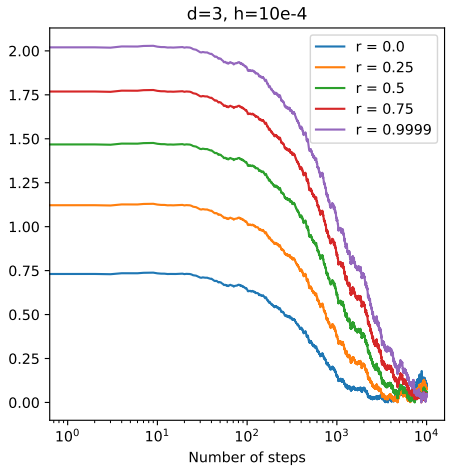
We approximate the entropy using

$$\widehat{J}_k^N = \frac{1}{N}\sum_{i=1}^{N} \log\left[\frac{\widehat{m}_k^N(\theta_{kh}^i)}{\mu_r(\theta_{kh}^i)}\right] = \frac{1}{N}\sum_{i=1}^{N}\left[\log(\widehat{m}_k^N(\theta_{kh}^i)) + U_r(\theta_{kh}^i)\right] + \log(\mathcal{Z}_r).$$

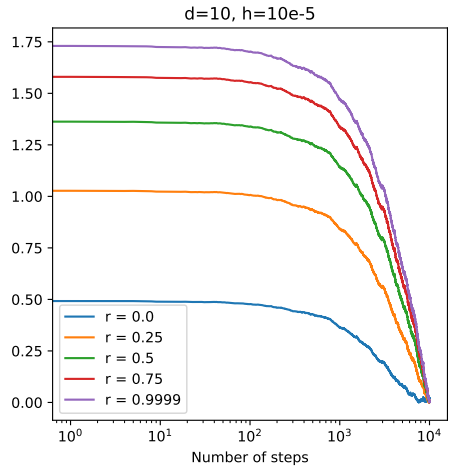
Of course we expect $\widehat{J}_k^N$ to be biased, since for fixed (even if small) h, m_k is the distribution of a Markov chain with invariant measure $\mu_h \neq \mu$. Therefore, in the simulations we will focus on the order of convergence of $\widehat{J}_k^N$ when the number of iterations k increases, which should be approximately proportional to $k^{-1}d^{1+2(1+r)^2}$ when $0 < r < 1$ and $k^{-1}d$ when $r = 0$.

We consider three dimensions of space $d = 3$, $d = 10$ and $d = 30$ with step sizes $h = 10^{-4}$, $h = 10^{-5}$ and $h = 10^{-7}$ respectively, initial standard deviation factor $\sigma = 0.2$ and fives values of the parameter r. We simulate $N = 100$ trajectories of $(\theta_{kh})_{k\geq 0}$ and for each k we estimate $\widehat{J}_k^N$. The results are presented in Figure 4.3.3.

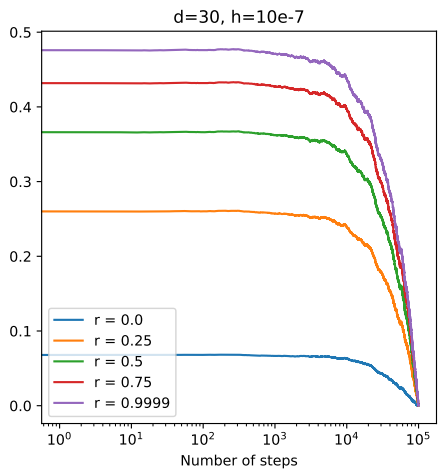
We observe that the entropy estimator decreases as the number of steps increases, with the convergence of the curve apparently being faster for $r = 0$. The curves are arranged according to the value of r, as expected, while the increasing dimension d is reflected in the need to use a decreasing value of h and in the slope of the curve.



(a)



(b)



(c)

Figure 4.3.3: Convergence of $\widehat{J}_k^N$ where the initial standard deviation factor is $\sigma = 0.2$ and the number of trajectories is $N = 100$.

Chapter 5

Perspectives

Some questions remain open, for example:

- After studying a continuous-time version of the SGLD in a $\mathcal{H}^{\mathbf{r}}_{\mathbf{KL}}(\mathfrak{c}, L)$ setting, we could find the order of convergence in the entropy of the discrete-time version. Furthermore, we could also consider the Rénji divergence, which is a stronger distance.

- It would be interesting to study the convergence of penalized versions of SGLD and ULA in the $\mathcal{H}^{\mathbf{r}}_{\mathbf{KL}}(\mathfrak{c}, L)$ framework.

- Another possible direction for future work is to prove the inequality

$$m_k(\theta) \leq A_d e^{-bU(\theta)}, \quad \theta \in \mathbb{R}^d,$$

  when U satisfies condition $\mathcal{H}^{\mathbf{r}}_{\mathbf{KL}}(\mathfrak{c}, L)$. Moreover, if we improve the bound of the Poincaré constant, it may be possible to achieve a better convergence rate for ULA in our $\mathcal{H}^{\mathbf{r}}_{\mathbf{KL}}(\mathfrak{c}, L)$ setting.

- The inequality p-WLSI, defined in Chapter 1, is valid for functions in $\mathbb{L}^p_m(\mathbb{R}^d)$, for $p > 2$, and we have proven that a semigroup $(P_t)_{t\geq 0}$ converges in entropy towards its invariant measure m, if m satisfies a p-WLSI, $p > 2$. This functional inequality was used in our work to obtain explicit bounds for the convergence in entropy of the ULA towards a weakly log-concave measure. It will be interesting to study the link between the functional inequality and the Modified LSI or the Latała-Oleszkiewicz inequality, which have been used in a similar setting.

Publications

Title: Stochastic Gradient Langevin Dynamics for (weakly) log-concave posterior distributions

Status: published in *Electronic Journal of Probability*, Vol. 29, 1-40.
DOI: 10.1214/24-EJP1235

Abstract: In this paper, we investigate a continuous time version of the Stochastic Gradient Langevin Dynamics, introduced in [Welling and Teh, 2011], that incorporates a stochastic sampling step inside the traditional over-damped Langevin diffusion. This method is popular in machine learning for sampling a posterior distribution. We will pay specific attention to the computational cost in terms of n (the number of observations that produces the posterior distribution), and d (the dimension of the ambient space where the parameter of interest is living). We derive our analysis in the weakly convex framework, which is parameterized with the help of the Kurdyka-Łojasiewicz (KL) inequality, that permits to handle a vanishing curvature settings, which is far less restrictive when compared to the simple strongly convex case. We establish that the final horizon of simulation to obtain an ε approximation (in terms of entropy) is of the order $(d\log^2(n))^{(1+r)^2}[\log^2(\varepsilon^{-1}) + n^2 d^{2(1+r)}\log^{4(1+r)}(n)]$ with a Poissonian sub-sampling, where the parameter r is involved in the KL inequality and varies between 0 (strongly log-concave case) and 1 (limiting Laplace situation).

Title: Discretisation of Langevin Diffusion in the weak log-concave case

Status: Preprint

Abstract: The Euler discretisation of Langevin diffusion, also known as Unadjusted Langevin Algorithm, is commonly used in machine learning for sampling from a given distribution $\mu \propto e^{-U}$. In this paper we investigate a potential $U : \mathbb{R}^d \longrightarrow \mathbb{R}$ which is a weakly convex function and has Lipschitz gradient. We parameterize the weak convexity with the help of the Kurdyka-Łojasiewicz (KL) inequality, that permits to handle a vanishing curvature settings, which is far less restrictive when compared to the simple strongly convex case. We prove that the final horizon of simulation to obtain an ε approximation (in terms of entropy) is of the order $\varepsilon^{-1}d^{3+2(1+r)^2}\mathrm{Poly}(\log(d), \log(\varepsilon^{-1}))$, where the parameter r is involved in the KL inequality and varies between 0 (strongly convex case) and 1 (limiting Laplace situation).

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